Industrial Capacity as a Military Deterrent*

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Abstract

Military outcomes depend not only on initial military strength but also on the capacity to expand it during war. We present a model in which industrial capacity can have a strategic role as a military deterrent. We study how a government can sustain peace through military deterrence at the lowest cost. We show that when wars are short, peacetime procurement of military equipment suffices because wartime production does not affect military outcomes. When wars are long, however, efficient deterrence also requires subsidies to industrial capacity because private investors do not internalize its strategic value as a military deterrent. Such industrial policy should favor technologies that allow the defense industrial base to scale output efficiently during war. We develop an Integrated Defense Economics Model to assess the welfare gains and identify priority sectors. Industrial policy can raise welfare by reducing the procurement expenditure required to sustain a given level of deterrence. Industrial priorities include munitions, guided missiles, and propulsion units, and the model shows how these priorities vary with the type of war.

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1 Introduction

The economic study of optimal defense policy—the allocation of goods, capital and labor to strengthen national defense—occupies a relatively small place in contemporary economic research. This relative neglect is surprising: national defense is among the state's most fundamental responsibilities and commands substantial public resources. Although early work thought about the distinctively economic foundations of national security such as the role of the defense industrial base, theoretical research on these issues has largely lain dormant since at least the Cold War. Therefore, economic theory provides little guidance on how policy should support the economic foundations of national security.

The need for such guidance is growing. Governments are not just responding to heightened security concerns by increasing their defense budgets but also pursuing policies to strengthen their defense industrial bases. The U.S. National Defense Industrial Strategy, for example, calls for investment in additional capacity and in the ability to expand production rapidly U.S. Department of Defense (2023). But the theoretical argument for such policies is unclear. Is it efficient for governments to support industrial capacity directly, or should they simply procure military equipment? What externality would justify direct support, and how does the case for intervention depend on the type of war a government expects to fight? And if support is warranted, how should it be targeted across sectors and technologies?

The usual neoclassical model of public-good provision does not provide the guidance needed to address these questions. It treats public-good provision as a production problem: the government procures military equipment and other inputs to provide military services, which enter household preferences. When embedded in a dynamic general equilibrium model with investment, this approach implies that, in the absence of other distortions, optimal procurement resolves the public-good problem and induces efficient investment in the industrial base.

We therefore develop a framework that embeds a strategic model of public-good provision in a dynamic general equilibrium model with investment. We model national security as peace sustained through military deterrence. In peacetime, the military thus provides a strategic service by deterring war. The key idea is that this role extends to the industrial base: beyond supplying military equipment, industrial capacity can itself act as a military deterrent. We begin with a simple baseline model to show how this strategic role depends on the expected duration of war and when it creates a national security externality that justifies direct peacetime support for the industrial base. We then use a more general model to study how support should be targeted across sectors.

Baseline Model Our model of military deterrence places industrial capacity at the center. Consider a country that faces an adversary who contemplates starting a war to challenge the status quo through military force. Such a war can either be long or short, and for simplicity we assume both parties know which it will be. During a war, territory changes hands depending on the relative strength of the militaries. Military strength includes both initial strength and any increases throughout the war. To sustain the peace, the country needs to make war sufficiently costly for its adversary. It can do so during peace by investing in both the size of its armed forces and the size of the defense industrial base.

The key point is that industrial capacity can act as a form of deterrence. Deterrence does not just depend on military equipment a country *has* at the start of the war but also on what it *can produce* over the duration of a longer war. An adversary understands that a country with larger industrial capacity will leverage it to expand military capacity throughout the war. In a way, the peacetime defense budget determines the army a country starts with; industrial capacity determines what it can end up with. Because it affects an adversary's calculus of aggression, industrial capacity is not just a supplier to the military but can directly contribute to deterrence.

In this framework, we ask how a government can sustain peace through deterrence at the lowest possible cost. We study two types of peacetime policy: *procurement* and *industrial* policy. Procurement consists of purchases of military equipment such as ammunition and tanks. Industrial policy influences the inputs firms use to produce that equipment. Since private firms choose investment in the defense industrial base, we focus on subsidies through which the government can encourage capital accumulation across sectors.[1]

Economists have traditionally distinguished sharply between these two types of policy. The classical policy prescription to solve the underprovision of public goods is to replace private provision with public provision—through procurement of military equipment in our case. Economists have often been skeptical of additional interventions such as industrial policies beyond such procurement policies. The seminal production efficiency result of Diamond and Mirrlees (1971) justifies this skepticism. It shows that industrial policy, as defined above, is indeed not needed for efficient public good provision in standard neoclassical models in which public goods are provided by the government by combining goods with capital or labor.

Throughout the paper, we emphasize the dynamic aspect of war. We have two reasons. First, as we will see, it matters for the argument for industrial policy. Second, assessments of the conflicts countries are likely to face are changing. Much of the eco-

[1] In practice, procurement contracts can combine what we call procurement and industrial policy by paying producers both for equipment and to retain additional production capacity.

nomic research on defense was conducted during the Cold War, when some influential analyses placed greater emphasis on forces available at the outset than on the capacity to expand production during war. For example, in *The Economics of Defense in the Nuclear Age* (1960), Hitch and McKean argued that technological advances had made wars requiring prolonged industrial mobilization relatively unlikely. In their assessment, the principal threats had shifted toward thermonuclear exchange and limited conflicts, placing greater weight on forces available at the outset. Capacity to expand wartime production therefore played a much smaller role in their account than it had during the world wars.[2]

The war in Ukraine has brought renewed attention to the possibility of prolonged conventional conflict, including conflicts involving a nuclear power. Such wars can quickly exhaust munitions stocks and destroy the equipment accumulated before the war. Wartime production during such conflicts can dwarf prewar inventories. Before World War II, the U.S. Army was the eighteenth largest, yet American industry subsequently produced roughly two-thirds of Allied military equipment (Herman, 2012). Industrial capacity—and the ability to expand it—therefore becomes central to military success. Reflecting renewed recognition of its role in deterrence, William LaPlante, while serving as Under Secretary of Defense for Acquisition and Sustainment, put it succinctly: "production itself is deterrence."[3] This renewed emphasis helps explain the widespread policy interest in expanding the defense industrial base and making it more scalable.[4]

Results Our first result provides a useful baseline: when wars are short, efficient deterrence only requires procurement policy. Wars end before wartime production can affect military outcomes, and so the defense industrial base has no direct role in deterrence. The

[2]This emphasis on forces available at the outset was especially pronounced in analyses of nuclear war. In a passage citing Hitch and McKean, Rowen (1960) argued that a nuclear war would be decided before the year or more required for mobilization, making forces maintained in peacetime much more important. NATO's 1957 strategic concept likewise prioritized existing forces, anticipating that nuclear destruction would severely restrict subsequent mobilization (NATO Military Committee, 1957). The emphasis also appeared in later Army doctrine: the 1976 edition of *FM 100-5: Operations* warned that a future war could be decided by initial combat, contrasting this possibility with America's historical reliance on resources mobilized after hostilities began (Department of the Army, 1976).

[3]LaPlante made this remark when discussing lessons from Ukraine (Clark, 2023). Similarly, NATO Secretary General Jens Stoltenberg argued that Ukraine's war of attrition had exposed insufficient Western industrial capacity to sustain such conflicts, prompting efforts to expand weapons and ammunition production (Stoltenberg, 2024).

[4]The U.S. National Defense Industrial Strategy calls for incentives to build spare capacity and develop flexible production (U.S. Department of Defense, 2023). The United Kingdom's Defence Industrial Strategy calls for maintaining flexible production capacity and assessing scalability in major procurements (Ministry of Defence, 2025). The European Defence Industrial Strategy proposes funding spare industrial capacity to enable rapid expansion during crises (European Commission and High Representative of the Union for Foreign Affairs and Security Policy, 2024). NATO's Updated Defence Production Action Plan likewise calls for industrial planning that addresses long-term scalability and surge requirements (NATO, 2025).

defense industrial base acts as a producer of military equipment, but this by itself is not a reason to intervene because market prices lead to the efficient investment in capacity. The short war view is an interesting benchmark because thinking about defense as a neoclassical public goods provision problem, while ignoring its strategic nature, essentially leads to the right economic policy recommendation.

When wars are long, this is no longer the case. Our second result shows that industrial policy will also be needed for efficient deterrence. When wars are long, industrial capacity acts as a military deterrent, but the private sector does not internalize this strategic benefit. This national security externality therefore generates underinvestment in the defense industrial base and hence justifies industrial policy that increases industrial capacity. Without industrial policy, firms choose capacity to meet relatively stable peacetime procurement. But a production structure that serves this demand at low cost need not be well suited to expanding output during war.

Our third result shows how this mismatch affects the composition of investment: industrial policy should encourage technologies that make the defense industrial base more scalable. We extend the baseline to include two production technologies that differ in their scalability—the extent to which they can expand output at low costs in the short run. By tilting the composition of investment towards the more scalable technology, industrial policy increases the capacity to expand wartime production. By changing the composition of technologies, industrial policy can raise deterrence even if the aggregate capital stock in the defense industrial base remains fixed.

The interaction between the two key model ingredients—deterrence as a public good and war duration—is central to these results. The former introduces a national security externality that can overturn the production-efficiency conclusion of Diamond and Mirrlees (1971) and create a role for industrial policy; the latter determines whether industrial capacity actually contributes to deterrence. Together, these ingredients provide a framework for understanding why economists working under the Cold War view of short wars were justified in paying relatively little attention to industrial policy for the defense industrial base. The same framework explains why the growing emphasis on prolonged wars provides a rationale for renewed support for industrial capacity and the accompanying emphasis on scalability.

Quantitative Exercise In the final part of our paper, we study how industrial policy should be targeted. When wars are long, policymakers must decide which industries to support alongside military procurement. These decisions depend on both the structure of the economy and how war is expected to unfold. We therefore develop an Integrated

Defense Economics Model (IDEM) to study how industrial capacity contributes to deterrence and study how policy should be targeted.

IDEM combines a dynamic general equilibrium economic block with a military block. The economic block captures military producers' reliance on other industries, both for intermediate inputs and for the capital goods needed to expand production during war. The military block combines different types of equipment into military power, allowing their relative importance to vary across conflicts. War duration is uncertain and depends on the military balance: a sufficiently strong aggressor may end a war quickly, while a prolonged conflict gives industrial capacity time to shape the outcome. We calibrate the model using U.S. production and investment networks and historical evidence on war durations and military buildups.

We find that industrial policy raises welfare relative to procurement policy alone. It raises steady-state consumption by reducing military expenditure and raising the aggregate capital stock, holding deterrence fixed. Optimal support is concentrated in a relatively narrow set of industries, providing quantitative support for the view that the national security argument for industrial policy applies to only a limited set of industries. Priority sectors include guided missiles, propulsion units, and ammunition. Missiles and ammunition can be stockpiled but are expended in combat, so demand for them can rise especially sharply during war. Our exercise also suggests that industrial policy should depend on the conflict being deterred. A land war in Europe places greater emphasis on tank production, while a war over Taiwan favors aircraft and shipbuilding. When wars are long, economic policymaking therefore cannot be separated from military analysis. We conclude by discussing how IDEM could support closer integration between the two.

Related Literature The framework developed in this paper brings together two literatures that study different aspects of war or military buildups: their economic consequences and the strategic decisions that generate them. The theoretical argument for industrial policy follows from the interaction between the strategic nature of public-good provision and the fact that it is often the private sector that invests in industrial capacity. We draw particular inspiration from Hitch and McKean (1960), which offers the clearest account of the interplay between the defense industrial base, military capacity, deterrence, and the nature of warfare. Our contribution is to provide a formal framework of these interactions, characterize the resulting rationale for industrial policy, and operationalize it through IDEM.

The neoclassical literature in macroeconomics and public finance uses general equilibrium models to study investment, production, and government policy in response to war

or military buildups. It typically treats these events as exogenous shocks and abstracts from the strategic decisions behind them. Empirical work in macroeconomics has used military spending to estimate fiscal multipliers (Ramey, 2011; Barro and Redlick, 2011; Nakamura and Steinsson, 2014).[5] Recent work studies military buildups in their own right (see Ilzetzki (2025) for a review), examining their effects on military-goods prices Antonova et al. (2025) and their scale during major wars Marzian and Trebesch (2026). We draw on this evidence in our quantitative exercise.

The strategic literature in political economy and international relations uses game-theoretic models to study war or military buildups as endogenous outcomes of countries' assessments of the costs and benefits of conflict. It typically abstracts from private-sector investment decisions and from features of the economy such as input–output networks. Building on Schelling (1960, 1966), it recognizes that because war is costly, national security policy is often about preventing conflict through deterrence. Work in this tradition includes Mearsheimer (1983); Baliga et al. (2020); Baliga and Sjöström (2020); Luo and Wolitzky (2025); Herrera et al. (2022); Di Lonardo and Tyson (2022, 2026).

By providing a rationale for supporting the defense industrial base, our paper also contributes to the industrial policy literature (reviewed in Juhász et al. (2024)). This literature has studied the empirical effects of defense-related policies, including military procurement, public R&D, and direct investment in manufacturing capacity Lichtenberg (1988); Gross and Sampat (2023); Ilzetzki (2024); Moretti et al. (2025); Garin and Rothbaum (2025), but formal arguments for prioritizing the defense industrial base in peacetime remain less developed.[6] Thompson (1979) provides a rationale in a model in which wartime price controls generate underinvestment during peace. Kooi (2025) formalizes and quantifies a strategic rationale based on the benefits of lowering the domestic cost of war. Lower costs can strengthen bargaining power or, in a deterrence model, make a country's commitment to fight if the status quo is challenged more credible. Our baseline takes this commitment as given, so this motive is absent. Section 3.5 extends our model to incorporate this additional strategic rationale and clarify how it is distinct from the one developed here.

[5]There is also a related public finance literature that initially focused on war finance Pigou (1921); Keynes (1940) but now more commonly studies fiscal policy over the business cycle Barro (1979); Lucas and Stokey (1983); Bhandari et al. (2017).

[6]Standard rationales for industrial policy, such as knowledge spillovers and credit constraints, can in principle justify support for the defense industrial base. To justify prioritizing defense over other sectors, however, these rationales would need to be quantitatively more important in defense. This is not obvious: knowledge spillovers can also pose national security risks, while defense firms are often large and established, potentially limiting their exposure to credit constraints. An argument grounded in the distinctive nature of public-good provision in national defense therefore has particular appeal.

Finally, our paper relates to the nascent geoeconomics literature (see Mohr and Trebesch (2025) for a recent review). This literature also studies the interaction between economic structure and geopolitics, with much recent work emphasizing trade and financial relationships as sources of geopolitical power rather than military production. Our model shares structural features with work combining national-level bargaining and private-sector investment in irreversible capital McLaren (1997); Kooi (2025); Becko and O'Connor (2025). Kooi (2025) highlights similarities between the policy implications of bargaining in the shadow of trade and military conflict. Other work examines the links between military security and international economic relationships Alekseev and Lin (2025); Mayer et al. (2025); Adão and Kooi (2026). Further contributions to geoeconomics include Clayton et al. (2024, 2026) and Liu and Yang (2025).

2 Model

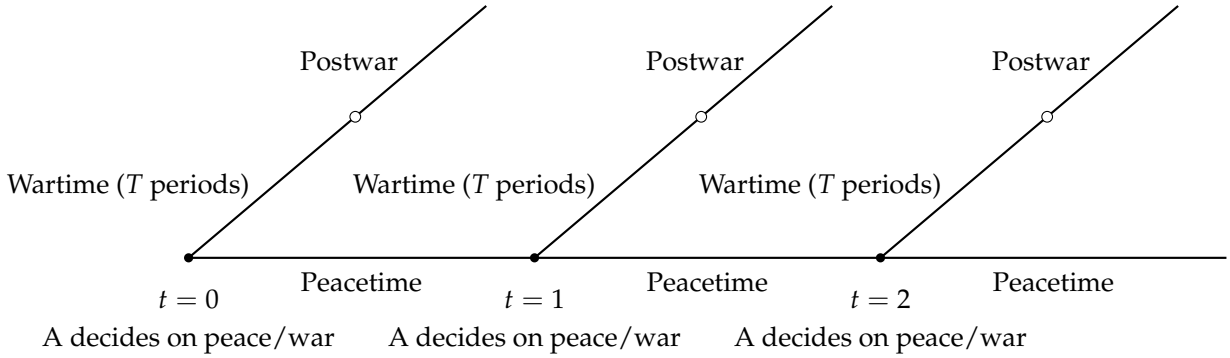
Overview There are two countries, Sovereign and Adversary, indexed by $i \in \{S, A\}$, each inhabited by a representative agent. Time is discrete and infinite, with periods denoted $t = 0, 1, \ldots$.

The countries compete over a divisible public good, which we interpret as territory. Both countries value territory, but each unit can belong to only one of them. Its allocation remains at the status quo during peace and can change only through war. At the end of a war, territory is reallocated according to the countries' relative military strength over the course of the conflict. This strength reflects both their initial military stocks and the additional military equipment accumulated during the war. The resulting territorial allocation remains fixed thereafter.

The economy begins in peace. At the beginning of each period, Adversary decides whether to maintain the peace or start a war. A war lasts one or two periods, $T \in \{1, 2\}$, which we refer to as short and long wars, respectively. Its duration is exogenous and known to both countries. The key difference is that during long wars a country can produce additional military equipment through its defense industrial base but during short wars they cannot. War can occur only once: after it ends, economic activity continues but no further war is possible.

Figure 1 illustrates this timing. We study how Sovereign can use military procurement and industrial policies that increase the capital stock of the defense industrial base to sustain the peace, so the economy follows the horizontal peacetime path in the equilibria we consider. The wartime and postwar branches describe what would happen if Adversary attacked. These possible continuations are central to the analysis because they determine Adversary's willingness to break the peace and challenge the status quo through military force.

Figure 1: Timing of peace, war, and the postwar period.



2.1 Economic Block

We begin by describing the economic block of the model, which is common to both countries. Households supply labor and own firms, while the government procures military goods and can subsidize capital in the defense industrial base. We denote the geopolitical state by $z_t \in \{P, C\}$, where $z_t = P$ if no war has begun and $z_t = C$ once a war has begun, including the postwar period. In what follows, all equilibrium variables are, in general, functions of the history $z^t = (z_0, z_1, \ldots, z_t)$, but whenever no confusion is possible we leave this dependence implicit.

Preferences The representative agent in each country has preferences given by

$$U^i = \sum_{t=0}^{\infty} \beta^t \left(G_t^i + c_t^i \right), \tag{1}$$

where $\beta \in (0, 1)$ is the discount factor and $\{c_t^i\}_{t=0}^{\infty}$ is the sequence of consumption. Here G_t^i is the public good introduced above, which is not chosen by households but is instead determined by geopolitics.

Capital Accumulation There are two capital stocks in this economy. There is a stock of military capital, M_t^i, which represents the strength of the armed forces and depreciates at rate δ_M. There is also capital in the defense industrial base, K_t^i, which serves as an input into the production of military capital and depreciates at rate δ_K. Let m_t^i denote government purchases of new military equipment and i_t^i investment in industrial capital during period t. The two capital stocks therefore evolve according to

$$M_{t+1}^i = M_t^i(1 - \delta_M) + m_t^i \tag{2}$$

$$K_{t+1}^i = (1 - \delta_K)K_t^i + i_t^i. \tag{3}$$

Both stocks are predetermined at the beginning of period t. Purchases and investment during period t augment the stocks available in period $t + 1$.

Production There are two production technologies in this economy. Output of the numeraire good is denoted by $y_{n,t}^i$ and is produced through a linear technology that uses labor $l_{n,t}^i$ as input. The defense sector produces military goods, denoted by $y_{m,t}^i$, which are used to accumulate military capital. Military goods are produced using a Cobb–Douglas technology with $\alpha \in (0, 1)$ that takes K_t^i and labor, $l_{m,t}^i$, as inputs. We normalize produc-

tivity in military goods production to one. The two technologies are given by

$$y_{n,t}^i = l_{n,t}^i \tag{4}$$

$$y_{m,t}^i = \left(K_t^i\right)^\alpha \left(l_{m,t}^i\right)^{1-\alpha}. \tag{5}$$

Market Clearing The numeraire good can be used for consumption or investment in the defense industrial base. The military good is used only to accumulate military capital. The economy is endowed with $\bar{l}^i$ units of labor, which are allocated between the numeraire sector and the defense sector. The three market-clearing conditions are therefore given by

$$c_t^i + i_t^i = y_{n,t}^i \tag{6}$$

$$m_t^i = y_{m,t}^i \tag{7}$$

$$l_{n,t}^i + l_{m,t}^i = \bar{l}^i. \tag{8}$$

Competitive Equilibrium Households choose consumption to maximize utility subject to their budget constraint, taking as given the price of the numeraire good $p_{n,t}^i$, the wage w_t^i, firm profits Π_t^i, and the lump-sum tax T_t^i levied by the government. They have access to a risk-free bond with return r_t^i between periods t and $t+1$, which is available in zero net supply. Let $R_t^i = \prod_{u=0}^{t-1}(1+r_u^i)$ denote the cumulative gross return from date 0 to date t, with $R_0^i = 1$. The household's problem is

$$\max_{\{c_t^i\}_{t=0}^\infty} \left\{ \sum_{t=0}^\infty \beta^t c_t^i \,\middle|\, \sum_{t=0}^\infty \frac{1}{R_t^i}\left(p_{n,t}^i c_t^i - w_t^i \bar{l}^i - \Pi_t^i + T_t^i\right) = 0 \right\}. \tag{9}$$

The producers of the numeraire good solve a static problem in which they maximize profits by hiring labor:

$$\max_{y_{n,t}^i, l_{n,t}^i} \left\{ p_{n,t}^i y_{n,t}^i - w_t^i l_{n,t}^i \,\middle|\, (4),\ l_{n,t}^i \geq 0 \right\}. \tag{10}$$

The defense industrial base maximizes profits by choosing investment and labor to produce military goods. It takes as given the price of military goods, $p_{m,t}^i$, and the subsidy on its capital stock, s_t^i, which the government uses to affect investment decisions. The subsidy is a payment of s_t^i per unit of installed capital in period t. Given its initial capital

stock K_0^i, its optimization problem is therefore

$$\max_{\{y_{m,t}^i, l_{m,t}^i, i_t^i, K_{t+1}^i\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \frac{1}{R_t^i} \left(p_{m,t}^i y_{m,t}^i + s_t^i K_t^i - w_t^i l_{m,t}^i - p_{n,t}^i i_t^i \right) \middle| (3),\ (5),\ l_{m,t}^i \geq 0,\ i_t^i \geq 0 \right\}. \quad (11)$$

The government raises revenue to fund the subsidies it pays to the defense industrial base and to procure military goods. It uses a lump-sum tax to balance its budget period by period and starts with no initial debt. Its budget constraint is therefore given by

$$T_t^i = p_{m,t}^i m_t^i + s_t^i K_t^i. \quad (12)$$

Definition 1 (Competitive Equilibrium)**.** *Taking as given the sequence of geopolitical states $\{z_t\}_{t=0}^{\infty}$, the sequence of government purchases and subsidies $\{m_t^i, s_t^i\}_{t=0}^{\infty}$, and the initial capital stocks K_0^i and M_0^i, a competitive equilibrium in country i is a sequence of prices $\{r_t^i, p_{n,t}^i, p_{m,t}^i, w_t^i\}_{t=0}^{\infty}$, quantities $\{c_t^i, y_{n,t}^i, y_{m,t}^i, i_t^i, K_t^i, M_t^i, l_{n,t}^i, l_{m,t}^i\}_{t=0}^{\infty}$, and taxes and profits $\{T_t^i, \Pi_t^i\}_{t=0}^{\infty}$ such that households and firms solve their optimization problems, markets clear, and the government budget constraint holds in every period.*

2.2 Deterrence Block

We now describe the deterrence block. We use t to index calendar time and write peacetime quantities without a regime superscript. When considering a war that begins at date t, we index its continuation by $\tau = 0, 1, \ldots$, where $\tau = 0$ is the first period of war. For a generic variable X, we write $X_\tau^{i,C}$ for its value τ periods after the outbreak, including the postwar period. Since capital stocks are predetermined when Adversary decides whether to attack, the initial stocks in this continuation satisfy $K_0^{i,C} = K_t^i$ and $M_0^{i,C} = M_t^i$.

The Allocation of the Public Good Countries are endowed with an initial allocation of the public good, $\bar{G}^i$. During peacetime, the allocation of the public good is fixed at its initial level. This implies that

$$G_t^i = \bar{G}^i. \quad (13)$$

The allocation of the public good can only be changed through war. At the end of the war, the public good is reallocated based on the parties' relative military capacity throughout the war. The duration of the war, $T \in \{1, 2\}$, is exogenous and known to both parties. The technology is linear, and the parameter $\chi > 0$ captures the effectiveness of military power in obtaining territory. The new allocation applies from the final period of war onward and

remains fixed thereafter:

$$G_{\tau}^{i,C} = \begin{cases} \bar{G}^i, & \tau < T-1, \\ \bar{G}^i + \chi \sum_{u=0}^{T-1} \left(M_u^{i,C} - M_u^{-i,C} \right), & \tau \geq T-1, \end{cases} \tag{14}$$

where $-i$ denotes the other country.

Adversary's Policy Rules Adversary's government is assumed to follow simple policy rules that keep its military and industrial capital stocks at their initial levels, denoted by $\bar{M}^A$ and $\bar{K}^A$, respectively. It replaces depreciated military equipment and industrial capital in every period, during peace, war, and the postwar period.

The reason for restricting attention to such simple rules is that the focus is on the strategic interaction between Sovereign's military procurement and industrial policies, on the one hand, and Adversary's aggression decision, on the other. Allowing for richer behavioral responses by Adversary would introduce additional strategic interactions. Section 3.5 explains how to incorporate these and discusses their implications for policy.

Adversary's Challenge Decision At the beginning of any peacetime period, Adversary can challenge the status-quo allocation of territory by starting a war. Let $W_t^{A,P}(\mathbf{x}_t)$ denote its discounted welfare from maintaining peace at date t, and let $W_t^{A,C}(\mathbf{x}_t)$ denote its discounted welfare from starting a war at that date. Both values include consumption and public-good flows from date t onward. Here $\mathbf{x}_t = \{G_t^i, K_t^i, M_t^i\}_{i \in \{S,A\}}$ denotes the vector of state variables at the time of the decision. Choosing peace at date t leaves open the possibility of starting a war at a later date; the superscript P therefore refers to the current decision, rather than a commitment to permanent peace.

Let η_t^A denote Adversary's additional political payoff from going to war. Adversary starts a war whenever

$$W_t^{A,C}(\mathbf{x}_t) + \eta_t^A > W_t^{A,P}(\mathbf{x}_t). \tag{15}$$

A higher η_t^A therefore makes Adversary more willing to attack. When indifferent, it maintains the peace.

2.3 Deterrence Equilibrium

We consider equilibria in which Adversary maintains the peace at every point in time. We define deterrence as

$$\mathcal{D}_t(\mathbf{x}_t) \equiv W_t^{A,P}(\mathbf{x}_t) - W_t^{A,C}(\mathbf{x}_t). \tag{16}$$

This measures Adversary's welfare advantage from maintaining peace, before including the additional political payoff from attacking. Adversary is deterred when this advantage is at least as large as η_t^A. We therefore define a deterrence equilibrium as follows.

Definition 2 (Deterrence equilibrium)**.** *A deterrence equilibrium is a deterministic sequence of quantities, prices, subsidies, taxes, and profits that (i) is consistent with the competitive equilibrium and (ii) sustains the peace by satisfying the deterrence constraint*

$$\mathcal{D}_t(\mathbf{x}_t) \geq \eta_t^A, \quad \forall t. \tag{17}$$

There may be many allocations that satisfy these equilibrium conditions.[7] Specifically, there may be several feasible combinations of military and defense-industrial capital that satisfy the deterrence constraint (17) and can be implemented in a competitive equilibrium. Section 3 studies which policies Sovereign needs in order to attain the deterrence equilibrium that maximizes domestic welfare. First, however, it is useful to characterize the deterrence constraint more explicitly.

2.4 Simplifying the Deterrence Constraint

Our definition of equilibrium is expressed in terms of $\mathcal{D}_t(\mathbf{x}_t)$. As will become clear in the next section, the characterization of optimal subsidies applies to general deterrence functions. However, our comparative statics use the microfoundations of $\mathcal{D}_t(\mathbf{x}_t)$. In this section, we show how our assumptions simplify this function.

Adversary's Welfare during Peace We begin by deriving an expression for Adversary's welfare during peace. Because Adversary maintains its military and industrial capital stocks at $\bar{M}^A$ and $\bar{K}^A$, its consumption is constant. It equals the total labor endowment minus the labor used to produce replacement military equipment and industrial capital:

$$c_t^A = \bar{c}^A \equiv \bar{l}^A - \delta_K \bar{K}^A - \left(\frac{\delta_M \bar{M}^A}{(\bar{K}^A)^\alpha}\right)^{\frac{1}{1-\alpha}}. \tag{18}$$

[7] We assume that a feasible deterrence equilibrium exists even in the absence of industrial policy.

In a deterrence equilibrium, Adversary expects peace to continue at every future date. Its peacetime welfare is therefore

$$W_t^{A,P} = \frac{\bar{G}^A + \bar{c}^A}{1-\beta}. \tag{19}$$

Sovereign's Wartime Problem Adversary's welfare if it chooses war depends in part on how much military equipment Sovereign produces during the conflict. We therefore begin by studying Sovereign's optimal policy during war.

Consider a war beginning at date t. Sovereign enters the conflict with capital stocks K_t^S and M_t^S and the status-quo territorial allocation $\bar{G}^S$. Substituting the conflict technology (14) into the household objective function yields the following policy problem:

$$\max_{\substack{\{c_\tau^{S,C}, y_{n,\tau}^{S,C}, y_{m,\tau}^{S,C}, l_{n,\tau}^{S,C}, l_{m,\tau}^{S,C}, \\ i_\tau^{S,C}, m_\tau^{S,C}, K_{\tau+1}^{S,C}, M_{\tau+1}^{S,C}\}_{\tau=0}^{\infty}}} \left\{ \sum_{\tau=0}^{\infty} \beta^\tau \left(c_\tau^{S,C} + \bar{G}^S \right) + \frac{\chi \beta^{T-1}}{1-\beta} \sum_{\tau=0}^{T-1} \left(M_\tau^{S,C} - \bar{M}^A \right) \,\middle|\, (2)\text{–}(8) \right\}. \tag{20}$$

The constraints are evaluated along the conflict continuation, with initial stocks inherited from date t and all allocations subject to non-negativity constraints. Since $\bar{G}^S$ and $\bar{M}^A$ enter the objective additively and separably, Sovereign's optimal military stocks depend only on its inherited capital stocks, holding parameters fixed. We denote this response by $M_\tau^{S,C}(K_t^S, M_t^S)$.

Adversary's Welfare during War Having characterized Sovereign's military stocks as a function of its inherited capital stocks, we can now characterize Adversary's welfare during war. Adversary maintains the same consumption during peace and conflict, so $c_\tau^{A,C} = \bar{c}^A$. Substituting Sovereign's military stocks into the conflict technology (14), Adversary's welfare if it attacks at date t is

$$W_t^{A,C} = \frac{\bar{G}^A + \bar{c}^A}{1-\beta} + \frac{\chi \beta^{T-1}}{1-\beta} \sum_{\tau=0}^{T-1} \left(\bar{M}^A - M_\tau^{S,C}(K_t^S, M_t^S) \right). \tag{21}$$

The Simplified Deterrence Constraint We now combine (19) and (21) to simplify the deterrence constraint. The common consumption and status-quo territorial terms cancel. Since the capital stocks available when war begins are predetermined, the wartime response is evaluated at Sovereign's peacetime stocks K_t^S and M_t^S. Keeping Adversary's fixed military stock $\bar{M}^A$ implicit as a parameter, we can therefore write deterrence as

$$\mathcal{D}_t(K_t^S, M_t^S) = \frac{\chi \beta^{T-1}}{1-\beta} \sum_{\tau=0}^{T-1} \left(M_\tau^{S,C}(K_t^S, M_t^S) - \bar{M}^A \right) \tag{22}$$

where $M_0^{S,C} = M_t^S$ for both short and long wars. For long wars,

$$M_1^{S,C} = (1-\delta_M)M_t^S + m_0^{S,C}(K_t^S), \quad m_0^{S,C}(K) = \min\left\{ \left[\tfrac{\chi\beta(1-\alpha)}{1-\beta}\right]^{\frac{1-\alpha}{\alpha}} K,\ K^{\alpha}(\bar{l}^S)^{1-\alpha} \right\}. \tag{23}$$

The first term in the minimum describes production when some labor remains in the numeraire sector; the second is the maximum output attainable with full labor mobilization. Military output as a function of capital is therefore continuous but kinked at a single point. For expositional ease, the main text focuses on allocations at which the deterrence function is differentiable, without further comment; the proofs also treat the kink.[8]

Deterrence is increasing in the amount of military power deployed throughout the war.[9] The deterrence constraint used going forward is therefore

$$\mathcal{D}_t(K_t^S, M_t^S) \geq \eta_t^A. \tag{24}$$

2.5 Discussion

Before turning to optimal economic policy, we briefly comment on some of the modelling assumptions.

The Asymmetries of Deterrence In our model, there is a clear asymmetry in the security problem for the two countries. Adversary is the challenger to the status quo, whereas Sovereign is its defender. We view this as a useful simplification that allows us to develop the strategic argument for industrial policy. But such asymmetries are also quite natural from a historical perspective. When the balance of geopolitical power changes—for example, because the economies of rising powers grow faster than those of existing powers (Kennedy, 1987)—existing powers find themselves trying to preserve a status quo that favors them for historical reasons. The assymetry thus arises endogenously from changes in the balance of power. We view this as a useful description of the security problem facing Western countries seeking to deter challenges to the geopolitical status quo by China and Russia.

[8] We could instead restrict attention to cases in which the wartime labor constraint is slack. The strictly binding case is extreme: all labor is devoted to military production, yet Sovereign would like to expand military output further. We retain this case to clarify the theoretical point in Section 3.2 that capital is not merely a commitment device: it continues to strengthen deterrence even under full labor mobilization, the wartime allocation Sovereign would promise under full commitment.

[9] The model can be extended to allow Adversary's stocks to follow exogenous paths that differ between peace and conflict. These paths would change the level of $\mathcal{D}_t$ but leave its marginal effects with respect to K_t^S and M_t^S unchanged. Such shifts in the deterrence constraint can be represented through changes in η_t^A.

Policy Instruments We study a specific form of industrial policy in the form of subsidies to the capital stock. In these types of models, there are in general many policies the government could use to implement its desired allocation of capital, including investment subsidies or direct government control of the defense industrial base. The subsidies are therefore best interpreted as wedges that clarify when the government should deviate from a competitive equilibrium without industrial policy rather than as a prescription for the use of a particular policy instrument.

Deterrence and Beliefs We have emphasized that the duration of war matters for deterrence. In the baseline model, countries have rational expectations in the sense that they agree on T, and their beliefs coincide with the true duration of war. But because war never occurs in a deterrence equilibrium, what matters for deterrence is how Adversary expects a war to unfold if it attacks. Sovereign must therefore take into account Adversary's beliefs about the duration of war, which may not always coincide with its true duration. For example, Mearsheimer (1983) argues that deterrence is more likely to fail when an adversary believes that a war will be short—that its troops will be home by Christmas. Misperceptions can also go in the opposite direction: leaders may expect wars to be long even when they turn out to be short.[10] The argument for industrial policy does not require Adversary's beliefs about war duration to be correct. What matters is that Adversary assigns some positive probability to a war lasting long enough for Sovereign's wartime production to affect the outcome. Of course, when Adversary believes that wars are likely to be short, it places less weight on Sovereign's ability to produce additional equipment, weakening the deterrent value of industrial capital.

3 Optimal Economic Policy to Sustain the Peace

In this section, we study how the government should use procurement and industrial policy to maximize domestic welfare while sustaining peace through deterrence. After stating the peacetime planning problem, we explain when and why capital in the defense industrial base contributes to deterrence. We then characterize the optimal policy and explain why, in the absence of policy, the market and the planner solve different cost-minimization problems. Finally, we show why optimal industrial policy can encourage investment in technologies that allow military production to scale during war.

[10] For example, Hitler was preparing during the fall of 1939 for a long, drawn-out war against France, yet the campaign ended after only six weeks. See chapter 10 in Tooze (2007).

3.1 Sovereign's Peacetime Planning Problem

The central question is how Sovereign should choose investment in military capital and in the defense industrial base in order to attain the deterrence equilibrium that maximizes domestic welfare. Given initial capital stocks K_0^S and M_0^S, we therefore focus on the following planning problem:

$$\max_{\{c_t^S, y_{n,t}^S, y_{m,t}^S, l_{n,t}^S, l_{m,t}^S, i_t^S, m_t^S, K_{t+1}^S, M_{t+1}^S\}_{t=0}^{\infty}} \quad \sum_{t=0}^{\infty} \beta^t \left(\bar{G}^S + c_t^S\right)$$

s.t. (i) Technology and market-clearing constraints (2)–(8), for all t,

(ii) Deterrence constraint (24), for all $t \geq 1$,

(iii) Non-negativity constraints, for all t.

The planner chooses only peacetime allocations. In the deterrence equilibria we study, wars are off the equilibrium path, and the planner is not allowed to commit to off-path allocations. If war begins, Sovereign solves the wartime problem in Section 2.4 given its inherited capital stocks. This response is already incorporated into the deterrence constraint.

Note that this is a primal planning problem. The planner chooses the entire sequence of peacetime allocations directly. Household and firm optimization constraints are not imposed explicitly. Instead, we later verify that the allocations solving this problem can be decentralized as a competitive equilibrium using procurement and capital subsidies.

3.2 The Optimal Deterrence Decision

In this section we study the optimal investment decisions and explain when and why capital in the defense industrial base contributes to deterrence.

Optimal Investment At an interior peacetime allocation, the optimality conditions for investment in military capital and defense capital are, respectively,

$$\lambda_{m,t}^S = \beta \left(\gamma_{t+1}^S \frac{\partial \mathcal{D}_{t+1}}{\partial M_{t+1}^S} + (1 - \delta_M) \lambda_{m,t+1}^S \right) \tag{25}$$

$$1 = \beta \left(\gamma_{t+1}^S \frac{\partial \mathcal{D}_{t+1}}{\partial K_{t+1}^S} + (1 - \delta_K) + \lambda_{m,t+1}^S \alpha \left(K_{t+1}^S\right)^{\alpha-1} \left(l_{m,t+1}^S\right)^{1-\alpha} \right) \tag{26}$$

where γ_t^S is the multiplier on (24) and $\lambda_{m,t}^S$ is the multiplier on (2), both at date t.

Equation (25) characterizes optimal investment in the stock of military capital. Its value is determined by two forces: its deterrent value next period, given by the shadow value of deterrence γ_{t+1}^S multiplied by its marginal contribution to deterrence, and the value of the equipment that remains after depreciation. Equation (26) characterizes optimal investment in the defense industrial base. The planner values defense capital both because it contributes directly to deterrence and because it raises military-goods output, as captured by the final term. As we show below, the productive role of defense capital during peace is reflected in market prices, whereas its role as a deterrent is not.

The Deterrent Value of Military and Industrial Capacity Before turning to the analysis of optimal policy, it is useful to clarify the determinants of the deterrent value of capital.

Proposition 1 (Capital and Deterrence)**.** *Suppose wars are short ($T = 1$). Then the marginal deterrent values of military and industrial capacity at a possible attack date t are*

$$\frac{\partial \mathcal{D}_t}{\partial M_t^S} = \frac{\chi}{1-\beta} > 0, \qquad \frac{\partial \mathcal{D}_t}{\partial K_t^S} = 0.$$

Suppose instead that wars are long ($T = 2$) and some labor remains in numeraire production in the first period of war. Then the deterrent values of military and industrial capacity are

$$\frac{\partial \mathcal{D}_t}{\partial M_t^S} = \frac{\chi\beta}{1-\beta}\Big(1 + (1-\delta_M)\Big) > 0, \qquad \frac{\partial \mathcal{D}_t}{\partial K_t^S} = \left(\frac{\chi\beta}{1-\beta}\right)^{\frac{1}{\alpha}} (1-\alpha)^{\frac{1-\alpha}{\alpha}} > 0. \tag{27}$$

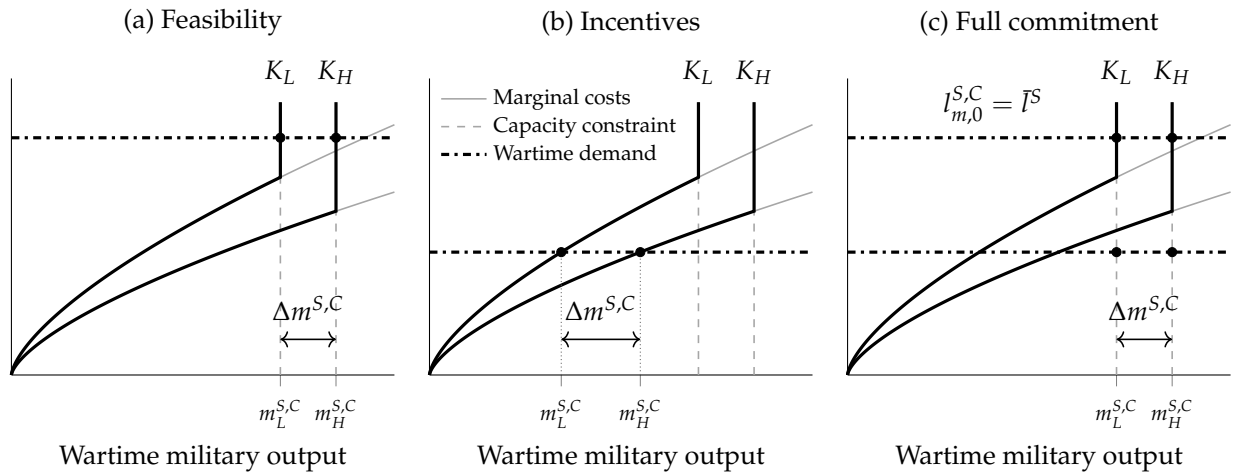
Suppose instead that wars are long ($T = 2$) and the labor constraint binds strictly in the first period of war, so all labor is allocated to military production. Then the deterrent values of military and industrial capacity are

$$\frac{\partial \mathcal{D}_t}{\partial M_t^S} = \frac{\chi\beta}{1-\beta}\Big(1 + (1-\delta_M)\Big) > 0, \qquad \frac{\partial \mathcal{D}_t}{\partial K_t^S} = \frac{\chi\beta}{1-\beta}\alpha \left(K_t^S\right)^{\alpha-1} \left(\bar{l}^S\right)^{1-\alpha} > 0. \tag{28}$$

Proof. Found in Appendix A.2. □

The proposition makes three points. First, military capacity deters in both short and long wars. Second, holding the initial military stock fixed, industrial capacity acts as a deterrent only when wars are long. The reason is that, in this model, wartime production can affect battlefield outcomes only if there is time to build before the war ends. Industrial capacity can still contribute to building the peacetime military stock before a short war. Third, the deterrent value of the defense industrial base depends on whether wartime production is limited by incentives or feasibility. We discuss these cases next.

Figure 2: Industrial capital and wartime military production



Note: All panels use the same supply curves; $m_L^{S,C}$ and $m_H^{S,C}$ denote wartime output at $K_L < K_H$. Under full commitment, Sovereign promises full labor mobilization at either demand level.

Incentive Compatibility versus Feasibility Why does capital contribute to deterrence? In this model, it does so during long wars because, holding initial military stocks fixed, an increase in K raises wartime military production $m^{S,C}(K)$ and thereby reduces Adversary's gain from attacking. There are two reasons why capital can raise wartime production, illustrated in Figure 2.

First, capital can relax a *feasibility constraint*, as shown in panel (a). Sovereign would like to expand military production further, but all available labor is already allocated to it. Production is therefore limited by the capacity of the defense industrial base. Raising capital from K_L to K_H allows the same labor endowment to produce more military equipment, shifting the capacity constraint outward and strengthening deterrence.

Second, capital can relax a *wartime incentive compatibility constraint* arising from limited commitment. During peace, Sovereign would like to commit to allocating all available labor to military production if war begins. The prospect of a larger wartime buildup strengthens deterrence, while the costs of carrying it out are never incurred in a deterrence equilibrium. Once war begins, however, Sovereign must weigh the military benefits of production against the loss of consumption. It may therefore prefer to retain some labor in numeraire production rather than carry out the promised buildup. This difference between what Sovereign would like to promise during peace and what it would choose during war is the time-inconsistency problem.

Panel (b) illustrates this case: wartime demand intersects the upward-sloping part of the supply curve, so some labor remains in numeraire production. Here, an increase in capital raises wartime military output through two effects. First, it raises output for

a given allocation of labor, just as in the feasibility-constrained case. Second, it raises the marginal product of military labor, inducing Sovereign to reallocate additional labor toward military production. This latter effect brings the labor allocation closer to the full-commitment solution in which all available labor is mobilized. It is this crowding-in of military labor that we refer to as relaxing the wartime incentive compatibility constraint.

Finally, panel (c) shows that capital continues to deter even when full commitment removes the wartime incentive compatibility constraint and the associated time-inconsistency problem. Sovereign promises full labor mobilization at either demand level, raising wartime production relative to panel (b) but leaving it unchanged relative to panel (a). With labor already fully mobilized, additional capital cannot crowd in more military labor. It nevertheless raises the output that the same labor endowment can produce. Capital therefore continues to strengthen deterrence by relaxing the feasibility constraint.

3.3 Optimal Industrial Policy

In this section we first provide a sufficient statistic for optimal industrial policies. We then explain why industrial policy can be needed to efficiently deter long but not short wars. Lastly, we clarify the market failure by explaining why procurement policy alone does not minimize the cost of deterrence during long wars.

Optimal Subsidies We begin by studying the firm's decision to invest in capital in the defense industrial base. At an interior allocation, its first-order condition for optimal investment can be expressed as

$$1 = \beta\Big((1-\delta_K) + p^S_{m,t+1}\alpha(K^S_{t+1})^{\alpha-1}(l^S_{m,t+1})^{1-\alpha} + s^S_{t+1}\Big). \tag{29}$$

It shows that, in the absence of industrial policy, the firm invests because it increases its output, which it gets to sell at market prices. It does not consider the effect of its investment on the deterrence constraint. Each competitive firm takes prices and government procurement as given. Slackening the deterrence constraint reduces the resources needed to sustain deterrence and benefits all households, but the firm receives no payment for this contribution. At the allocation being implemented, $p^S_{m,t} = \lambda^S_{m,t}$, so comparison with (26) gives the following result.

Proposition 2. *An optimal deterrence equilibrium can be implemented by choosing procurement*

optimally and setting subsidies equal to

$$s_t^S = \gamma_t^S \frac{\partial \mathcal{D}_t}{\partial K_t^S}. \tag{30}$$

Proof. Delegated to Appendix A.3. □

The expression shows that industrial policy is used because capital has a strategic value as a deterrent. The subsidy equals its marginal contribution to deterrence multiplied by the shadow value of deterrence. The other reason for a firm to value capital – because it uses it to produce output – is valued appropriately because it receives the market price for that output and hence does not appear in the expression.

We now turn to a key comparative static. The next proposition shows how optimal industrial policy depends on the length of the war.

Proposition 3. *Suppose wars are short ($T = 1$). Then the planner does not use industrial policy, $s_t^S = 0$. Suppose instead wars are long ($T = 2$) and the deterrence constraint binds, $\gamma_t^S > 0$. Then optimal subsidies are positive, $s_t^S > 0$, regardless of whether it is the incentive compatibility or feasibility constraint that binds.*

Proof: Immediate from the above propositions.

The proposition shows that whether there is a role for industrial policy is endogenous to the nature of the war. When wars are short, the defense industrial base does not add to deterrence holding the initial military stock fixed, and so industrial policy is not needed. When wars are long and the deterrence constraint has a positive shadow value, it does, and hence industrial policy is needed.[11]

On the Difference between the Market and the Planner Lastly, we explain why the competitive equilibrium can be inefficient in the absence of industrial policy. We show that the planner minimizes the cost of deterrence while the market minimizes the cost of military equipment. These two problems happen to coincide when wars are short but not when wars are long.

We illustrate this point by studying the steady state of the model when $\beta \to 1$.[12] This case is convenient because the steady-state problems of both the planner and the market

[11] The deterrence constraint may not always bind. The reason it may not is that an economy could start with very high M_0^S and K_0^S. In an optimal steady state with positive military capital, the deterrence constraint always binds because otherwise a country could lower the steady-state military capital stock and enjoy higher consumption.

[12] To ensure that the value of military equipment during conflict does not explode as $\beta \to 1$, the marginal benefit of military equipment is rescaled accordingly, i.e., $\chi = (1-\beta)\hat{\chi}$ for some $\hat{\chi} > 0$.

are effectively static. The solution of the optimal static problem then coincides with the limiting steady-state allocation of the dynamic model.[13] We then obtain the following result.

Proposition 4. *Consider a steady state of the Ramsey problem when $\beta \to 1$ with a positive shadow value of deterrence. The input choices in the defense industrial base of the competitive equilibrium in the absence of industrial policy minimize the cost of producing the steady-state level of military equipment. The planner instead minimizes the cost of providing deterrence. These problems coincide when wars are short but not when they are long. This implies that conditional on government procurement there is too little capital and hence too much labor used in military production when wars are long.*

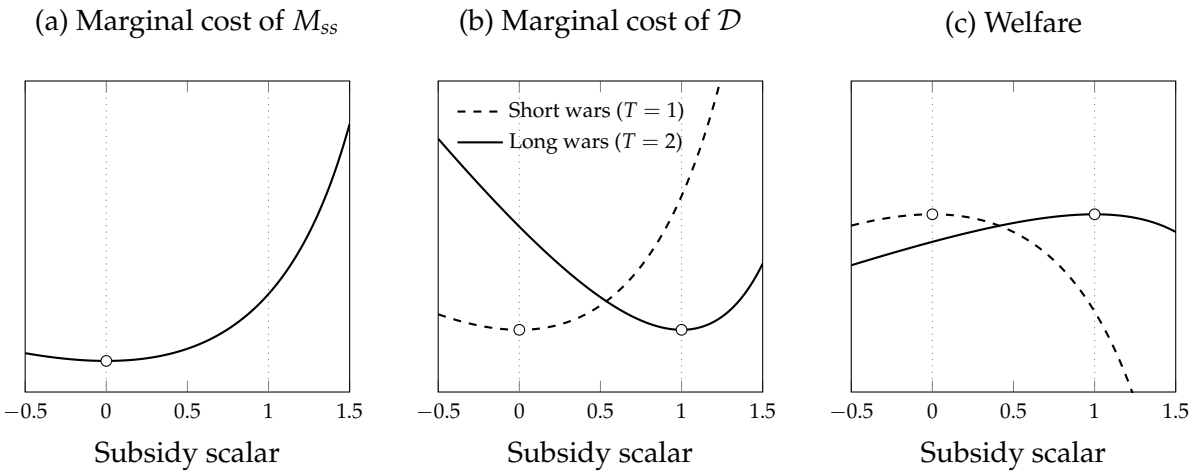
The intuition is best explained graphically. Figure 3(a) plots the steady-state marginal cost of maintaining a larger military stock. This cost is minimized when the government does not use industrial policy to distort the private sector's choice of inputs. Panel (b) instead plots the steady-state marginal cost of providing additional deterrence. For short wars, this cost is also minimized without industrial policy. For long wars, however, it is minimized with positive subsidies, because industrial capacity itself contributes to deterrence. Panel (c) shows the corresponding welfare implications: welfare is maximized at the subsidy level that minimizes the cost of providing deterrence.

3.4 Encouraging the Capacity to Scale

As policymakers revisit the role of industrial policy in the defense industrial base, they have emphasized investments that allow production to scale during war. We now study the sense in which policy should encourage this capacity to scale. In the baseline model, only a single technology is used in the defense industrial base, so there is no sense in which policy can explicitly encourage "scalability". We therefore introduce a second technology to capture the idea that some production technologies allow output to expand more readily than others. Our notion of scalability builds on Murphy and Topel (2013), who think of a technology as scalable if it has a high output elasticity with respect to a flexible input. We augment this idea by also modelling the elasticity of supply of that input. For a technology to be scalable in practice, output must respond strongly to flexible inputs, and those inputs must be available to the defense sector without sharply increasing their cost.

[13] The logic is analogous to the neoclassical growth model, where the Ramsey capital stock coincides with the consumption-maximizing capital stock when $\beta \to 1$.

Figure 3: Deterrence equilibria and optimal subsidies



Note: The subsidy scalar multiplies the planner's optimal subsidy for long wars: 1 is the optimum and 0 implies no subsidies. Panel (a) plots the steady-state marginal cost of M_{ss}. Panels (b) and (c) plot the steady-state marginal cost of $\mathcal{D}$ and welfare for short and long wars. Cost curves are normalized at their own minima and welfare curves at their own maxima.

The baseline model is therefore extended by allowing military goods to be produced using two technologies $g \in \{1, 2\}$. Each technology has a Cobb–Douglas production function with constant returns to scale, $m_g = K_g^{\alpha_g} l_g^{1-\alpha_g}$, and total output of the military good is $m = \sum_g m_g$. One can think of the two technologies as representing different ways to produce the same equipment, or as different types of equipment that serve a similar function on the battlefield. Labor is supplied separately to each technology according to $w_g(l_g) = l_g^{1/\varepsilon_{\ell,g}}$, where $\varepsilon_{\ell,g}$ is the sectoral labor supply elasticity. The upward-sloping labor supply curves make the marginal cost of production rise as each technology expands, allowing both technologies to be used despite constant returns in production.[14]

We begin by establishing that the key results from the previous sections generalize.

Proposition 5. *When there are two technologies, optimal investment subsidies are given by*

$$s_{g,t}^S = \gamma_t^S \frac{\partial \mathcal{D}_t}{\partial K_{g,t}^S}. \tag{31}$$

If the deterrence constraint has a positive shadow value, the planner chooses strictly positive investment subsidies for each active technology in long wars but does not use them in short wars.

[14]We assume $0 < \alpha_g < 1$ and $0 < \varepsilon_{\ell,g} < \infty$. The labor supply curves are the same during peace and war. We interpret $w_g(l_g)$ as the marginal opportunity cost of labor in units of the numeraire. The total opportunity cost is therefore $C_g(l_g) = l_g^{1+1/\varepsilon_{\ell,g}}/(1 + 1/\varepsilon_{\ell,g})$, and the resource constraint becomes $c + \sum_g i_g + \sum_g C_g(l_g) = \bar{l}$. We focus on allocations with positive numeraire consumption during peace and in the first period of conflict, and retain a common capital depreciation rate δ_K.

Proof. Found in Appendix A.4. □

Next we ask how investment subsidies depend on the capacity to scale production. In this extension, this capacity depends on both the output elasticity of labor and the labor supply elasticity. Holding capital fixed, the short-run supply elasticity of technology g is

$$\varepsilon_g \equiv \left.\frac{\partial \log m_g}{\partial \log p_m}\right|_{K_g} = \frac{1-\alpha_g}{\alpha_g + 1/\varepsilon_{\ell,g}}. \tag{32}$$

A higher output elasticity of labor, $1-\alpha_g$, or a higher labor supply elasticity, $\varepsilon_{\ell,g}$, therefore makes production more scalable in the short run. The supply elasticity is constant, so the same expression applies during peace and war.

Proposition 6. *Consider optimal subsidies in the steady state when wars are long, both technologies are active, and the deterrence constraint has a positive shadow value. Suppose that military production rises during conflict relative to the peacetime steady state and that technology 1 is more scalable, $\varepsilon_1 > \varepsilon_2$. In this case the planner subsidizes technology 1 relative to technology 2:*

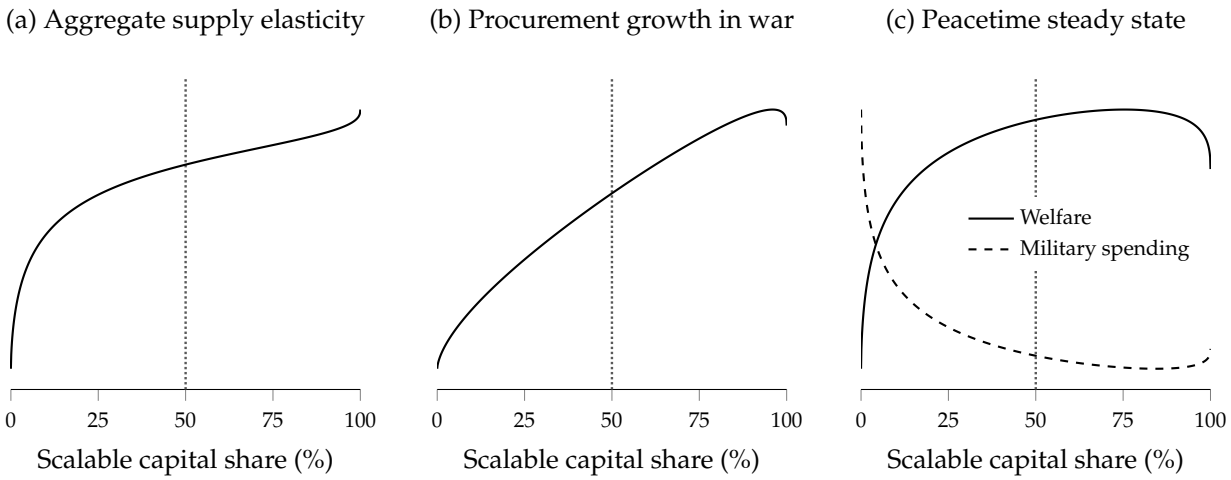
$$s^S_{1,t} > s^S_{2,t}. \tag{33}$$

Proof. Found in Appendix A.5. □

The optimal subsidies combine two effects: they expand the defense industrial base and change its composition. Having illustrated the first effect above, we use Figure 4 to illustrate the second. It plots a numerical example where in the no-subsidy steady state, both technologies have the same amount of capital. We then vary the share allocated to the more scalable technology, holding total industrial capital and deterrence fixed. Panel (a) shows that aggregate short-run supply elasticity rises as more capital is allocated to the scalable technology. Panel (b) illustrates how greater scalability can support a larger wartime expansion in procurement.[15] Panel (c) shows that changing the composition of capital alone can raise peacetime steady-state welfare and lower military expenditure: a greater capacity to expand wartime production reduces the peacetime buildup needed to sustain deterrence.

[15]The increase in procurement growth reverses at very high shares of the scalable technology. This reflects two opposing effects of changing the composition of capital. Below the no-subsidy benchmark, increasing the scalable technology's capital share makes supply more elastic and lowers the cost of producing the benchmark peacetime quantity. Above the benchmark, further increases make supply more elastic but raise that cost. Initially, the scalability effect dominates. At very high shares, however, the cost effect dominates: the marginal cost of producing a given wartime quantity rises and wartime output falls. This explains why procurement growth rises monotonically below the benchmark but not above it.

Figure 4: Scalability and the composition of industrial capital



Note: The dotted line marks the no-subsidy benchmark. Supply elasticity is evaluated at wartime output; procurement growth is relative to peacetime output. In panel (c), welfare is measured by consumption-equivalent gains and military spending by $p_m^P m^P$, on separate vertical scales.

3.5 Extensions

In this section, we discuss a number of extensions to the baseline model. We provide an overview here and delegate details to Appendix B.

Traditional Rational Deterrence Theory Traditional rational deterrence theory has long emphasized the problem of credibility: a country may threaten to fight but prefer to back down once challenged. Our baseline model abstracts from this problem by assuming that Sovereign always fights when Adversary challenges the status quo. But there are historical examples in which countries have not responded to such challenges with war. For example, France did not respond to Germany's reoccupation of the Rhineland with military force. Similarly, Ukraine in 2014 did not attempt to retake Crimea.

To study how this credibility problem affects optimal industrial policy, we study an extended model in which we explicitly model the possibility of backing down. A country that wants to deter needs to commit to using military force when the status quo is challenged. This implies that countries also need to satisfy a limited commitment constraint. The cost of war needs to be sufficiently low for a country to be able to commit to going to war. This constraint generates another argument for industrial policy. Countries choose capital stocks in order to lower the cost of war. The optimal industrial policy that implements these capital stocks subsidizes capital that is valuable during war. The expression is identical to the one found in Kooi (2025). That paper develops many of its implications

in an open-economy general-equilibrium model. It shows that this argument for industrial policy can also support investment in scalable technologies. Because many of the implications are already developed there, we do not develop them again here.

Equilibrium Wars In our baseline model, we study a deterrence equilibrium in which war never occurs. We now add shocks to Adversary's willingness to attack, so that it sometimes chooses war despite Sovereign's deterrence efforts. We study how industrial and procurement policy can attain a generalized deterrence equilibrium—a given sequence of war probabilities—most efficiently. The deterrence constraint now depends on capital and military stocks in all future periods. The intuition is that Adversary compares attacking today with waiting and retaining the option to attack later. Anticipated future stocks therefore affect its current incentives to attack. For example, stronger future defenses can make waiting less attractive and encourage an earlier attack. The basic principle therefore carries over, but the optimal investment subsidy at date t now reflects how the anticipated capital stock at that date affects attack incentives both at date t and at earlier dates.

Endogenous War Duration Our baseline model assumes that the duration of war is exogenous. We therefore study an extended model in which the probability of reaching the second period depends on the initial military balance, assuming that a larger Sovereign military stock makes a long war more likely. We show that this does not change the optimal subsidy formula in Proposition 2. But it does change the shape of the deterrence function in two ways. First, industrial capital contributes to deterrence only in long wars, so its deterrent value is weighted by the probability of reaching the second period. Second, there is now an additional effect of initial military equipment, namely that it changes the probability that the war lasts into the second period. If long wars are at least as favorable for deterrence as short wars, then in the steady state Sovereign chooses a larger military stock relative to capital in the defense industrial base, and therefore chooses lower subsidies than in the case where wars are always long.

Strategic Interactions In our baseline model, Adversary chooses whether to attack, but its military stock is fixed. We extend the model to allow for a more general form of strategic interaction in which Adversary adjusts its military stock in response to Sovereign's peacetime choices. Specifically, Adversary's military stock is allowed to respond separately to Sovereign's military stock and to Sovereign's capital in the defense industrial base. In the steady state, when wars are long, a stronger response to Sovereign's military

stock makes that stock less effective at providing deterrence, so Sovereign relies more on the defense industrial base and optimal capital subsidies rise. A stronger response to Sovereign's defense-industrial capital makes that capital less effective at providing deterrence, so optimal capital subsidies fall.

4 Quantitative Model

In our baseline model, a single sector represents the defense industrial base. In practice, the defense industrial base and the civilian sector consist of multiple industries, all relying on other industries' output to produce and invest. To study how industrial policy should be targeted, we develop an Integrated Defense Economics Model (IDEM). It combines an economic block describing how the economy responds to military buildups and industrial policy with a military block describing how wars unfold. Together, these blocks determine how capital investment contributes to deterrence.

We present the model below and study optimal policy in the next section. To simplify the exposition, we describe Sovereign's economy, suppressing the country index and dependence on the history of geopolitical states.

4.1 Economic Block

Let $\mathcal{G}$ denote the set of N sectors. All aggregators below have constant returns to scale. Each time period represents a year.

Preferences The representative household consumes sectoral goods $c_{g,t}$, supplies labor $L_{g,t}$ to each sector, and values the public good G_t. The discount factor is β, consumption weights are ϵ_g, and σ_C is the elasticity of substitution across goods in consumption. Following Vom Lehn and Winberry (2022), τ is the elasticity of labor supply across sectors and ξ is the Frisch elasticity of the aggregate labor index. Preferences are

$$\sum_{t=0}^{\infty} \beta^t \Big(\log C_t - \frac{\tilde{L}_t^{1+1/\xi}}{1+1/\xi} + G_t \Big), \quad C_t = \Big[\sum_{g \in \mathcal{G}} \epsilon_g^{\frac{1}{\sigma_C}} c_{g,t}^{\frac{\sigma_C - 1}{\sigma_C}} \Big]^{\frac{\sigma_C}{\sigma_C - 1}}, \quad \tilde{L}_t = \Big[\sum_{g \in \mathcal{G}} L_{g,t}^{\frac{\tau+1}{\tau}} \Big]^{\frac{\tau}{\tau+1}}.$$

Production Sector g uses capital $K_{g,t}$, labor $L_{g,t}$, and intermediate goods $x_{g,r,t}$ purchased from sectors r. Let $V_{g,t}$ denote value added, $X_{g,t}$ the intermediate composite, and $Y_{g,t}$ gross output. The α's are sector-specific technology weights. The elasticities σ_{VA}, σ_X, and σ_Y govern substitution between capital and labor, across intermediate goods, and

between value added and intermediates, respectively, and are common across sectors:

$$V_{g,t} = \left[\alpha_{K,g}\, K_{g,t}^{\frac{\sigma_{VA}-1}{\sigma_{VA}}} + \alpha_{L,g}\, L_{g,t}^{\frac{\sigma_{VA}-1}{\sigma_{VA}}}\right]^{\frac{\sigma_{VA}}{\sigma_{VA}-1}}, \qquad X_{g,t} = \left[\sum_{r\in\mathcal{G}} \alpha_{g,r}\, x_{g,r,t}^{\frac{\sigma_X-1}{\sigma_X}}\right]^{\frac{\sigma_X}{\sigma_X-1}},$$
$$Y_{g,t} = \left[\alpha_{V,g}\, V_{g,t}^{\frac{\sigma_Y-1}{\sigma_Y}} + \alpha_{X,g}\, X_{g,t}^{\frac{\sigma_Y-1}{\sigma_Y}}\right]^{\frac{\sigma_Y}{\sigma_Y-1}}.$$

Capital Accumulation Sector-specific capital depreciates at rate δ_g. Investment is irreversible, $I_{g,t} \geq 0$, and adjustment costs governed by $\varphi > 0$ reduce effective investment, following Vom Lehn and Winberry (2022). The weights $\vartheta_{g,r}$ describe the composition of sector g's investment purchases $i_{g,r,t}$:

$$K_{g,t+1} = (1-\delta_g)K_{g,t} + I_{g,t} - \frac{\varphi}{2}\left(\frac{I_{g,t}}{K_{g,t}} - \delta_g\right)^2 K_{g,t}, \qquad I_{g,t} = \prod_{r\in\mathcal{G}} i_{g,r,t}^{\vartheta_{g,r}}, \qquad g \in \mathcal{G}.$$

Military Power The government accumulates J military stocks $M_{j,t}$, indexed by $j \in \mathcal{J}$. Suppliers are partitioned into sets $\mathcal{G}_j \subset \mathcal{G}$; weights $\psi_{j,g}$ aggregate purchases $m_{j,g,t}$ into non-negative procurement $m_{j,t}$. Depreciation $\delta_{j,t}$ equals δ_j in peace and after war, and $\delta_j^w \geq \delta_j$ during war to reflect combat losses. The weights ϕ_j determine each stock's contribution to military power M_t:[16]

$$m_{j,t} = \prod_{g\in\mathcal{G}_j} m_{j,g,t}^{\psi_{j,g}}, \qquad M_{j,t+1} = (1-\delta_{j,t})M_{j,t} + m_{j,t}, \qquad M_t = \prod_{j\in\mathcal{J}} \left(\frac{M_{j,t}}{\phi_j}\right)^{\phi_j}.$$

Market Clearing Sectoral output meets consumption, intermediate, investment, and military demand, and each sector's labor market clears at the sectoral wage $w_{g,t}$:

$$Y_{g,t} = c_{g,t} + \sum_{b\in\mathcal{G}} x_{b,g,t} + \sum_{b\in\mathcal{G}} i_{b,g,t} + \sum_{j\in\mathcal{J}} m_{j,g,t}, \qquad g \in \mathcal{G}.$$

Competitive Equilibrium The competitive equilibrium is analogous to that in Section 2.1. The key change is that subsidies apply at rate $s_{g,t}$ to capital's replacement value $P_{g,t}^I K_{g,t}$, where $P_{g,t}^I$ is the price of sector g's investment basket. The per-unit subsidy of the simple model corresponds to $s_{g,t}P_{g,t}^I$. Let $R_t = \prod_{v=0}^{t-1}(1+r_v)$, with $R_0 = 1$, denote the cumulative gross return. With goods prices $p_{g,t}$ and wages $w_{g,t}$, the producer's objective

[16] We collect military stocks in $\mathbf{M}_t$ and sectoral capital stocks in $\mathbf{K}_t$.

now reads

$$\sum_{t=0}^{\infty} \frac{1}{R_t}\Big(p_{g,t}Y_{g,t} + s_{g,t}P^I_{g,t}K_{g,t} - w_{g,t}L_{g,t} - \sum_{r\in\mathcal{G}} p_{r,t}\big(x_{g,r,t} + i_{g,r,t}\big)\Big).$$

Appendix C.1 defines equilibrium and derives the optimality conditions.

4.2 Deterrence Block

Timing and War Duration The initial balance of military power determines the probability that a war lasts more than one period. If Sovereign avoids a quick defeat, the conflict becomes an "industrial war of attrition," whose duration follows a fixed distribution. Adversary decides whether to attack before the duration is known; if war breaks out, the duration is drawn and revealed to both parties at the outbreak. Let π_T denote the baseline probabilities of wars lasting $T \in \{1, \ldots, \bar{T}\}$ periods. The initial balance of power scales the probability of a long war by $a(M_t)$: a stronger Adversary is more likely to defeat Sovereign quickly. Let M^E_0 denote Adversary's initial military power and $\kappa \geq 0$ the sensitivity to its share of military power. At parity ($M_t = M^E_0$) the probabilities are the baseline frequencies π_T; the probability of a short war rises by κ per unit of Adversary's share of military power. The probability of reaching the attrition phase is therefore $a(M_t)(1-\pi_1)$; conditional on reaching it, duration follows the fixed distribution $\pi_T/(1-\pi_1)$ for $T \geq 2$. We set:

$$\tilde{\pi}_T(M_t) = a(M_t)\,\pi_T \quad (T \geq 2), \qquad \tilde{\pi}_1(M_t) = 1 - a(M_t)\,(1-\pi_1),$$
$$a(M_t) = 1 - \frac{\kappa}{1-\pi_1}\Big(\frac{M^E_0}{M_t + M^E_0} - \frac{1}{2}\Big).$$

The Allocation of the Public Good Re-index dates so that war begins at date zero. The parameters $\chi^{(T)}_s$ measure the effectiveness of military power in period s of a war lasting T periods; M^E_s denotes Adversary's military power. The postwar settlement changes the public good from its initial level G_0 according to

$$G_t = G_0 + \sum_{s=0}^{T-1} \chi^{(T)}_s\big(M_s - M^E_s\big) \quad \text{if } t \geq T, \qquad \text{and} \qquad G_t = G_0 \quad \text{otherwise.}$$

Adversary's Policy and Challenge Decision Adversary's military power is constant in peace and follows an exogenous path $\{M^E_s\}_{s=0}^{T-1}$ in a war of length T. Mobilization is costly for Adversary as well: from the outbreak of a war of length T, its consumption and

labor supply follow exogenous paths that depart from their peacetime levels. Adversary attacks when $W_t^{A,C} + \eta_t > W_t^{A,P}$, where η_t is its political preference for war. It decides before the duration is known, so its welfare from war averages $W_t^{A,C,(T)}$ across durations:

$$W_t^{A,C} = \sum_{T=1}^{\bar{T}} \tilde{\pi}_T(M_t)\, W_t^{A,C,(T)}.$$

Sovereign's Wartime and Postwar Policy As in the simple model, Sovereign optimally chooses wartime military expenditure, trading off consumption against military gains in the settlement. After war, peace is permanent and the public good is fixed. The departure from the simple model is that postwar expenditure follows an exogenous rule: the government spends a share θ of GDP on procurement, allocated across baskets with shares θ_j. This rule allows the model to capture a realistic postwar reduction in military procurement. Under the rule, the military stocks carried out of a war have no value to Sovereign, so purchases in the last period of the war, which arrive only after the settlement, do not affect it; they follow the postwar rule. Industrial policy is not needed during or after war. Appendices C.2 and C.1 give the wartime problem and the postwar equilibrium.

Deterrence Equilibrium As in Definition 2, a deterrence equilibrium is a competitive equilibrium (Definition 3) satisfying $\mathcal{D}_t(\mathbf{K}_t, \mathbf{M}_t) \geq \eta_t$ in every period. We present $\mathcal{D}_t$ below.

The Simplified Deterrence Constraint Let $M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t)$ be Sovereign's optimal military power in period s of a war lasting T periods, starting from date-t stocks $(\mathbf{K}_t, \mathbf{M}_t)$. Let $\mathcal{E}^{(T)}$ denote the present value, at the outbreak, of the utility Adversary loses in war relative to peace, due to lower consumption and labor disutility. Deterrence requires Adversary's expected loss from war—the discounted settlement plus the economic cost of its mobilization—to offset its preference for war (Appendix C.2). Sovereign's stocks affect Adversary's economic cost only through the war-length probabilities: a militarily stronger Sovereign makes a long war, which is costlier for Adversary, more likely. The simplified deterrence constraint is:

$$\mathcal{D}_t(\mathbf{K}_t, \mathbf{M}_t) = \sum_{T=1}^{\bar{T}} \tilde{\pi}_T(M_t) \left[\frac{\beta^T}{1-\beta} \sum_{s=0}^{T-1} \chi_s^{(T)} \left(M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t) - M_s^E \right) + \mathcal{E}^{(T)} \right] \geq \eta_t, \qquad \forall t.$$

4.3 Calibration

We calibrate the model to the United States, with $N = 370$ sectors and $J = 8$ military baskets. The economic structure and military stocks are disciplined by sectoral accounts and defense procurement data; war durations and wartime spending discipline the conflict parameters. Appendix C.3 details the data construction and calibration procedure.

Economic Structure Consumption and production weights reproduce spending shares in the 2017 BEA input-output accounts. Investment weights $\vartheta_{g,r}$ and sectoral depreciation rates δ_g come from Iovino et al. (2025), who extend the investment network of Vom Lehn and Winberry (2022) to the 2017 BEA detail level. Appendix C.3 explains how we choose the weights to match the observed shares.

Table 1 reports the common economic parameters and their sources.

Table 1: Economic parameters.

Parameter	Value	Source
β, discount factor	0.96	Vom Lehn and Winberry (2022)
ξ, aggregate Frisch elasticity	0.4	Reichling and Whalen (2012)
τ, sectoral labor supply elasticity	4.5	Vom Lehn and Winberry (2022)
φ, capital adjustment cost	0.5	Vom Lehn and Winberry (2022)
σ_Y, value added–intermediates elasticity	0.5	Baqaee and Farhi (2019)
σ_{VA}, capital–labor elasticity	0.5	Oberfield and Raval (2021)
σ_X, intermediate-input elasticity	0.001	Baqaee and Farhi (2019)
σ_C, consumption-goods elasticity	0.9	Baqaee and Farhi (2019)

Military Stocks and Procurement The postwar spending share is $\theta = 3.37\%$ of GDP, the US average over 2015–2025 in the SIPRI data. Basket shares θ_j and within-basket weights $\psi_{j,g}$ match 2017 US defense procurement. We choose military-power weights ϕ_j so that a government deterring in peacetime without using industrial policy reproduces this spending composition. The no-industrial-policy deterrence problem of Sovereign is described in Appendix C.5. Table 6 in Appendix C.3 reports the basket shares and the military-power weights.

Peacetime depreciation rates come from BEA national-defense fixed assets, with ammunition depreciation measured by purchases relative to stocks in DoD inventory reports. Wartime rates add combat losses from the Russia–Ukraine war for land systems and US World War II experience for air and naval systems. Structures, other equipment,

and intellectual property retain their peacetime rates; non-durables are consumed within the year, as is ammunition during war. Table 5 in Appendix C.3 reports the resulting rates. This calibration describes a high-intensity combined-arms war fought abroad; Section 5.4 considers alternatives.

War Duration Baseline probabilities π_T match annual duration frequencies of inter-state wars with at least 1,000 battle deaths in the Correlates of War data, 1816–2007. Wars lasting six years or more are pooled at $\bar{T} = 6$. We set $\kappa = 0.33$: in the National Material Capabilities data, a one-unit rise in the initiator's prewar share of military personnel lowers the probability of a war longer than one year by 0.33.

Wartime Spending We recover combat efficiency parameters $\chi_s^{(T)}$ from the procurement conditions that make the wartime spending path estimated by Marzian and Trebesch (2026) optimal. Starting from the no-industrial-policy steady state, we target spending increases above the prewar level of $(5.3, 15, 19, 21.5, 20.5)\%$ of steady-state GDP in the first five years. A war of length T matches the first $T-1$ targets: final-period procurement cannot contribute to wartime military power and follows the postwar rule. Table 7 in the appendix reports the fitted parameters. Outbreak efficiency χ_0 does not affect wartime procurement, so the spending targets do not identify it; by convention, we set $\chi_0 = 0.438$, the probability-weighted mean of subsequent efficiencies across war lengths.

Peacetime Military Expenditure We calibrate Adversary's political preference for war, η_t, so that military expenditure in the no-industrial-policy peacetime steady state matches the US average of 3.37% of GDP. Under our baseline symmetry assumption, Adversary's economy mirrors Sovereign's: in a war of each length, its military power, consumption, and labor supply follow Sovereign's own paths in that war started from this steady state. The settlement terms of the deterrence constraint then vanish at the observed expenditure level, and the constraint binds there when η_t equals Adversary's expected economic cost of war, $\eta_t = \sum_T \pi_T \mathcal{E}^{(T)}$, which is our calibrated value. In units of lifetime utility $\eta_t = 0.090$: Adversary's expected economic cost of war equals that of giving up 0.36% of its peacetime consumption in every period.

5 Quantitative Results

We first state Sovereign's peacetime planning problem and the capital subsidies that decentralize its solution, the optimal deterrence equilibrium. We then present the quantita-

tive results.

5.1 Optimal Deterrence Equilibrium

Sovereign's peacetime planning problem generalizes that in Section 3.1. We explain how we solve this numerically in Appendix C.6. Given initial stocks, the planner solves

$$\begin{aligned} &\max_{\{c_{g,t},L_{g,t},x_{g,r,t},i_{g,r,t},I_{g,t},K_{g,t+1},m_{j,g,t},m_{j,t},M_{j,t+1}\}_{t=0}^{\infty}} \quad \sum_{t=0}^{\infty} \beta^t \left(\log C_t - \frac{\tilde{L}_t^{1+1/\xi}}{1+1/\xi} + G_0 \right) \\ &\text{s.t. (i) the technology and market-clearing conditions (113), for all } t, \\ &\quad \text{(ii) the laws of motion of capital and military stocks (114), for all } t, \\ &\quad \text{(iii) the deterrence constraint } \mathcal{D}_t(\mathbf{K}_t, \mathbf{M}_t) \geq \eta_t, \text{ for all } t \geq 1, \\ &\quad \text{(iv) the non-negativity constraints } I_{g,t} \geq 0 \text{ and } m_{j,t} \geq 0, \text{ for all } t. \end{aligned} \tag{34}$$

Proposition 2 likewise generalizes to this economy. Let $\gamma_t \geq 0$ denote the shadow value of relaxing the date-t deterrence constraint, expressed in units of date-t consumption.

Proposition 7. *An optimal deterrence equilibrium can be implemented with sectoral capital subsidy rates*

$$s_{g,t} = \frac{\gamma_t}{P^I_{g,t}} \frac{\partial \mathcal{D}_t}{\partial K_{g,t}}, \qquad g \in \mathcal{G}.$$

Proof. See Appendix C.4. □

The subsidy compensates firms for capital's marginal deterrence value, which they do not internalize. Dividing by the investment-basket price $P^I_{g,t}$ converts this value into a rate on capital's replacement value. Capital can affect deterrence through military production and through supplies of intermediate and investment goods to other sectors; the derivative captures these effects through the wartime equilibrium. When $\gamma_t > 0$, positive marginal deterrence effects warrant subsidies and negative effects warrant taxes.

5.2 Aggregate Outcomes

Industrial policy lowers the cost of sustaining peace. Relative to the no-industrial-policy benchmark, in which the government deters through procurement alone, optimal industrial policy raises steady-state welfare by 2.50% in consumption-equivalent terms and lowers military procurement from 3.37% to 2.46% of GDP, while providing the same level

of deterrence.[17] Table 2 compares the two steady states.

Table 2: Aggregate outcomes in the steady state.

$\mathcal{D}$	0.00
welfare	2.50
consumption	3.51
military procurement share	−27.12
real GDP	4.37
aggregate capital	8.76
labor supply	1.11

All entries are percent deviations from the no-industrial-policy benchmark. $\mathcal{D}$ is the value of the deterrence function, at its common binding level $\mathcal{D} = \eta$ in both steady states. Welfare is expressed as the equivalent permanent increase in benchmark consumption. The military procurement share is procurement as a share of GDP, 3.37% in the benchmark. GDP and capital are Fisher quantity indices, and labor supply is the sum of sectoral labor.

Welfare rises mainly because consumption rises, by 3.5%. A lower procurement burden and higher real GDP make room for it; GDP rises because most sectors receive positive subsidies (Section 5.3), which raise the aggregate capital stock. Labor supply also rises, and its disutility makes the welfare gain smaller than the rise in consumption.

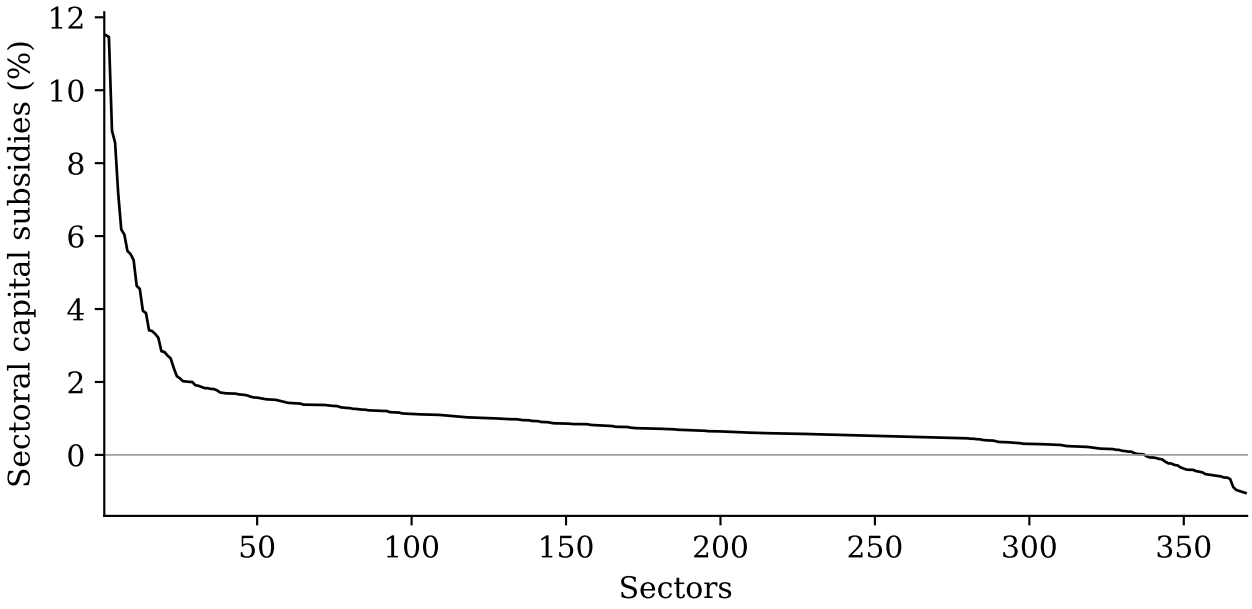
5.3 Industrial Priorities in the Baseline

We now use the quantitative model to study which industries a government should encourage and how much their subsidies should differ. Figure 5 shows the cross-section of optimal subsidy rates, which decline rapidly down the sectoral ranking. The largest rates are concentrated in a small group of industries: they fall from 11.51% in the highest-ranked sector to 2.82% in the twentieth, a decline of 75%.

Next, we present the ranking of industrial priorities. Table 3 reports the twenty industries with the highest subsidy rates; the full ranking appears in Appendix C.7. Three broad categories dominate the ranking. First are producers of munitions and their components, including guided missiles, propulsion units, and ammunition. Second are manufacturers of heavy military equipment—ships, aircraft, and armored vehicles—and their input suppliers. Third are producers of advanced electronics, including search, detection, and navigation instruments, electrical and signal testing equipment, and communications equipment.

[17] Eliminating the deterrence constraint altogether would raise welfare by 4.65% relative to the same benchmark. Industrial policy therefore eliminates 54% of the steady-state welfare cost of deterrence.

Figure 5: The cross-section of optimal capital subsidies.



The line traces the subsidy rates of the 370 sectors, ordered by decreasing subsidy. The subsidy s_g is a rate on the replacement value of the sector's capital stock.

What determines whether a sector is an industrial priority? Our model highlights two channels. First, military stocks differ in how quickly they are depleted in war, and high wartime depletion raises the value of the capacity to replenish a stock, especially when its losses rise sharply relative to peacetime. A shell can be fired only once, so munitions are replenished more often than tanks, which in turn suffer battlefield losses that military structures do not. The capacity to produce munitions and equipment with high wartime loss rates is therefore particularly valuable.

Second, a sector can become an industrial priority through its links to military demand along the input-output and investment networks: component suppliers lower the cost of military equipment, and suppliers of investment goods help expand production capacity during war. Appendix C.8 illustrates both links by raising the capital stock of each of the twenty sectors in Table 3 by 10% at the outbreak of a six-year war. Additional capacity in propulsion units serves direct military purchases and lowers the cost of inputs to missile production, while additional capacity in switchgear or in communications equipment supplies investment goods that expand production capacity.

Table 3: The 20 sectors with the highest optimal capital subsidies.

rank	sector	s_g (%)
1	Guided missile and space vehicle manufacturing	11.51
2	Propulsion units and parts for space vehicles and guided missiles	11.46
3	Ship building and repairing	8.89
4	Search, detection, and navigation instruments manufacturing	8.56
5	Military armored vehicle, tank, and tank component manufacturing	7.18
6	Ammunition, arms, ordnance, and accessories manufacturing	6.18
7	Other aircraft parts and auxiliary equipment manufacturing	6.04
8	Aircraft manufacturing	5.59
9	Truck transportation	5.51
10	Architectural, engineering, and related services	5.34
11	Electronic and precision equipment repair and maintenance	4.63
12	Electricity and signal testing instruments manufacturing	4.55
13	Other computer related services, including facilities management	3.95
14	Broadcast and wireless communications equipment	3.89
15	Water transportation	3.41
16	Switchgear and switchboard apparatus manufacturing	3.40
17	Computer systems design services	3.32
18	Air transportation	3.21
19	Aircraft engine and engine parts manufacturing	2.84
20	Other electronic component manufacturing	2.82

5.4 Industrial Priorities and the Nature of Conflict

We next ask how industrial priorities depend on the war a country expects to fight, using three stylized scenarios: a land war, an air war, and a naval war. In each scenario, we set the military-power weights of the other two platform categories to zero, rescale the remaining weights proportionally so that they still sum to one, and solve again for the optimal deterrence equilibrium. The combat efficiency parameters and all other parameters keep their baseline values, except that, as in the baseline, Adversary's wars mirror Sovereign's and η is set so that the deterrence constraint binds at the postwar steady state. Common military inputs, including ammunition, continue to contribute to military power in all three scenarios.

Table 4 reports baseline subsidy rates and the changes under each scenario for five industries whose subsidies vary substantially across scenarios. Shipbuilding receives a subsidy of 8.89% in the baseline, which rises slightly, to 8.96%, in the naval scenario but falls to 0.8% and 1.0% in the land and air scenarios. The subsidy to armored vehicles and tanks is almost unchanged in the land scenario and falls by roughly four percentage

points in the other two. Subsidies to aircraft and their components remain close to their baseline levels in the air scenario but fall substantially in the land and naval scenarios.

Table 4: Changes in optimal capital subsidies across conflict scenarios.

	Baseline	Change from baseline (pp)		
Sector	(%)	Land	Air	Naval
Shipbuilding	8.89	−8.12	−7.84	+0.07
Armored vehicles, tanks, and components	7.18	−0.22	−4.11	−4.44
Other aircraft parts and auxiliary equipment	6.04	−5.09	−0.07	−5.05
Aircraft manufacturing	5.59	−4.54	−0.09	−4.49
Aircraft engines and parts	2.84	−1.80	+0.09	−1.80

Baseline entries are subsidy rates s_g in percent; scenario entries are changes in percentage points. Sector names are abbreviated. Appendix C.9 reports subsidy levels for all twenty industries in Table 3.

Other industries remain priorities across all three scenarios. As the fuller comparison in Appendix C.9 shows, guided missile and propulsion unit manufacturing retain subsidy rates around 11%, while ammunition and search, detection, and navigation instruments stay highly subsidized. These industries supply military inputs that every scenario requires.

Optimal industrial policy therefore depends not only on economic fundamentals, such as the input-output and investment networks, but also on the war a country expects to fight: different conflicts require different mixes of military equipment and give rise to different industrial priorities. When wars may be prolonged, economic policymaking and military analysis need to be closely linked.

6 Conclusion

We develop a dynamic model of rational deterrence to study how procurement and industrial policy can jointly sustain peace at the lowest cost. When wars are short, procurement alone suffices. When wars last long enough for wartime production to affect military outcomes, industrial capacity itself contributes to deterrence. Private investors do not internalize this strategic benefit, creating a national security externality that justifies support for industrial investment and more scalable production technologies. We develop an Integrated Defense Economics Model (IDEM) to operationalize this framework and show how it can generate quantitative rankings of industries to guide the targeting of industrial policy.

An important implication of our analysis is that economic policy for national security should be closely linked to military analysis. IDEM takes a step in this direction, but there is much more that could be done. Contemporaneous work by Grassi and Markó (2026) integrating combat models that generate Lanchester laws with general equilibrium models of production could further enrich IDEM's military block. Introducing geography would allow us to study how breakthroughs expose industrial capacity to capture and how the scope for retreat—strategic depth—in larger countries can prolong conflict. It would also allow a richer treatment of air power. Our current model is closest to its tactical role in supporting other armed forces on the front line, while a spatial model could also capture its strategic role in disrupting an adversary's economy.

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A Proofs

A.1 Derivation of (25) and (26)

Proof. We derive the planner optimality conditions at an interior peacetime allocation, allowing the wartime labor constraint to bind. We use two smooth functions, $\mathcal{D}_t^I$ and $\mathcal{D}_t^F$, where I denotes incentives and F denotes feasibility.

For short wars ($T = 1$), wartime production cannot affect the military outcome, so we define both functions identically:

$$\mathcal{D}_t^I(K, M) = \mathcal{D}_t^F(K, M) = \frac{\chi}{1-\beta}(M - \bar{M}^A).$$

For long wars ($T = 2$), let $B \equiv \chi\beta/(1-\beta)$ and $a \equiv [B(1-\alpha)]^{(1-\alpha)/\alpha}$, and define

$$\begin{aligned}\mathcal{D}_t^I(K,M) &= B\big[(2-\delta_M)M + aK - 2\bar{M}^A\big],\\ \mathcal{D}_t^F(K,M) &= B\big[(2-\delta_M)M + K^\alpha(\bar{l}^S)^{1-\alpha} - 2\bar{M}^A\big].\end{aligned} \tag{35}$$

Here $\mathcal{D}_t^I$ uses the wartime production Sovereign would choose in the absence of the labor constraint, while $\mathcal{D}_t^F$ uses the maximum output attainable with full labor mobilization. For either war length, the deterrence function is

$$\mathcal{D}_t(K,M) = \min\left\{\mathcal{D}_t^I(K,M), \mathcal{D}_t^F(K,M)\right\}.$$

Thus $\mathcal{D}_t \geq \eta_t^A$ is equivalent to imposing both $\mathcal{D}_t^I \geq \eta_t^A$ and $\mathcal{D}_t^F \geq \eta_t^A$. Throughout the derivation, $j \in \{I, F\}$. This formulation covers differentiable allocations and the kink using ordinary multipliers.

Suppress the argument $z_t = P$ and write the constraints as

$$c_t^S + i_t^S = l_{n,t}^S, \tag{36}$$

$$K_{t+1}^S = (1-\delta_K)K_t^S + i_t^S, \tag{37}$$

$$l_{n,t}^S + l_{m,t}^S = \bar{l}^S, \tag{38}$$

$$m_t^S = (K_t^S)^\alpha (l_{m,t}^S)^{1-\alpha}, \tag{39}$$

$$M_{t+1}^S = (1-\delta_M)M_t^S + m_t^S, \tag{40}$$

$$\mathcal{D}_{t+1}^j(K_{t+1}^S, M_{t+1}^S) \geq \eta_{t+1}^A, \qquad j \in \{I, F\}. \tag{41}$$

Let $\lambda_{n,t}^S$ be the multiplier on (36), $\lambda_{K,t}^S$ on (37), $\lambda_{l,t}^S$ on (38), $\lambda_{y,t}^S$ on (39), $\lambda_{m,t}^S$ on (40), and $\gamma_{j,t+1}^S$ on branch j of (41). Define $\gamma_t^S \equiv \sum_j \gamma_{j,t}^S$. For short wars, the two constraints coincide, so only the sum of their multipliers matters. The deterrence multipliers satisfy, for $t \geq 1$,

$$\gamma_{j,t}^S \geq 0, \qquad \gamma_{j,t}^S\big[\mathcal{D}_t^j(K_t^S, M_t^S) - \eta_t^A\big] = 0, \qquad j \in \{I, F\}.$$

The Lagrangian is

$$\begin{aligned}\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big(& c_t^S + \lambda_{n,t}^S (l_{n,t}^S - c_t^S - i_t^S) + \lambda_{K,t}^S ((1-\delta_K) K_t^S + i_t^S - K_{t+1}^S) \\ & + \lambda_{l,t}^S (\bar{l}^S - l_{n,t}^S - l_{m,t}^S) + \lambda_{y,t}^S ((K_t^S)^\alpha (l_{m,t}^S)^{1-\alpha} - m_t^S) \\ & + \lambda_{m,t}^S ((1-\delta_M) M_t^S + m_t^S - M_{t+1}^S) \\ & + \beta \sum_j \gamma_{j,t+1}^S (\mathcal{D}_{t+1}^j (K_{t+1}^S, M_{t+1}^S) - \eta_{t+1}^A) \Big).\end{aligned}$$

First-Order Conditions (within Period). For each t, the FOCs w.r.t. $c_t^S, i_t^S, l_{n,t}^S, l_{m,t}^S, m_t^S$ are

$$[c_t^S]: \quad 1 - \lambda_{n,t}^S = 0, \tag{42}$$

$$[i_t^S]: \quad -\lambda_{n,t}^S + \lambda_{K,t}^S = 0, \tag{43}$$

$$[l_{n,t}^S]: \quad \lambda_{n,t}^S - \lambda_{l,t}^S = 0, \tag{44}$$

$$[l_{m,t}^S]: \quad -\lambda_{l,t}^S + \lambda_{y,t}^S (1-\alpha)(K_t^S)^\alpha (l_{m,t}^S)^{-\alpha} = 0, \tag{45}$$

$$[m_t^S]: \quad -\lambda_{y,t}^S + \lambda_{m,t}^S = 0. \tag{46}$$

First-Order Conditions for the Stocks M_{t+1}^S and K_{t+1}^S. Differentiating $\mathcal{L}$ w.r.t. M_{t+1}^S yields

$$[M_{t+1}^S]: \quad -\beta^t \lambda_{m,t}^S + \beta^{t+1} (1-\delta_M) \lambda_{m,t+1}^S + \beta^{t+1} \sum_j \gamma_{j,t+1}^S \frac{\partial \mathcal{D}_{t+1}^j}{\partial M_{t+1}^S} = 0. \tag{47}$$

Differentiating $\mathcal{L}$ w.r.t. K_{t+1}^S yields

$$\begin{aligned}[K_{t+1}^S]: \quad & -\beta^t \lambda_{K,t}^S + \beta^{t+1} (1-\delta_K) \lambda_{K,t+1}^S \\ & + \beta^{t+1} \lambda_{y,t+1}^S \alpha (K_{t+1}^S)^{\alpha-1} (l_{m,t+1}^S)^{1-\alpha} + \beta^{t+1} \sum_j \gamma_{j,t+1}^S \frac{\partial \mathcal{D}_{t+1}^j}{\partial K_{t+1}^S} = 0.\end{aligned} \tag{48}$$

Optimality Conditions and Differentiable Allocations. Equations (42)–(43) imply $\lambda_{n,t}^S = \lambda_{K,t}^S = 1$, and (46) gives $\lambda_{y,t}^S = \lambda_{m,t}^S$. Every branch has the same derivative with respect to M, which equals $\partial \mathcal{D}_t / \partial M$ even at the kink. Dividing (47) by β^t therefore gives

$$\lambda_{m,t}^S = \beta \Big(\gamma_{t+1}^S \frac{\partial \mathcal{D}_{t+1}}{\partial M_{t+1}^S} + (1-\delta_M) \lambda_{m,t+1}^S \Big),$$

which is (25).

Similarly, (48) becomes

$$1 = \beta\Big(\sum_j \gamma^S_{j,t+1} \frac{\partial \mathcal{D}^j_{t+1}}{\partial K^S_{t+1}} + (1-\delta_K) + \lambda^S_{m,t+1} \alpha (K^S_{t+1})^{\alpha-1} (l^S_{m,t+1})^{1-\alpha} \Big). \tag{49}$$

For short wars, both branches have zero derivative with respect to capital. For long wars away from the kink, only the branch attaining the minimum can have a positive multiplier. In either case, the sum in (49) reduces to $\gamma^S_{t+1}\, \partial \mathcal{D}_{t+1}/\partial K^S_{t+1}$, recovering (26). At the long-war kink, both branches can carry positive multipliers and (49) continues to apply. The main-text derivative formulas are thus the differentiable special cases of these conditions. □

A.2 Proof of Proposition 1

Fix a possible challenge date t. The deterrence constraint compares peace with the off-equilibrium continuation in which Adversary challenges at t. By (22), deterrence is

$$\mathcal{D}_t(K^S_t, M^S_t) = \frac{\chi \beta^{T-1}}{1-\beta} \sum_{\tau=0}^{T-1} \left(M^{S,C}_\tau(K^S_t, M^S_t) - \bar{M}^A \right).$$

Short Wars. Suppose first that $T = 1$. Then only the military stock available at the possible challenge date matters for the outcome:

$$\mathcal{D}_t(K^S_t, M^S_t) = \frac{\chi}{1-\beta} \left(M^S_t - \bar{M}^A \right).$$

Military production in that continuation affects $M^{S,C}_1$, but this stock is reached only after a short war would have ended. It therefore has no effect on the allocation of the public good. Hence

$$\frac{\partial \mathcal{D}_t}{\partial M^S_t} = \frac{\chi}{1-\beta} > 0, \qquad \frac{\partial \mathcal{D}_t}{\partial K^S_t} = 0.$$

This gives the first part of the proposition.

Long Wars. Suppose now that $T = 2$, and let $B \equiv \chi\beta/(1-\beta)$ denote the marginal value of first-period conflict production in a long war. The deterrence function is

$$\mathcal{D}_t(K^S_t, M^S_t) = B \left(M^S_t + M^{S,C}_1 - 2\bar{M}^A \right).$$

In this off-equilibrium continuation, the date-t stocks are predetermined, so

$$M_1^{S,C} = (1-\delta_M)M_t^S + m_0^{S,C}.$$

The wartime choice of military production depends on K_t^S, but not on M_t^S, because M_t^S enters the wartime objective additively and separably. Therefore

$$\frac{\partial \mathcal{D}_t}{\partial M_t^S} = B\Big(1 + (1-\delta_M)\Big) > 0.$$

It remains to characterize the derivative with respect to the defense-industrial capital stock. In the first period of the off-equilibrium long-war continuation, capital is predetermined and Sovereign chooses labor in the defense sector. Using the resource constraint to substitute out consumption, and noting that from the final period of the war all labor is allocated to producing consumption goods, the relevant static problem is

$$\max_{0 \le l_{m,0}^{S,C} \le \bar{l}^S} \left\{ -l_{m,0}^{S,C} + B(K_t^S)^\alpha \big(l_{m,0}^{S,C}\big)^{1-\alpha} \right\}.$$

Some Labor Remains in Numeraire Production. If some labor remains in numeraire production in the first period of war, the choice of $l_{m,0}^{S,C}$ is interior. The first-order condition is

$$B(1-\alpha)(K_t^S)^\alpha \big(l_{m,0}^{S,C}\big)^{-\alpha} = 1,$$

so

$$l_{m,0}^{S,C}(K_t^S) = \big(B(1-\alpha)\big)^{1/\alpha} K_t^S.$$

Substituting this policy into the production function gives

$$m_0^{S,C}(K_t^S) = \big(B(1-\alpha)\big)^{\frac{1-\alpha}{\alpha}} K_t^S.$$

It follows that

$$\frac{\partial \mathcal{D}_t}{\partial K_t^S} = B\frac{\partial m_0^{S,C}}{\partial K_t^S} = \left(\frac{\chi\beta}{1-\beta}\right)^{\frac{1}{\alpha}} (1-\alpha)^{\frac{1-\alpha}{\alpha}} > 0.$$

All Labor Is Allocated to Military Production. If instead the labor constraint binds strictly in the first period of war, all labor is allocated to the defense sector: $l_{m,0}^{S,C} = \bar{l}^S$. In that case

$$m_0^{S,C} = (K_t^S)^\alpha (\bar{l}^S)^{1-\alpha},$$

and therefore

$$\frac{\partial \mathcal{D}_t}{\partial K_t^S} = \frac{\chi\beta}{1-\beta}\alpha(K_t^S)^{\alpha-1}(\bar{l}^S)^{1-\alpha} > 0.$$

The Boundary between the Two Cases. Let $a \equiv [B(1-\alpha)]^{(1-\alpha)/\alpha}$ and $K^\dagger \equiv \bar{l}^S/[B(1-\alpha)]^{1/\alpha}$. Combining the two cases gives

$$m_0^{S,C}(K) = \min\left\{aK,\ K^\alpha(\bar{l}^S)^{1-\alpha}\right\}.$$

The labor constraint is slack for $K < K^\dagger$ and binds strictly for $K > K^\dagger$. At $K = K^\dagger$, it binds with a zero multiplier. Output is continuous there, but the left and right derivatives of deterrence with respect to capital are

$$\mathcal{D}^-_{K,t}(K^\dagger, M_t^S) = Ba, \qquad \mathcal{D}^+_{K,t}(K^\dagger, M_t^S) = \alpha Ba.$$

Thus the ordinary derivative does not exist at this point, although both one-sided marginal deterrent values are positive. The derivative with respect to military capital remains $B(2-\delta_M)$. This is the point at which the two branches in (35) coincide; Appendix A.1 treats it by retaining both deterrence constraints.

A.3 Proof of Proposition 2

Private-Sector Equilibrium Conditions. Consider the defense-industry firm in country S. Taking as given prices $(p_{m,t}^S, p_{n,t}^S, w_t^S)$ and a per-unit subsidy s_t^S on the capital stock K_t^S, it chooses $\{i_t^S, l_{m,t}^S\}_{t\geq 0}$ to maximize discounted profits subject to the technology (5) and the law of motion (3). Let $\hat{\lambda}_{K,t}^S$ denote the firm's multiplier on the capital accumulation constraint. The firm's first-order conditions are

$$[l_{m,t}^S]: \qquad w_t^S = p_{m,t}^S(1-\alpha)(K_t^S)^\alpha(l_{m,t}^S)^{-\alpha}, \tag{50}$$

$$[i_t^S]: \qquad p_{n,t}^S = \hat{\lambda}_{K,t}^S, \tag{51}$$

$$[K_{t+1}^S]: \qquad \hat{\lambda}_{K,t}^S = \frac{1}{1+r_t^S}\left(s_{t+1}^S + p_{m,t+1}^S\alpha(K_{t+1}^S)^{\alpha-1}(l_{m,t+1}^S)^{1-\alpha} + (1-\delta_K)\hat{\lambda}_{K,t+1}^S\right). \tag{52}$$

The price of the numeraire good is normalized to one, $p_{n,t}^S = 1$. With the linear technology (4), profit maximization by numeraire producers gives $w_t^S = p_{n,t}^S = 1$ whenever $l_{n,t}^S > 0$, and with linear utility and positive consumption the household's first-order conditions give $1/(1+r_t^S) = \beta$. Using (51), this implies $\hat{\lambda}_{K,t}^S = 1$ for all t. Substituting these restrictions into (52) yields the firm Euler equation (29).

Decentralization We look for a decentralization of the planner allocation using a supporting price sequence for military goods satisfying

$$p^S_{m,t} = \lambda^S_{m,t} \qquad \forall t, \tag{53}$$

where $\lambda^S_{m,t}$ is the planner multiplier on (2). With (53), the firm Euler equation (29) becomes

$$1 = \beta\Big((1-\delta_K) + \lambda^S_{m,t+1}\alpha(K^S_{t+1})^{\alpha-1}(l^S_{m,t+1})^{1-\alpha} + s^S_{t+1}\Big). \tag{54}$$

Wedge Relative to the Planner and Closing It. Comparing (54) with the planner condition (49) shows that the capital-stock subsidy

$$s^S_t = \sum_j \gamma^S_{j,t}\frac{\partial \mathcal{D}^j_t}{\partial K^S_t}, \qquad t \geq 1, \tag{55}$$

exactly closes the deterrence wedge. The firm's optimal accumulation of K^S_{t+1} therefore coincides with the planner's for every $t \geq 0$. Given the same initial conditions and with lump-sum taxes available to finance the subsidy, the remaining equilibrium conditions coincide by construction, and the resulting competitive equilibrium implements the planner allocation.

For short wars, both capital derivatives vanish, so $s^S_t = 0$. For long wars at differentiable allocations, complementary slackness reduces (55) to $s^S_t = \gamma^S_t\,\partial\mathcal{D}_t/\partial K^S_t$, as stated in the proposition. At the long-war kink, both branch multipliers enter, giving $s^S_t = Ba\,\gamma^S_{I,t} + \alpha Ba\,\gamma^S_{F,t}$. These multipliers are determined jointly with the planner allocation. For long wars, the subsidy remains positive whenever $\gamma^S_t > 0$, including at the kink.

A.4 Proof of Proposition 5

The Two-Technology Deterrence Constraint. Fix a possible challenge date t. With two technologies, military output is $m^S_t = \sum_{g=1}^2 (K^S_{g,t})^{\alpha_g}(l^S_{g,t})^{1-\alpha_g}$ and $M^S_{t+1} = (1-\delta_M)M^S_t + m^S_t$. By (22), deterrence is

$$\mathcal{D}_t(K^S_{1,t}, K^S_{2,t}, M^S_t) = \frac{\chi\beta^{T-1}}{1-\beta}\sum_{\tau=0}^{T-1}\left(M^{S,C}_\tau - \bar{M}^A\right).$$

If wars are short, $T = 1$, only $M^{S,C}_0 = M^S_t$ enters this sum. Since technology-specific capital affects military production only after the possible challenge date, $\partial\mathcal{D}_t/\partial K^S_{g,t} = 0$

for each g.

If wars are long, $T = 2$, then, with $B \equiv \chi\beta/(1-\beta)$ as in Appendix A.2,

$$\mathcal{D}_t = B\left(M_t^S + M_1^{S,C} - 2\bar{M}^A\right), \quad M_1^{S,C} = (1-\delta_M)M_t^S + \sum_{g=1}^{2}(K_{g,t}^S)^{\alpha_g}\left(l_{g,0}^{S,C}\right)^{1-\alpha_g}.$$

Therefore technology-g capital contributes to deterrence through its effect on first-period conflict production $m_{g,0}^{S,C} = (K_{g,t}^S)^{\alpha_g}(l_{g,0}^{S,C})^{1-\alpha_g}$. Since numeraire consumption is positive in the first period of war, labor in technology g equates the marginal value of its output to its marginal cost, $B(1-\alpha_g)(K_{g,t}^S)^{\alpha_g}(l_{g,0}^{S,C})^{-\alpha_g} = w_g(l_{g,0}^{S,C})$. With $w_g(l) = l^{1/\varepsilon_{\ell,g}}$, conflict labor is proportional to $(K_{g,t}^S)^{\alpha_g/(\alpha_g+1/\varepsilon_{\ell,g})}$, so conflict production has elasticity $\alpha_g(1+\varepsilon_g)$ with respect to $K_{g,t}^S$. Hence

$$\frac{\partial \mathcal{D}_t}{\partial K_{g,t}^S} = B\alpha_g(1+\varepsilon_g)\frac{m_{g,0}^{S,C}}{K_{g,t}^S} > 0$$

for each active technology.

Firm and Planner FOCs. Consider technology g. Under the supporting price $p_{m,t}^S = \lambda_{m,t}^S$, the firm's Euler equation for $K_{g,t+1}^S$ and the corresponding planner Euler equation are

$$\begin{aligned}
\text{firm:} \quad & 1 = \beta\Big((1-\delta_K) + \lambda_{m,t+1}^S \alpha_g (K_{g,t+1}^S)^{\alpha_g - 1}(l_{g,t+1}^S)^{1-\alpha_g} + s_{g,t+1}^S\Big), \\
\text{planner:} \quad & 1 = \beta\Big((1-\delta_K) + \lambda_{m,t+1}^S \alpha_g (K_{g,t+1}^S)^{\alpha_g - 1}(l_{g,t+1}^S)^{1-\alpha_g} + \gamma_{t+1}^S \frac{\partial \mathcal{D}_{t+1}}{\partial K_{g,t+1}^S}\Big).
\end{aligned}$$

The labor conditions coincide under the supporting price, since the firm pays the wage $w_g(l_{g,t}^S)$ and the planner faces the same marginal cost. The only wedge is the deterrence value of technology-g capital. Setting

$$s_{g,t}^S = \gamma_t^S \frac{\partial \mathcal{D}_t}{\partial K_{g,t}^S}, \qquad t \geq 1,$$

closes this wedge for each $g \in \{1,2\}$ and implements the planner allocation.

Short and Long Wars. The shape of the deterrence constraint gives the rest of the result. If wars are short, the derivative with respect to $K_{g,t}^S$ is zero, so $s_{g,t}^S = 0$. If wars are long and the deterrence constraint has a positive shadow value, $\gamma_t^S > 0$, the derivative is strictly positive for each active technology, so $s_{g,t}^S > 0$. This proves the proposition.

A.5 Proof of Proposition 6

Setup. Consider the steady state of the two-technology economy when wars are long, and suppress country superscripts and time subscripts. Technology g produces $m_g = K_g^{\alpha_g} l_g^{1-\alpha_g}$ with labor supplied at the marginal cost $w_g(l_g) = l_g^{1/\varepsilon_{\ell,g}}$; l_g^C and m_g^C denote its labor and output in the first period of a war. By Proposition 5, $s_g = \gamma\, \partial \mathcal{D}/\partial K_g$. Since the deterrence constraint has a positive shadow value, $\gamma > 0$, so the comparison of subsidies reduces to a comparison of deterrence derivatives.

Conflict versus Peace. Let $B \equiv \chi\beta/(1-\beta)$ be the marginal value of first-period conflict production in a long war, as in Appendix A.2, and let p_m be the peacetime price of military goods. The conflict and peacetime labor FOCs for technology g are

$$B(1-\alpha_g)K_g^{\alpha_g}(l_g^C)^{-\alpha_g} = (l_g^C)^{1/\varepsilon_{\ell,g}}, \qquad p_m(1-\alpha_g)K_g^{\alpha_g} l_g^{-\alpha_g} = l_g^{1/\varepsilon_{\ell,g}}.$$

Hence $l_g^C/l_g = (B/p_m)^{1/(\alpha_g + 1/\varepsilon_{\ell,g})}$ and $m_g^C/m_g = (l_g^C/l_g)^{1-\alpha_g} = (B/p_m)^{\varepsilon_g}$. The assumption that military production rises during conflict implies $B/p_m > 1$.

The Deterrence Derivative. In a long war, technology-g capital affects deterrence by changing first-period conflict production. By the conflict labor FOC, l_g^C is proportional to $K_g^{\alpha_g/(\alpha_g + 1/\varepsilon_{\ell,g})}$, so conflict production has elasticity $\alpha_g(1+\varepsilon_g)$ with respect to K_g and

$$\frac{\partial \mathcal{D}}{\partial K_g} = B\alpha_g(1+\varepsilon_g)\frac{m_g^C}{K_g}.$$

Using the peacetime capital Euler equation, $p_m \alpha_g m_g/K_g = \beta^{-1} - (1-\delta_K) - s_g$, and substituting for m_g^C/m_g gives

$$\frac{\partial \mathcal{D}}{\partial K_g} = \Phi(\varepsilon_g)\left(\beta^{-1} - (1-\delta_K) - s_g\right), \qquad \Phi(\varepsilon) \equiv (1+\varepsilon)\left(\frac{B}{p_m}\right)^{1+\varepsilon}.$$

Because $B/p_m > 1$, $\Phi(\varepsilon)$ is strictly increasing in ε.

Comparing Subsidies. Combining $s_g = \gamma\, \partial \mathcal{D}/\partial K_g$ with the previous display gives

$$s_g = \gamma\Phi(\varepsilon_g)\left(\beta^{-1} - (1-\delta_K) - s_g\right), \quad \text{so} \quad s_g = \frac{\gamma\Phi(\varepsilon_g)}{1+\gamma\Phi(\varepsilon_g)}\left(\beta^{-1} - (1-\delta_K)\right).$$

Since $\beta^{-1} - (1 - \delta_K) > 0$, the right-hand side is strictly increasing in $\Phi(\varepsilon_g)$, and Φ is strictly increasing in ε_g. Therefore $\varepsilon_1 > \varepsilon_2$ implies $s_1 > s_2$. Restoring time subscripts gives $s^S_{1,t} > s^S_{2,t}$.

B Extensions

B.1 Traditional Rational Deterrence Theory

Geopolitical States. Fix a possible challenge date t. The baseline model assumes that if Adversary challenges the status quo, Sovereign fights. This extension makes that response explicit. There are three possible geopolitical states at t:

$$z_t \in \{P, B, C\}.$$

The first is peace, denoted by P, in which Adversary does not challenge and the status quo allocation of the public good remains in place. The second is backing down, denoted by B, in which Adversary challenges and Sovereign does not contest the challenge. The letter B stands for backing down. The third is conflict, denoted by C, in which Adversary challenges and Sovereign fights.

The desired equilibrium outcome is still peace. To sustain it, two incentive constraints must hold. First, Adversary must prefer not to challenge, taking as given that a challenge would be met by conflict. This is the deterrence constraint already studied in the baseline model. Second, if Adversary were to challenge, Sovereign must prefer fighting to backing down. This is the new commitment constraint. Throughout this extension, Sovereign cannot commit to its own future decisions: neither to fighting after a challenge nor to its future peacetime allocation, which it chooses anew at every date.

Peace. Let K^S_t and M^S_t denote the peacetime defense-industrial and military stocks at the possible challenge date, and let $V^{S,P}_t$ denote the consumption value of the peacetime continuation. Since Sovereign cannot commit to its future peacetime allocation, the continuation is chosen by Sovereign at date t and later, so $V^{S,P}_t$ is a function of the stocks it inherits. If the state is P, Sovereign's continuation welfare is

$$W^{S,P}_t(K^S_t, M^S_t) = \frac{\bar{G}^S}{1-\beta} + V^{S,P}_t(K^S_t, M^S_t). \tag{56}$$

Backing Down. If the state is B, Sovereign keeps the peacetime economic allocation but loses a present value $\Delta_t^S \geq 0$ of the public good:

$$W_t^{S,B}(K_t^S, M_t^S) = W_t^{S,P}(K_t^S, M_t^S) - \Delta_t^S. \tag{57}$$

This captures the idea that backing down preserves peace but concedes part of the disputed public good.

Conflict. If the state is C, the off-equilibrium continuation is the same wartime continuation as in the baseline model. Sovereign starts the conflict with the peacetime stocks (K_t^S, M_t^S), solves the wartime problem (20), and obtains

$$\begin{aligned} W_t^{S,C}(K_t^S, M_t^S) = \frac{\bar{G}^S}{1-\beta} + V_t^{S,C}(K_t^S, M_t^S) \\ + \frac{\chi \beta^{T-1}}{1-\beta} \sum_{\tau=0}^{T-1} \left(M_\tau^{S,C}(K_t^S, M_t^S) - \bar{M}^A \right), \end{aligned} \tag{58}$$

where $V_t^{S,C}$ is the consumption value of the optimal wartime continuation. The term with the summation is the present value of the change in Sovereign's public good generated by the conflict technology.

Deterrence Constraint. The deterrence constraint is unchanged. Since the conjectured response to a challenge is conflict, Adversary compares peace with the conflict continuation. Define

$$\mathcal{D}_t(K_t^S, M_t^S) \equiv W_t^{A,P} - W_t^{A,C}(K_t^S, M_t^S). \tag{59}$$

Adversary is deterred when

$$\mathcal{D}_t(K_t^S, M_t^S) \geq \eta_t^A. \tag{60}$$

Commitment Constraint. The new object is Sovereign's commitment constraint. Let $\bar{V}_t^S$ denote any exogenous commitment value, such as resolve, reputation, domestic audience costs, or other political costs of backing down. Sovereign is willing to fight after a challenge when

$$W_t^{S,C}(K_t^S, M_t^S) + \bar{V}_t^S \geq W_t^{S,B}(K_t^S, M_t^S). \tag{61}$$

Equivalently, define

$$\mathcal{C}_t(K_t^S, M_t^S) \equiv W_t^{S,C}(K_t^S, M_t^S) - W_t^{S,P}(K_t^S, M_t^S) + \Delta_t^S + \bar{V}_t^S. \tag{62}$$

The commitment constraint is

$$\mathcal{C}_t(K_t^S, M_t^S) \geq 0. \tag{63}$$

The two constraints play different roles. The deterrence constraint is Adversary's no-challenge condition. The commitment constraint is Sovereign's no-back-down condition. A deterrence equilibrium with limited commitment requires both.

Planning Problem. Sovereign chooses a peacetime allocation intended to implement the P outcome. The allocation must be feasible, must deter Adversary from challenging, and must make Sovereign's threatened response credible. As in the baseline problem, conflict is off the equilibrium path. The planner nevertheless has to choose peacetime stocks that make the off-equilibrium threat credible, because Adversary's no-challenge condition is evaluated under the conjecture that a challenge would be met by conflict. The planner takes the continuation value $V_t^{S,P}$ in the commitment constraint as given: it is determined by Sovereign's future decisions, to which Sovereign cannot commit, and in equilibrium it is the value of the peacetime path that the planning problem generates from the date-t stocks. The primal planning problem is therefore

$$\begin{aligned} \max_{\{c_t^S, y_{n,t}^S, y_{m,t}^S, l_{n,t}^S, l_{m,t}^S, i_t^S, m_t^S, K_{t+1}^S, M_{t+1}^S\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \left(\bar{G}^S + c_t^S \right) \\ \text{s.t.} \quad & \text{technology and market-clearing constraints (2)–(8)}, \\ & \mathcal{D}_t(K_t^S, M_t^S) \geq \eta_t^A, \quad \forall t \geq 1, \\ & \mathcal{C}_t(K_t^S, M_t^S) \geq 0, \quad \forall t, \\ & \text{non-negativity constraints, for all } t. \end{aligned} \tag{64}$$

First-Order Conditions. Let $\gamma_t^S \geq 0$ and $\kappa_t^S \geq 0$ be the multipliers on the date-t deterrence and commitment constraints. The optimality condition for military capital is

$$\lambda_{m,t}^S = \beta \left((1 - \delta_M) \lambda_{m,t+1}^S + \gamma_{t+1}^S \frac{\partial \mathcal{D}_{t+1}}{\partial M_{t+1}^S} + \kappa_{t+1}^S \frac{\partial \mathcal{C}_{t+1}}{\partial M_{t+1}^S} \right). \tag{65}$$

The optimality condition for defense-industrial capital is

$$1 = \beta \left((1 - \delta_K) + \lambda_{m,t+1}^S \alpha \left(K_{t+1}^S \right)^{\alpha - 1} \left(l_{m,t+1}^S \right)^{1-\alpha} + \gamma_{t+1}^S \frac{\partial \mathcal{D}_{t+1}}{\partial K_{t+1}^S} + \kappa_{t+1}^S \frac{\partial \mathcal{C}_{t+1}}{\partial K_{t+1}^S} \right). \tag{66}$$

Relative to the baseline model, each stock now has two strategic values. It can deter Adversary by lowering Adversary's payoff from conflict, and it can strengthen commitment by raising Sovereign's payoff from fighting relative to backing down.

Optimal Subsidy. The private Euler equation for defense-industrial capital is unchanged because firms only price the peacetime use of capital: it is (29). Using the same supporting price as in the baseline, $p^S_{m,t+1} = \lambda^S_{m,t+1}$, comparison with the planner's Euler equation gives the implementing subsidy, for $t \geq 1$,

$$s^S_t = \gamma^S_t \frac{\partial \mathcal{D}_t}{\partial K^S_t} + \kappa^S_t \frac{\partial \mathcal{C}_t}{\partial K^S_t}. \tag{67}$$

The first term is the baseline deterrence externality: firms do not internalize that capital raises the cost of war for Adversary. The second term is the commitment externality: firms also do not internalize that capital can make Sovereign more willing to fight after a challenge.

Characterizing the Commitment Derivative. Since Δ^S_t and $\bar{V}^S_t$ are exogenous, the marginal commitment value of defense-industrial capital is $\partial \mathcal{C}_t / \partial K^S_t = \partial W^{S,C}_t / \partial K^S_t - \partial W^{S,P}_t / \partial K^S_t$. Both terms follow from the envelope theorem, because each continuation value is the value of Sovereign's own problem from date t on, with the date-t stocks as parameters, so no derivative of Sovereign's policy rules appears. Let $p^{S,C}_{K,t} \equiv \partial W^{S,C}_t / \partial K^S_t$ denote the value of installed capital in the conflict continuation. In the peace continuation, an extra unit of installed capital adds its marginal product in military production at date t and leaves $1 - \delta_K$ units that replace investment, so by (29) $\partial W^{S,P}_t / \partial K^S_t = p^S_{m,t} \alpha (K^S_t)^{\alpha-1} (l^S_{m,t})^{1-\alpha} + 1 - \delta_K = \beta^{-1} - s^S_t$. Hence $\partial \mathcal{C}_t / \partial K^S_t = p^{S,C}_{K,t} - \beta^{-1} + s^S_t$. Substituting this into the subsidy formula and solving for s^S_t gives, for $t \geq 1$ and provided $\kappa^S_t < 1$,

$$s^S_t = \frac{1}{1 - \kappa^S_t} \left[\gamma^S_t \frac{\partial \mathcal{D}_t}{\partial K^S_t} + \kappa^S_t \left(p^{S,C}_{K,t} - \frac{1}{\beta} \right) \right]. \tag{68}$$

The Value of Capital in War. This expression separates the two reasons for subsidizing defense-industrial capital. The deterrence term reflects the effect of capital on Adversary's incentive to challenge. The commitment term compares the value of capital in the conflict continuation with its market value in peace, β^{-1}, the return on a unit installed at $t-1$ for one unit of the numeraire; it is positive when capital is worth more for fighting the challenge.

B.2 Equilibrium Wars

In this appendix, we extend the baseline model to allow for shocks to adversary's willingness to start a war. We study how Sovereign can attain some desired set of war probabilities most efficiently. Effectively, we generalize our notion of a best deterrence equilibrium to a more general case.

B.2.1 Economic Block

States. First we introduce notation to keep track of the states of geopolitics. As in the baseline case war is game-ending in that it can only be started once. Our convention will be that state P_t means that peace is reached at date t. The state $C_{t,\tau}$ means that war started at date t and the economy is $\tau \geq 0$ periods into the corresponding conflict continuation, at date $t + \tau$. As in the main text, peacetime quantities carry no regime superscript, and quantities in the conflict continuation carry the superscript C and are indexed by τ; here they are also indexed by the date t at which the war started. Let Q_t^P and $Q_{t,\tau}^C$ denote the date-0 Arrow-Debreu prices of the numeraire in these states. Note that we use the language of Arrow-Debreu prices to refer to states of geopolitics rather than states of nature.

We let $\bar{P}_t^C$ be the conditional probability that war starts at date t, given that peace has survived until the date-t challenge. Let $\bar{P}_t^P \equiv 1 - \bar{P}_t^C$. The implied unconditional probabilities are

$$\mathbb{P}_{0,t}^P \equiv \mathbb{P}_{0,t-1}^P \bar{P}_t^P, \qquad \mathbb{P}_{0,t}^C \equiv \mathbb{P}_{0,t-1}^P \bar{P}_t^C.$$

where we use the convention that $\mathbb{P}_{0,-1}^P \equiv 1$. Thus $\mathbb{P}_{0,t}^P$ is the probability that the date-t peacetime allocation is reached, and $\mathbb{P}_{0,t}^C$ is the probability that war first starts at date t.

Household. The household in Sovereign chooses consumption in all states to maximize

$$\sum_{t=0}^{\infty} \beta^t \mathbb{P}_{0,t}^P \left(\bar{G}^S + c_t^S \right) + \sum_{t=0}^{\infty} \sum_{\tau=0}^{\infty} \beta^{t+\tau} \mathbb{P}_{0,t}^C \left(G_{t,\tau}^{S,C} + c_{t,\tau}^{S,C} \right)$$

subject to

$$\begin{aligned} &\sum_{t=0}^{\infty} Q_t^P c_t^S + \sum_{t=0}^{\infty} \sum_{\tau=0}^{\infty} Q_{t,\tau}^C c_{t,\tau}^{S,C} \\ &\leq \sum_{t=0}^{\infty} Q_t^P \left(w_t^S \bar{l}^S + \Pi_t^S - T_t^S \right) + \sum_{t=0}^{\infty} \sum_{\tau=0}^{\infty} Q_{t,\tau}^C \left(w_{t,\tau}^{S,C} \bar{l}^S + \Pi_{t,\tau}^{S,C} - T_{t,\tau}^{S,C} \right). \end{aligned} \tag{69}$$

Firms. The numeraire firm solves the same static problem as in the baseline in each state. The defense firm chooses state-contingent production, labor, investment, and capital to maximize

$$\sum_{t=0}^{\infty} Q_t^P \left[p_{m,t}^S y_{m,t}^S + s_t^S K_t^S - w_t^S l_{m,t}^S - i_t^S \right] + \sum_{t=0}^{\infty} \sum_{\tau=0}^{\infty} Q_{t,\tau}^C \left[p_{m,t,\tau}^{S,C} y_{m,t,\tau}^{S,C} + s_{t,\tau}^{S,C} K_{t,\tau}^{S,C} - w_{t,\tau}^{S,C} l_{m,t,\tau}^{S,C} - i_{t,\tau}^{S,C} \right] \tag{70}$$

subject to

$$y_{m,t}^S = (K_t^S)^\alpha (l_{m,t}^S)^{1-\alpha}, \qquad K_{t+1}^S = (1-\delta_K) K_t^S + i_t^S,$$
$$y_{m,t,\tau}^{S,C} = (K_{t,\tau}^{S,C})^\alpha (l_{m,t,\tau}^{S,C})^{1-\alpha}, \qquad K_{t,\tau+1}^{S,C} = (1-\delta_K) K_{t,\tau}^{S,C} + i_{t,\tau}^{S,C},$$

and

$$K_{t,0}^{S,C} = K_t^S.$$

Competitive Equilibrium. Given a probability sequence $\{\bar{P}_t^C\}_{t\geq 0}$, government policies, and initial stocks, a competitive equilibrium is a state-contingent allocation, prices, profits, taxes, and transfers such that households and firms solve the problems above, the government budget constraint holds state by state, markets clear state by state, and conflict continuations inherit the peacetime stocks at the war-start date.

B.2.2 Challenge Decision and Probabilistic Deterrence

Adversary's Conflict Value. At a date t reached in peace, let $W_t^{A,C}(K_t^S, M_t^S)$ denote Adversary's continuation value from starting a war, before adding the attack shock. It is identical to that given in the main body of the paper (21). When starting a war adversary receives additional utility (possibly negative) given by η_t^A which it draws from a distribution denoted by F_t. The draw is realized at the beginning of period before it makes an attack decision. We assume that F_t is continuous and strictly increasing over the relevant range, has a finite conditional mean, and that the target probabilities are interior, $\bar{P}_t^C \in (0,1)$.

Adversary's Peace Value. Let $\mathbf{K}_t^S \equiv \{K_u^S\}_{u=t}^\infty$ and $\mathbf{M}_t^S \equiv \{M_u^S\}_{u=t}^\infty$ denote Sovereign's peacetime stock sequences from date t onward. Let $X_t^{A,P} = \bar{G}^A + \bar{c}^A$, where $\bar{c}^A$ is given by (18), denote Adversary's flow payoff in peace, excluding the attack shock. If Adversary

does not attack at date t, its continuation value solves

$$W_t^{A,P}(\mathbf{K}_t^S, \mathbf{M}_t^S) = X_t^{A,P} + \beta \int \max\left\{W_{t+1}^{A,P}(\mathbf{K}_{t+1}^S, \mathbf{M}_{t+1}^S), W_{t+1}^{A,C}(K_{t+1}^S, M_{t+1}^S) + \eta\right\} dF_{t+1}(\eta). \tag{71}$$

Challenge Decision. At date t, Adversary attacks if

$$W_t^{A,C}(K_t^S, M_t^S) + \eta_t^A > W_t^{A,P}(\mathbf{K}_t^S, \mathbf{M}_t^S).$$

Thus the conditional probability of war is

$$P_t^C = \Pr\left(\eta_t^A > W_t^{A,P}(\mathbf{K}_t^S, \mathbf{M}_t^S) - W_t^{A,C}(K_t^S, M_t^S)\right). \tag{72}$$

Probabilistic Deterrence Equilibrium. Fix $\{\bar{P}_t^C, \bar{P}_t^P\}_{t\geq 0}$ with $\bar{P}_t^P + \bar{P}_t^C = 1$. A $\{\bar{P}_t^C\}_{t\geq 0}$-deterrence equilibrium is a competitive equilibrium in which Adversary's optimal challenge decision generates

$$P_t^C = \bar{P}_t^C, \qquad P_t^P = \bar{P}_t^P$$

at every date t reached in peace.

Restrictions Imposed by Equilibrium. The notion of a probabilistic deterrence equilibrium allows us to set up the planning problem analogously to the non-probabilistic deterrence of problem in the body of the paper. We first define deterrence as in (16),

$$\mathcal{D}_t(\mathbf{K}_t^S, \mathbf{M}_t^S) \equiv W_t^{A,P}(\mathbf{K}_t^S, \mathbf{M}_t^S) - W_t^{A,C}(K_t^S, M_t^S). \tag{73}$$

A probabilistic deterrence equilibrium is consistent with a particular sequence of deterrence values. Note that since $P_t^C = 1 - F_t(\mathcal{D}_t)$, a $\{\bar{P}_t^C\}_{t\geq 0}$-deterrence equilibrium requires

$$\mathcal{D}_t(\mathbf{K}_t^S, \mathbf{M}_t^S) = \bar{\eta}_t, \qquad \text{where} \qquad \bar{\eta}_t \equiv F_t^{-1}(1 - \bar{P}_t^C). \tag{74}$$

It also simplifies (71). Let $\Omega_t \equiv \mathbb{E}[\eta_t^A \mid \eta_t^A > \bar{\eta}_t]$. Then

$$W_t^{A,P}(\mathbf{K}_t^S, \mathbf{M}_t^S) = X_t^{A,P} + \beta\left[\bar{P}_{t+1}^P W_{t+1}^{A,P}(\mathbf{K}_{t+1}^S, \mathbf{M}_{t+1}^S) + \bar{P}_{t+1}^C\left(W_{t+1}^{A,C}(K_{t+1}^S, M_{t+1}^S) + \Omega_{t+1}\right)\right]. \tag{75}$$

B.2.3 Peacetime Planning Problem

We now turn to the full statement of the planning problem.

Wartime Problem. If war starts at t, Sovereign enters the conflict continuation with $K_{t,0}^{S,C} = K_t^S$ and $M_{t,0}^{S,C} = M_t^S$ and solves the wartime problem (20). Let $W_t^{S,C}(K_t^S, M_t^S)$ denote its value, Sovereign's welfare from date t onward in the conflict continuation.

Peacetime Planning Problem. We can now state the full peacetime planning problem. For a fixed probability sequence $\{\bar{P}_t^C\}_{t\geq 0}$, the inner peacetime planning problem is

$$\begin{aligned}
\max_{\{c_t^S, l_{m,t}^S, i_t^S, m_t^S, K_{t+1}^S, M_{t+1}^S\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \left[\mathbb{P}_{0,t}^P \left(\bar{G}^S + c_t^S \right) + \mathbb{P}_{0,t}^C W_t^{S,C} \left(K_t^S, M_t^S \right) \right] \\
\text{s.t.} \quad & c_t^S + i_t^S + l_{m,t}^S = \bar{l}^S, \quad \forall t, \\
& m_t^S = \left(K_t^S \right)^{\alpha} \left(l_{m,t}^S \right)^{1-\alpha}, \quad \forall t, \\
& K_{t+1}^S = (1-\delta_K) K_t^S + i_t^S, \quad \forall t, \\
& M_{t+1}^S = (1-\delta_M) M_t^S + m_t^S, \quad \forall t, \\
& \mathcal{D}_t \left(\mathbf{K}_t^S, \mathbf{M}_t^S \right) = \bar{\eta}_t, \quad \forall t.
\end{aligned} \tag{76}$$

The initial stocks K_0^S and M_0^S are given.

B.2.4 Planner FOCs

Let $\lambda_{n,t}^S, \lambda_{y,t}^S, \lambda_{K,t}^S, \lambda_{m,t}^S$ be the current-value multipliers on the resource constraint, military production, defense-industrial capital accumulation, and military-capital accumulation, respectively. Let γ_t^S be the multiplier on $\mathcal{D}_t(\mathbf{K}_t^S, \mathbf{M}_t^S) - \bar{\eta}_t = 0$. For an interior allocation,

$$\lambda_{n,t}^S = \mathbb{P}_{0,t}^P, \qquad \lambda_{K,t}^S = \lambda_{n,t}^S, \qquad \lambda_{n,t}^S = \lambda_{y,t}^S (1-\alpha)(K_t^S)^{\alpha} (l_{m,t}^S)^{-\alpha}, \qquad \lambda_{m,t}^S = \lambda_{y,t}^S. \tag{77}$$

The present-value FOC for K_{t+1}^S is

$$\beta^t \lambda_{K,t}^S = \beta^{t+1} \left[(1-\delta_K) \lambda_{K,t+1}^S + \lambda_{y,t+1}^S \alpha (K_{t+1}^S)^{\alpha-1} (l_{m,t+1}^S)^{1-\alpha} + \mathbb{P}_{0,t+1}^C \frac{\partial W_{t+1}^{S,C}}{\partial K_{t+1}^S} \right] \tag{78}$$

$$+ \sum_{u=0}^{t+1} \beta^u \gamma_u^S \frac{\partial \mathcal{D}_u(\mathbf{K}_u^S, \mathbf{M}_u^S)}{\partial K_{t+1}^S}. \tag{79}$$

The FOC for M_{t+1}^S is

$$\beta^t \lambda_{m,t}^S = \beta^{t+1} \left[(1-\delta_M) \lambda_{m,t+1}^S + \mathbb{P}_{0,t+1}^C \frac{\partial W_{t+1}^{S,C}}{\partial M_{t+1}^S} \right] + \sum_{u=0}^{t+1} \beta^u \gamma_u^S \frac{\partial \mathcal{D}_u(\mathbf{K}_u^S, \mathbf{M}_u^S)}{\partial M_{t+1}^S}. \tag{80}$$

B.2.5 Decentralization

Household FOC. For an interior allocation, the household FOCs imply that Arrow–Debreu prices are proportional to expected marginal utilities. Normalizing the multiplier on the household's date-0 budget constraint to one,

$$Q_t^P = \beta^t \mathbb{P}_{0,t}^P, \qquad Q_{t,\tau}^C = \beta^{t+\tau} \mathbb{P}_{0,t}^C. \tag{81}$$

Planner FOC in Prices. Use (77) and support the military-goods price in peace with

$$p_{m,t}^S = \frac{\lambda_{m,t}^S}{\mathbb{P}_{0,t}^P}.$$

Then (79) becomes

$$Q_t^P = Q_{t+1}^P \left[(1-\delta_K) + p_{m,t+1}^S \alpha (K_{t+1}^S)^{\alpha-1} (l_{m,t+1}^S)^{1-\alpha} \right] + Q_{t+1,0}^C \frac{\partial W_{t+1}^{S,C}}{\partial K_{t+1}^S} + \sum_{u=0}^{t+1} \beta^u \gamma_u^S \frac{\partial \mathcal{D}_u(\mathbf{K}_u^S, \mathbf{M}_u^S)}{\partial K_{t+1}^S}. \tag{82}$$

Firm FOC. Let $p_{K,t+1,0}^{S,C}$ denote the value, in state $C_{t+1,0}$, of one installed unit of defense-industrial capital at the start of the conflict continuation. The defense firm's peacetime Euler equation is

$$Q_t^P = Q_{t+1}^P \left[(1-\delta_K) + p_{m,t+1}^S \alpha (K_{t+1}^S)^{\alpha-1} (l_{m,t+1}^S)^{1-\alpha} + s_{t+1}^S \right] + Q_{t+1,0}^C p_{K,t+1,0}^{S,C}. \tag{83}$$

B.2.6 Optimal Subsidies

Wartime Envelope. Let $V_{K,t,\tau}^{S,C}$ denote the current-value marginal value of installed capital in the wartime problem. The envelope condition gives

$$\frac{\partial W_t^{S,C}}{\partial K_t^S} = V_{K,t,0}^{S,C}.$$

The envelope recursion is

$$V_{K,t,\tau}^{S,C} = \lambda_{y,t,\tau}^{S,C} \alpha (K_{t,\tau}^{S,C})^{\alpha-1} (l_{m,t,\tau}^{S,C})^{1-\alpha} + \beta(1-\delta_K) V_{K,t,\tau+1}^{S,C}.$$

Because there is no remaining deterrence externality once war has begun, set $s^{S,C}_{t,\tau} = 0$ for all $\tau \geq 0$. The firm's conflict-state stock price then satisfies

$$p^{S,C}_{K,t,\tau} = p^{S,C}_{m,t,\tau} \alpha (K^{S,C}_{t,\tau})^{\alpha-1} (l^{S,C}_{m,t,\tau})^{1-\alpha} + \frac{Q^C_{t,\tau+1}}{Q^C_{t,\tau}} (1-\delta_K) p^{S,C}_{K,t,\tau+1}.$$

Using $p^{S,C}_{m,t,\tau} = \lambda^{S,C}_{y,t,\tau}$ and $Q^C_{t,\tau+1}/Q^C_{t,\tau} = \beta$, the two recursions coincide. Hence

$$p^{S,C}_{K,t,0} = \frac{\partial W^{S,C}_t}{\partial K^S_t}. \tag{84}$$

Capital Subsidy. Compare (82) with (83). Using (84), the direct wartime-value terms cancel. The optimal peacetime capital subsidy is therefore, for $t \geq 1$,

$$Q^P_t s^S_t = \sum_{u=0}^{t} \beta^u \gamma^S_u \frac{\partial \mathcal{D}_u(\mathbf{K}^S_u, \mathbf{M}^S_u)}{\partial K^S_t}. \tag{85}$$

Using (81),

$$s^S_t = \frac{1}{\beta^t \mathbb{P}^P_{0,t}} \sum_{u=0}^{t} \beta^u \gamma^S_u \frac{\partial \mathcal{D}_u(\mathbf{K}^S_u, \mathbf{M}^S_u)}{\partial K^S_t}. \tag{86}$$

The direct value of capital in wartime is already priced by firms through the conflict-state stock price. A positive probability of war therefore does not remove the case for subsidies: firms value the wartime use of capital, but not its effect on Adversary's decision to attack. This cancellation relies on the government not committing to wartime policy: once war begins, the allocation solves the wartime problem (20) and is decentralized without subsidies, so the conflict-state stock price equals the planner's marginal value of capital in war.

B.3 Endogenous War Duration

This appendix extends the baseline model by allowing the duration of a conflict to depend on the military balance at the possible challenge date. The point is to show that the sufficient statistic for industrial policy is unchanged, but that the optimal subsidy is lower when Adversary assigns positive probability to a short war, under a natural condition.

Duration. Fix a possible challenge date t. In the baseline model, the duration of the off-equilibrium conflict continuation is exogenous. Here, the continuation lasts two periods

with probability

$$\pi_t = \Pr(T = 2 \mid M_t^S) = \pi(M_t^S), \tag{87}$$

where π is differentiable, and we focus on the interior region in which $\pi_t \in (0,1)$. We assume the duration shock is realized at the possible challenge date, so the conditional conflict continuation is solved as in the baseline model once the duration is known.

Deterrence. Let $\mathcal{D}_t^1$ denote deterrence if the conflict continuation lasts one period and $\mathcal{D}_t^2$ denote deterrence if it lasts two periods. By (22),

$$\mathcal{D}_t^1 = \frac{\chi}{1-\beta}\left(M_t^S - \bar{M}^A\right), \qquad \mathcal{D}_t^2 = \frac{\chi\beta}{1-\beta}\left[M_t^S + M_1^{S,C} - 2\bar{M}^A\right].$$

The deterrence constraint is therefore

$$\mathcal{D}_t(K_t^S, M_t^S) = (1-\pi_t)\,\mathcal{D}_t^1 + \pi_t \mathcal{D}_t^2. \tag{88}$$

This nests the baseline model as the limiting cases $\pi_t = 0$ and $\pi_t = 1$.

Marginal Deterrent Values. We focus on the case in which some labor remains in numeraire production in the first period of the long-war continuation. Conditional on the continuation lasting two periods, the first-period military production problem is the same as in the baseline long-war case, so

$$m_0^{S,C}(K_t^S) = \left(\frac{\chi\beta}{1-\beta}(1-\alpha)\right)^{\frac{1-\alpha}{\alpha}} K_t^S.$$

Using $M_1^{S,C} = (1-\delta_M)M_t^S + m_0^{S,C}$, the marginal deterrent values are

$$\frac{\partial \mathcal{D}_t}{\partial K_t^S} = \pi_t \left(\frac{\chi\beta}{1-\beta}\right)^{\frac{1}{\alpha}} (1-\alpha)^{\frac{1-\alpha}{\alpha}}, \tag{89}$$

$$\frac{\partial \mathcal{D}_t}{\partial M_t^S} = (1-\pi_t)\,\frac{\chi}{1-\beta} + \pi_t \frac{\chi\beta}{1-\beta}\left(1 + (1-\delta_M)\right) + \pi'(M_t^S)\left(\mathcal{D}_t^2 - \mathcal{D}_t^1\right). \tag{90}$$

The industrial-capital derivative is the long-war derivative scaled by the probability that the second period is reached. Military capital has an additional duration effect because it changes this probability.

Optimal Subsidy. The sufficient statistic for industrial policy is unchanged. Proposition 2 applies to any differentiable deterrence constraint, so the optimal capital subsidy is still

$$s_t^S = \gamma_t^S \frac{\partial \mathcal{D}_t}{\partial K_t^S}.$$

Endogenous duration changes the derivative that enters this expression, not the expression itself.

Subsidies and the Private Steady State. To obtain a sharper comparative static, we characterize the steady state of the planning problem, in which the deterrence constraint binds and the derivatives of $\mathcal{D}$ are evaluated at the steady-state stocks. Steady-state values carry no time subscript or country superscript, $\rho \equiv \beta^{-1} - 1$ is the steady-state interest rate, and $\omega \equiv l_m/K$ is the labor-to-capital ratio in the defense sector. The two results of this and the next paragraph hold for any differentiable deterrence function with $\partial \mathcal{D}/\partial M^S > 0$, so they also apply to the extension of Appendix B.4. In a steady-state competitive equilibrium, the firm's labor and capital conditions are $1 = p_m(1-\alpha)\omega^{-\alpha}$ and $\rho + \delta_K - s = p_m \alpha\, \omega^{1-\alpha}$. Combining them writes the subsidy as a decreasing function of ω; as $\beta \to 1$ it reduces to $s = \delta_K - \alpha\omega/(1-\alpha)$, the case used in the main text:

$$s = \rho + \delta_K - \frac{\alpha}{1-\alpha}\,\omega. \tag{91}$$

The Efficient Capital Mix. It remains to characterize how the deterrence function affects ω. At constant stocks, the planner's conditions of Appendix A.1 for defense-sector labor, military capital, and defense capital are $1 = \lambda_m(1-\alpha)\omega^{-\alpha}$, $(\rho + \delta_M)\lambda_m = \gamma\, \partial \mathcal{D}/\partial M^S$, and $\rho + \delta_K = \lambda_m \alpha\, \omega^{1-\alpha} + \gamma\, \partial \mathcal{D}/\partial K^S$. As $\beta \to 1$ these are the conditions for choosing K and ω to minimize the resource cost $(\delta_K + \omega)K$ subject to deterrence, the problem studied in the main text. Eliminating λ_m and γ gives a relation whose left-hand side is increasing in both ω and the ratio of the marginal deterrent values, $(\partial \mathcal{D}/\partial K^S)/(\partial \mathcal{D}/\partial M^S)$. Therefore ω is decreasing in this ratio, and by (91) the optimal subsidy is increasing in it:

$$(\rho + \delta_M)\,\frac{\partial \mathcal{D}/\partial K^S}{\partial \mathcal{D}/\partial M^S}\,\omega^{\alpha} + \alpha\,\omega = (1-\alpha)(\rho + \delta_K). \tag{92}$$

Comparative Static. Let the superscript L denote the long-war-for-sure case and E the endogenous-duration case. From (89),

$$\left(\frac{\partial \mathcal{D}}{\partial K^S}\right)^E = \pi \left(\frac{\partial \mathcal{D}}{\partial K^S}\right)^L.$$

For military capital, (90) implies

$$\left(\frac{\partial \mathcal{D}}{\partial M^S}\right)^E = \pi \left(\frac{\partial \mathcal{D}}{\partial M^S}\right)^L + (1-\pi)\frac{\chi}{1-\beta} + \pi'(M)\left(\mathcal{D}^2 - \mathcal{D}^1\right).$$

Thus

$$\frac{\left(\partial \mathcal{D}/\partial K^S\right)^E}{\left(\partial \mathcal{D}/\partial M^S\right)^E} < \frac{\left(\partial \mathcal{D}/\partial K^S\right)^L}{\left(\partial \mathcal{D}/\partial M^S\right)^L}$$

whenever

$$(1-\pi)\frac{\chi}{1-\beta} + \pi'(M)\left(\mathcal{D}^2 - \mathcal{D}^1\right) > 0.$$

This condition is automatically satisfied whenever $\pi'(M)$ and $\mathcal{D}^2 - \mathcal{D}^1$ have the same sign or one of them is zero; in particular, when a larger military stock makes a long war more likely and long wars are at least as good for deterrence as short wars, $\mathcal{D}^2 \geq \mathcal{D}^1$. In that case, endogenous duration lowers the ratio of marginal deterrent values. Equation (92) then implies that ω is higher than in the long-war-for-sure case. Finally, (91) implies that the optimal capital subsidy is lower.

B.4 Strategic Interactions

This appendix extends the baseline model by allowing for a limited strategic interaction in which Adversary responds to Sovereign's choices of K^S_{t+1} and M^S_{t+1}. We focus on a simple interaction in which Adversary chooses M^A_{t+1} according to a linear rule. This allows us to study Sovereign's optimal policy when Adversary responds strategically, while staying close to the baseline model in which there is no war on the equilibrium path.[18]

B.4.1 Setup

Response Rule. At date t, Sovereign chooses the stocks that will be available at the next possible challenge date, K^S_{t+1} and M^S_{t+1}. After observing these stocks, Adversary sets its next-period military stock according to

$$M^A_{t+1} = \bar{M}^A + \iota_K K^S_{t+1} + \iota_M M^S_{t+1}, \tag{93}$$

[18]Fully endogenizing Adversary's decision would require changing the model in several ways because, in a deterrence equilibrium, Adversary never uses its military stockpile in war and hence has no direct reason to accumulate it. One would therefore have to introduce other motives for holding a stockpile of military equipment, such as a positive probability of war or the possibility that Sovereign or third parties attack Adversary.

where $\iota_K, \iota_M \geq 0$. The coefficient ι_K measures how much Adversary expands its military stock in response to Sovereign's defense-industrial capital, while ι_M measures the response to Sovereign's military capital. We assume the parameters are such that it is feasible for Adversary to follow this rule, and that the equipment Adversary adds in response is financed from an exogenous endowment rather than from its consumption, so that Adversary's consumption remains constant, as in the baseline model. Without this assumption, Adversary's consumption would differ between the peace continuation, in which it keeps adjusting its stock, and the conflict continuation, in which its stock is fixed, and the deterrence function would include the resource cost of the response in addition to the military balance.

Wars last either one or two periods, $T \in \{1, 2\}$. During a conflict that starts at t, Adversary's military stock is fixed at its stock at the possible challenge date:

$$M_\tau^{A,C} = M_t^A \qquad \text{for all } \tau \in \{0, 1\} \text{ reached during the conflict.}$$

Response-Adjusted Deterrence. Let $\mathcal{D}_t(K_t^S, M_t^S; M_t^A)$ denote the baseline deterrence function when Adversary's military stock at the possible challenge date is treated as a parameter. With the response rule (93), the deterrence function is

$$\widetilde{\mathcal{D}}_t(K_t^S, M_t^S) \equiv \mathcal{D}_t\left(K_t^S, M_t^S; \bar{M}^A + \iota_K K_t^S + \iota_M M_t^S\right). \tag{94}$$

The deterrence constraint is therefore

$$\widetilde{\mathcal{D}}_t(K_t^S, M_t^S) \geq \eta_t^A. \tag{95}$$

The distinction from the baseline is that Sovereign now internalizes how its peacetime choices affect Adversary's military stock, and hence the tightness of the deterrence constraint.

Marginal Deterrent Values. The response rule changes the marginal deterrent values through a chain-rule term. Suppressing time subscripts and evaluating derivatives at the relevant possible challenge date,

$$\frac{\partial \widetilde{\mathcal{D}}}{\partial K^S} = \frac{\partial \mathcal{D}}{\partial K^S} + \iota_K \frac{\partial \mathcal{D}}{\partial M^A}, \tag{96}$$

$$\frac{\partial \widetilde{\mathcal{D}}}{\partial M^S} = \frac{\partial \mathcal{D}}{\partial M^S} + \iota_M \frac{\partial \mathcal{D}}{\partial M^A}. \tag{97}$$

Since a higher adversary military stock makes conflict more attractive to Adversary, $\partial \mathcal{D}/\partial M^A < 0$. Hence an increase in ι_K lowers the deterrent value of Sovereign's defense-industrial capital, and an increase in ι_M lowers the deterrent value of Sovereign's military capital.

B.4.2 Optimal Policy

Planning Problem. Sovereign chooses the peacetime allocation taking the response rule as given. Since the response rule is already embedded in $\widetilde{\mathcal{D}}_t$, Adversary's military stock does not need to be tracked as an additional constraint in the reduced-form primal problem. The planning problem is

$$\begin{aligned} \max_{\{c_t^S, y_{n,t}^S, y_{m,t}^S, l_{n,t}^S, l_{m,t}^S, i_t^S, m_t^S, K_{t+1}^S, M_{t+1}^S\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \left(\bar{G}^S + c_t^S \right) \\ \text{s.t.} \quad & \text{technology and market-clearing constraints (2)–(8)}, \\ & \widetilde{\mathcal{D}}_t(K_t^S, M_t^S) \geq \eta_t^A, \quad \forall t \geq 1, \\ & \text{non-negativity constraints, for all } t. \end{aligned} \tag{98}$$

First-Order Conditions. Let $\gamma_t^S \geq 0$ be the multiplier on the date-t deterrence constraint. The optimality conditions for military capital and defense-industrial capital are

$$\lambda_{m,t}^S = \beta \left((1-\delta_M)\lambda_{m,t+1}^S + \gamma_{t+1}^S \frac{\partial \widetilde{\mathcal{D}}_{t+1}}{\partial M_{t+1}^S} \right), \tag{99}$$

$$1 = \beta \left((1-\delta_K) + \lambda_{m,t+1}^S \alpha \left(K_{t+1}^S \right)^{\alpha-1} \left(l_{m,t+1}^S \right)^{1-\alpha} + \gamma_{t+1}^S \frac{\partial \widetilde{\mathcal{D}}_{t+1}}{\partial K_{t+1}^S} \right). \tag{100}$$

The only difference from the baseline FOCs is that the planner uses the total derivatives in (96) and (97), rather than the partial derivatives holding Adversary's military stock fixed.

Optimal Subsidy. The private Euler equation for defense-industrial capital, (29), is unchanged. Comparing it with (100), and using the same supporting price for military goods as in the baseline, gives the implementing subsidy, for $t \geq 1$,

$$s_t^S = \gamma_t^S \frac{\partial \widetilde{\mathcal{D}}_t}{\partial K_t^S} = \gamma_t^S \left(\frac{\partial \mathcal{D}_t}{\partial K_t^S} + \iota_K \frac{\partial \mathcal{D}_t}{\partial M_t^A} \right). \tag{101}$$

Thus the sufficient statistic is unchanged in form: the subsidy equals the shadow value of deterrence times the marginal contribution of capital to deterrence. The response rule

changes the relevant marginal contribution. Since $\partial \mathcal{D}/\partial M^A < 0$, a stronger adversary response to K^S directly lowers the marginal deterrent value of defense-industrial capital.

B.4.3 Comparative Static: Result

Statement. We now ask how the optimal steady-state capital subsidy changes with the reaction coefficients ι_K and ι_M. Focus on long wars ($T = 2$) and an interior steady-state region in which the labor constraint is slack in the first period of war and the response-adjusted marginal deterrent values of both K^S and M^S are positive. Let s denote the steady-state capital subsidy. In this region, increasing the response to Sovereign's defense-industrial capital lowers the optimal capital subsidy, while increasing the response to Sovereign's military capital raises it:

$$\frac{\partial s}{\partial \iota_K} < 0, \qquad \frac{\partial s}{\partial \iota_M} > 0. \tag{102}$$

The intuition is simple. A higher ι_K means Adversary arms more in response to Sovereign's defense-industrial capital. This makes defense-industrial capital less effective at providing deterrence, so the planner tilts away from it and the capital subsidy falls. A higher ι_M means Adversary arms more in response to Sovereign's military capital. This makes military capital less effective at providing deterrence relative to the industrial base, so the planner tilts toward defense-industrial capital and the capital subsidy rises.

B.4.4 Comparative Static: Derivation

This section derives the comparative static in (102). Throughout we focus on the case where wars are long.

Deterrent Values. In a long war, the baseline deterrence function (22), with Adversary's stock M_t^A in place of $\bar{M}^A$, is

$$\mathcal{D}_t(K_t^S, M_t^S; M_t^A) = \frac{\chi\beta}{1-\beta}\left[M_t^S + M_1^{S,C} - 2M_t^A\right].$$

Thus

$$\frac{\partial \mathcal{D}}{\partial M^A} = -\frac{2\chi\beta}{1-\beta}.$$

Equations (96) and (97) become

$$\frac{\partial \widetilde{\mathcal{D}}}{\partial K^S} = \frac{\partial \mathcal{D}}{\partial K^S} - \frac{2\chi\beta}{1-\beta}\iota_K, \tag{103}$$

$$\frac{\partial \widetilde{\mathcal{D}}}{\partial M^S} = \frac{\partial \mathcal{D}}{\partial M^S} - \frac{2\chi\beta}{1-\beta}\iota_M. \tag{104}$$

Signing the Ratio. The steady-state conditions of Appendix B.3 apply with $\widetilde{\mathcal{D}}$ in place of $\mathcal{D}$, so the optimal subsidy is increasing in the response-adjusted ratio of marginal deterrent values,

$$\frac{\partial \widetilde{\mathcal{D}}/\partial K^S}{\partial \widetilde{\mathcal{D}}/\partial M^S} = \frac{\partial \mathcal{D}/\partial K^S + \iota_K \partial \mathcal{D}/\partial M^A}{\partial \mathcal{D}/\partial M^S + \iota_M \partial \mathcal{D}/\partial M^A}.$$

Because $\partial \mathcal{D}/\partial M^A < 0$, and because we focus on the interior region in which the numerator and denominator are positive,

$$\frac{\partial}{\partial \iota_K}\left(\frac{\partial \widetilde{\mathcal{D}}/\partial K^S}{\partial \widetilde{\mathcal{D}}/\partial M^S}\right) = \frac{\partial \mathcal{D}/\partial M^A}{\partial \widetilde{\mathcal{D}}/\partial M^S} < 0,$$

$$\frac{\partial}{\partial \iota_M}\left(\frac{\partial \widetilde{\mathcal{D}}/\partial K^S}{\partial \widetilde{\mathcal{D}}/\partial M^S}\right) = -\frac{\left(\partial \widetilde{\mathcal{D}}/\partial K^S\right)\left(\partial \mathcal{D}/\partial M^A\right)}{\left(\partial \widetilde{\mathcal{D}}/\partial M^S\right)^2} > 0.$$

Equation (92) then implies that increasing ι_K raises ω, while increasing ι_M lowers ω. Finally, (91) implies that increasing ι_K lowers the optimal capital subsidy, while increasing ι_M raises it.

C Quantitative Model

C.1 Competitive Equilibrium

Throughout, $\Lambda_t \equiv 1/C_t$ is the marginal utility of consumption, $\rho \equiv 1/\beta - 1$, and $\rho_V = \frac{\sigma_{VA}-1}{\sigma_{VA}}$, $\rho_X = \frac{\sigma_X - 1}{\sigma_X}$, $\rho_Y = \frac{\sigma_Y - 1}{\sigma_Y}$.

Household's Problem The household holds a risk-free bond B_t in zero net supply and takes as given the goods prices $p_{g,t}$, the sectoral wages $w_{g,t}$, the interest rate r_t, the dividends Π_t of the producers it owns, and the lump-sum tax T_t. The price index of the consumption basket is the numeraire. With C_t and $\tilde{L}_t$ the aggregators of Section 4, $B_0 = 0$,

and a no-Ponzi condition, it solves:

$$\max_{\{c_{g,t},L_{g,t},B_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \Big(\log C_t - \frac{\tilde{L}_t^{\,1+1/\xi}}{1+1/\xi} + G_t \Big)$$
$$\text{s.t.} \quad \sum_{g\in\mathcal{G}} p_{g,t} c_{g,t} + \frac{B_{t+1}}{1+r_t} = \sum_{g\in\mathcal{G}} w_{g,t} L_{g,t} + \Pi_t - T_t + B_t.$$

Producers' Problem The producer of sector g owns its capital stock and pays out its period profit as a dividend. It takes as given the goods prices, the wage, the interest rate, and the subsidy rate $s_{g,t}$ on the replacement value $P^I_{g,t}K_{g,t}$ of its capital, where $P^I_{g,t}$ is the price of its investment basket. With $K_{g,0}$ given and $R_t = \prod_{v=0}^{t-1}(1+r_v)$ the cumulative gross return, it solves:

$$\max_{\{L_{g,t},x_{g,r,t},i_{g,r,t},I_{g,t},K_{g,t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \frac{1}{R_t} \Big(p_{g,t} Y_{g,t} + s_{g,t} P^I_{g,t} K_{g,t} - w_{g,t} L_{g,t} - \sum_{r\in\mathcal{G}} p_{r,t} \big(x_{g,r,t} + i_{g,r,t} \big) \Big)$$
$$\text{s.t.} \quad Y_{g,t} = \Big[\alpha_{V,g} V_{g,t}^{\rho_Y} + \alpha_{X,g} X_{g,t}^{\rho_Y} \Big]^{\frac{1}{\rho_Y}},$$
$$V_{g,t} = \Big[\alpha_{K,g} K_{g,t}^{\rho_V} + \alpha_{L,g} L_{g,t}^{\rho_V} \Big]^{\frac{1}{\rho_V}}, \qquad X_{g,t} = \Big[\sum_{r\in\mathcal{G}} \alpha_{g,r}\, x_{g,r,t}^{\rho_X} \Big]^{\frac{1}{\rho_X}},$$
$$K_{g,t+1} = (1-\delta_g) K_{g,t} + I_{g,t} - \frac{\varphi}{2} \Big(\frac{I_{g,t}}{K_{g,t}} - \delta_g \Big)^2 K_{g,t}, \qquad I_{g,t} = \prod_{r\in\mathcal{G}} i_{g,r,t}^{\vartheta_{g,r}}, \qquad I_{g,t} \geq 0.$$

Government The government buys each basket at minimum cost, pays the subsidies, and balances its budget with the lump-sum tax; in the postwar phase the spending rule of Section 4 replaces the choice of the baskets and $s_{g,t} = 0$. With $p_{m,j,t}$ the price index of basket j given below, the basket aggregators and GDP, final demand at market prices in the numeraire, are:

$$m_{j,t} = \prod_{g\in\mathcal{G}_j} m_{j,g,t}^{\psi_{j,g}}, \qquad \text{GDP}_t = C_t + \sum_{g\in\mathcal{G}} P^I_{g,t} I_{g,t} + \sum_{j\in\mathcal{J}} p_{m,j,t} m_{j,t}.$$

Definition 3 (Competitive equilibrium of the quantitative model)**.** *Given the government's purchases and subsidy rates* $\{m_{j,t}, s_{g,t}\}_{t=0}^{\infty}$ *and the initial stocks* $(\mathbf{K}_0, \mathbf{M}_0)$*, a competitive equilibrium is a sequence of quantities* $\{c_{g,t}, L_{g,t}, x_{g,r,t}, i_{g,r,t}, I_{g,t}, K_{g,t+1}, m_{j,g,t}, M_{j,t+1}\}_{t=0}^{\infty}$*, prices* $\{p_{g,t}, w_{g,t}, r_t, P^X_{g,t}, P^I_{g,t}, p_{m,j,t}\}_{t=0}^{\infty}$*, and taxes and dividends* $\{T_t, \Pi_t\}_{t=0}^{\infty}$ *such that the household and the producers solve their problems, the government buys each basket at minimum cost and its budget holds, and the goods and labor markets clear in every period.*

The rest of this subsection collects the conditions that characterize the equilibrium.

Household: Optimality Conditions The CES demands for goods and the first-order conditions for the bond and for sectoral labor supply are:

$$c_{g,t} = \epsilon_g \Big(\frac{p_{g,t}}{P_{C,t}}\Big)^{-\sigma_C} C_t, \qquad P_{C,t} = \Big[\sum_{g\in\mathcal{G}} \epsilon_g \, p_{g,t}^{1-\sigma_C}\Big]^{\frac{1}{1-\sigma_C}} = 1, \tag{105}$$

$$\frac{1}{1+r_t} = \beta \, \frac{\Lambda_{t+1}}{\Lambda_t}, \qquad \Lambda_t \, w_{g,t} = \tilde{L}_t^{\,1/\xi} \Big(\frac{L_{g,t}}{\tilde{L}_t}\Big)^{\frac{1}{\tau}}. \tag{106}$$

Producers: Static Optimality Conditions Labor and the intermediate composite are paid their marginal revenue products, and the demands for individual intermediate inputs and investment goods follow from cost minimization:

$$w_{g,t} = p_{g,t} \, \alpha_{V,g} \, \alpha_{L,g} \Big(\frac{Y_{g,t}}{V_{g,t}}\Big)^{\frac{1}{\sigma_Y}} \Big(\frac{V_{g,t}}{L_{g,t}}\Big)^{\frac{1}{\sigma_{VA}}}, \qquad P_{g,t}^X = p_{g,t} \, \alpha_{X,g} \Big(\frac{Y_{g,t}}{X_{g,t}}\Big)^{\frac{1}{\sigma_Y}}, \tag{107}$$

$$x_{g,r,t} = \alpha_{g,r}^{\sigma_X} \Big(\frac{p_{r,t}}{P_{g,t}^X}\Big)^{-\sigma_X} X_{g,t}, \qquad P_{g,t}^X = \Big[\sum_{r\in\mathcal{G}} \alpha_{g,r}^{\sigma_X} \, p_{r,t}^{1-\sigma_X}\Big]^{\frac{1}{1-\sigma_X}}, \tag{108}$$

$$i_{g,r,t} = \vartheta_{g,r} \, \frac{P_{g,t}^I I_{g,t}}{p_{r,t}}, \qquad P_{g,t}^I = \prod_{r\in\mathcal{G}} \Big(\frac{p_{r,t}}{\vartheta_{g,r}}\Big)^{\vartheta_{g,r}}. \tag{109}$$

Producers: Investment and Capital The producer values its capital at its marginal revenue product plus the subsidy, but not at its effect on deterrence. With $q_{g,t}$ the shadow value of installed capital, the first-order condition for irreversible investment and the Euler equation for capital are

$$\frac{P_{g,t}^I}{1-\varphi\big(I_{g,t}/K_{g,t}-\delta_g\big)} \;\geq\; q_{g,t}, \qquad I_{g,t} \geq 0, \qquad \text{with complementary slackness,} \tag{110}$$

$$\begin{aligned} q_{g,t} = \beta \, \frac{\Lambda_{t+1}}{\Lambda_t} \Big[&(1-\delta_g) \, q_{g,t+1} + s_{g,t+1} P_{g,t+1}^I + p_{g,t+1} \, \mathrm{MPK}_{g,t+1} \\ &+ q_{g,t+1} \, \frac{\varphi}{2} \Big(\Big(\frac{I_{g,t+1}}{K_{g,t+1}}\Big)^2 - \delta_g^2 \Big) \Big], \\ \mathrm{MPK}_{g,t} = \alpha_{V,g} \, \alpha_{K,g} &\Big(\frac{Y_{g,t}}{V_{g,t}}\Big)^{\frac{1}{\sigma_Y}} \Big(\frac{V_{g,t}}{K_{g,t}}\Big)^{\frac{1}{\sigma_{VA}}}. \end{aligned} \tag{111}$$

Government and Markets: Conditions Given the government's instruments $\{m_{j,t}, s_{g,t}\}$ and the initial stocks, the competitive equilibrium solves (105)–(114) together with the technology. The remaining conditions are the basket price indices and demands, the gov-

ernment budget, goods market clearing, and the laws of motion:

$$p_{m,j,t} = \prod_{g \in \mathcal{G}_j} \left(\frac{p_{g,t}}{\psi_{j,g}} \right)^{\psi_{j,g}}, \qquad m_{j,g,t} = \psi_{j,g} \frac{p_{m,j,t} m_{j,t}}{p_{g,t}}, \qquad T_t = \sum_{j \in \mathcal{J}} p_{m,j,t} m_{j,t} + \sum_{g \in \mathcal{G}} s_{g,t} P^I_{g,t} K_{g,t}, \tag{112}$$

$$Y_{g,t} = c_{g,t} + \sum_{b \in \mathcal{G}} x_{b,g,t} + \sum_{b \in \mathcal{G}} i_{b,g,t} + \sum_{j \in \mathcal{J}} m_{j,g,t}, \tag{113}$$

$$K_{g,t+1} = (1 - \delta_g) K_{g,t} + I_{g,t} - \frac{\varphi}{2} \left(\frac{I_{g,t}}{K_{g,t}} - \delta_g \right)^2 K_{g,t}, \qquad M_{j,t+1} = (1 - \delta_{j,t}) M_{j,t} + m_{j,t}. \tag{114}$$

C.2 Wartime Equilibrium and the Deterrence Function

This appendix derives the deterrence function $\mathcal{D}_t$ of Section 4. Suppose Adversary challenges at date t. The war is re-indexed to start at $s = 0$ from the peacetime stocks $(\mathbf{K}_t, \mathbf{M}_t)$, which are predetermined at the outbreak; its length T is drawn with probability $\tilde{\pi}_T(M_t)$ and then known to both parties.

Adversary's Economic Cost of War In peace Adversary's consumption and labor supply are constant; from the outbreak of a war of length T they follow the paths $\{C_s^{E,(T)}, \tilde{L}_s^{E,(T)}\}_{s \geq 0}$, which return to the peacetime levels after the war. Let U^E denote Adversary's peacetime period utility from consumption and labor, $U \equiv \log C - \tilde{L}^{1+1/\xi}/(1 + 1/\xi)$, and $U_s^{E,(T)}$ its value along the war paths. The economic cost of the war, which does not depend on Sovereign's stocks, is

$$\mathcal{E}^{(T)} = \sum_{s=0}^{\infty} \beta^s (U^E - U_s^{E,(T)}).$$

Adversary's Welfare Adversary's welfare differs between peace and war through the public good and through the economic cost of its mobilization. In peace the public good stays at its initial level; in a war of length T it is reduced from period T on by the settlement $\sum_{s=0}^{T-1} \chi_s^{(T)} (M_s - M_s^E)$, the territory Sovereign gains; here $M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t)$ is Sovereign's military power in period s of the war, derived below, with $M_0^{(T)} = M_t$. Adversary evaluates the war before its length is known, $W_t^{A,C} = \sum_T \tilde{\pi}_T(M_t) W_t^{A,C,(T)}$, so $\mathcal{D}_t = W_t^{A,P} - W_t^{A,C}$ is the expression of Section 4; the initial level of the public good cancels and does not enter the deterrence function. Discounting the settlement from its first

period, Adversary's welfare loss from a war of length T is

$$W_t^{A,P} - W_t^{A,C,(T)} = \frac{\beta^T}{1-\beta} \sum_{s=0}^{T-1} \chi_s^{(T)} \left(M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t) - M_s^E \right) + \mathcal{E}^{(T)}.$$

Sovereign's Wartime Problem Sovereign's public good rises by the settlement from period T on. Let V^{post} denote the household's welfare in the postwar competitive equilibrium under the spending rule, evaluated with the public good at its initial level G_0; the rise in the public good enters the objective through the settlement term. The wartime government takes the postwar purchases as given: they are an exogenous demand for sectoral goods, equal in equilibrium to the amounts the spending rule implies. Subject to the technology, market clearing, the laws of motion with the wartime depreciation rates δ_j^w, $m_{j,s} \geq 0$ and $I_{g,s} \geq 0$, the wartime government maximizes

$$\sum_{s=0}^{T-1} \beta^s \left(\log C_s - \frac{\tilde{L}_s^{1+1/\xi}}{1+1/\xi} + G_0 \right) + \frac{\beta^T}{1-\beta} \sum_{s=0}^{T-1} \chi_s^{(T)} \left(M_s - M_s^E \right) + \beta^T V^{\text{post}}(\mathbf{K}_T, \mathbf{M}_T).$$

Wartime Optimality Conditions Because the settlement depends on the military stocks only, and sectoral capital affects it only through production, the wartime Euler equation for capital carries no deterrence wedge. Given the postwar purchases, the postwar competitive equilibrium is efficient, so the capital carried out of the war has its competitive value. The government's within-period allocation therefore coincides with the competitive one, and the multipliers on the technology and market-clearing constraints equal the competitive prices. Conditions (105)–(114) hold with $s_{g,s} = 0$, since there is no industrial policy during the war, and with the wartime rates δ_j^w in the laws of motion of the military stocks. The only conditions specific to the wartime problem concern procurement. Let $\mu_{j,s}$ denote the shadow value, in period-s utility, of a unit of stock j at the beginning of period s. A unit of stock j is worth its contribution to the settlement in the current period, through the marginal product $\phi_j M_s / M_{j,s}$ of the stock in military power, plus its depreciated value next period. The terminal condition $\mu_{j,T} = 0$ follows from the postwar spending rule, under which the stocks carried out of the war have no value. The purchases of the last period of the war arrive after the settlement, so the procurement condition would set them to zero; we let them follow the postwar rule instead, $p_{m,j,T-1} m_{j,T-1} = \theta_j \theta \, \text{GDP}_{T-1}$, so that procurement does not drop to zero before the postwar phase. Solved from the stocks $(\mathbf{K}_t, \mathbf{M}_t)$, these conditions deliver Sovereign's wartime military power $M_s^{(T)}(\mathbf{K}_t, \mathbf{M}_t)$. The first-order condition for procurement, for

$s = 0, \ldots, T-2$, and the recursion for the shadow values are

$$\Lambda_s \, p_{m,j,s} \;\geq\; \beta \, \mu_{j,s+1}, \qquad m_{j,s} \geq 0, \qquad \text{with complementary slackness,} \qquad s = 0, \ldots, T-2, \tag{115}$$

$$\mu_{j,s} = \frac{\beta^{T-s}}{1-\beta} \, \chi_s^{(T)} \, \phi_j \, \frac{M_s}{M_{j,s}} + \beta \, (1 - \delta_j^w) \, \mu_{j,s+1}, \qquad s = 0, \ldots, T-1, \qquad \mu_{j,T} = 0. \tag{116}$$

C.3 Calibration

Sectors The sectoral data are the 2017 BEA input-output accounts at the detail level (U.S. Bureau of Economic Analysis, 2023b). Following Iovino et al. (2025), we drop finance and insurance, owner-occupied housing, and government industries, which leaves 378 private industries. We further drop eight retail trade industries (motor vehicle and parts dealers; food and beverage stores; general merchandise stores; building material and garden equipment and supplies dealers; health and personal care stores; gasoline stations; clothing and clothing accessories stores; all other retail) whose output has no buyer in the model: no consumption weight, no defense procurement, and no intermediate or investment purchase by another sector. The remaining $N = 370$ industries are the sectors of the model.

Technology and Preference Weights The consumption weights ϵ_g are set so that the expenditure shares $p_g c_g / C$ in the postwar steady state equal the shares $s_{C,g}$ of personal consumption expenditures, non-defense government purchases (the intermediate purchases of federal non-defense and of state and local general government, and their gross investment in structures, equipment, and intellectual property), and exports in the use table: by (105), $\epsilon_g = s_{C,g} \, p_g^{\sigma_C - 1}$ at the prices of that steady state. The CES weights of the production nests are calibrated to the cost shares in the data: the intermediate spending shares of the use table, the value-added share of gross output (compensation plus operating surplus), and the labor share of value added (compensation). Away from the Cobb–Douglas case a CES weight is not a cost share: the capital share of value added, for instance, is $\alpha_{K,g}(K_g/V_g)^{\rho_V}$. We therefore choose the weights jointly with the postwar steady state, so that its cost shares equal the data shares, and normalize the weights of every nest to sum to one. Without this normalization the share conditions would fix only the ratios of the weights within a nest, since the level of each nest's output is a choice of units (see also Carvalho et al. (2025)). The postwar steady state then reproduces every data share for any elasticities. The weights are held fixed thereafter, so the expenditure and cost shares move with relative prices during the wars and in the steady state with industrial policy.

Investment Network and Depreciation The investment weights $\vartheta_{g,r}$ are the shares of each producing industry in the investment of industry g. They are built from the BEA detailed fixed-asset investment by industry and asset type (U.S. Bureau of Economic Analysis, 2023a), bridging each asset type to its producing industries (for equipment with the BEA bridge table, U.S. Bureau of Economic Analysis, 2024) by the method of Vom Lehn and Winberry (2022), in the detail-level implementation of Iovino et al. (2025). The purchasing side is observed for 66 fixed-asset industries, so the detailed sectors within one fixed-asset industry share the same capital composition. The sectoral depreciation rate δ_g is the depreciation rate of the sector's steady-state capital stock implied by its investment mix over asset types and the depreciation rates by asset type that Iovino et al. (2025) take from Barro and Furman (2018): 8.8% for equipment, 2.0% for structures, 12.2% for research and development, and 19.6% for software and artistic originals.

Military Baskets Military procurement consists of the purchases of goods and services by national defense in the use table: the intermediate purchases of the national-defense government sector and defense investment in structures, equipment, and intellectual property. We first remove from these purchases the overhead services, such as facilities support, food service, and management consulting, which keep installations and personnel running but do not add to military power. The remaining 184 detailed industries are partitioned into the eight baskets: land systems (military armored vehicles), air systems (aircraft, aircraft parts and engines), naval systems (ship and boat building), ammunition (ammunition, arms and ordnance; guided missiles; missile and space propulsion), structures, intellectual property (research and development, engineering, software), nondurables (fuel, pharmaceuticals, transportation, and other goods and services consumed within the year, which BEA expenses), and the residual other equipment. The within-basket weights $\psi_{j,g}$ and the basket shares θ_j are shares of the remaining purchases. Spending on the baskets totals the postwar spending share $\theta = 3.37\%$ of GDP, the US average over 2015–2025 in the SIPRI data (Stockholm International Peace Research Institute, 2026), so by dropping the overhead we change the composition of modeled procurement but not its level. Table 8 lists every basket and its component sectors, and Table 9 the overhead services excluded from the baskets.

Depreciation of the Military Stocks The peacetime depreciation rates δ_j are the post-1970 means of current-cost depreciation over the net stock of Federal national-defense fixed assets in the BEA Fixed Assets tables (U.S. Bureau of Economic Analysis, 2025, Tables 7.1 and 7.3): structures, total equipment, and intellectual property for their bas-

kets, and vehicles, aircraft, and ships for land, air, and naval systems. Non-durables are consumed within the year. For ammunition the rate is the FY2014–2019 mean of procurement (U.S. Department of Defense, 2015–2020) over the DoD Class V inventory of conventional ammunition and missiles (U.S. Department of Defense, 2014–2019, Table 1.7A), 12.7%. The wartime rates δ_j^w add combat losses. For land systems we use the Russia–Ukraine war: visually confirmed annual losses (destroyed, captured, and abandoned; Oryx, 2022a,b, as compiled by Drake, 2022) over the stock at the start of the year (*The Military Balance*, International Institute for Strategic Studies, 2022, 2023, 2024, 2025), 2022–2025, give attrition of 18% for Russia and 27% for Ukraine; added to the peacetime rate these give wartime rates of 34 and 43%, and we use their mean, $\delta^w = 38.3\%$. For air and naval systems we use the losses of the US armed forces in the Second World War, the only war in which US air and naval forces fought a peer adversary at scale for several years. For aircraft, attrition is the Army Air Forces' losses overseas from all causes, combat and operational, over the aircraft on hand at the start of the year (U.S. Army Air Forces, 1945, Tables 83 and 99), 24.2% per year on average over 1942–45; adding the peacetime rate gives $\delta^w = 34.3\%$. For ships, it is major combatants sunk or damaged beyond repair (battleships, carriers, cruisers, destroyers, destroyer escorts, and submarines; Naval History and Heritage Command, 2004) over the average number in commission during the year (Naval History and Heritage Command, 2017), 6.5% per year on average; adding the peacetime rate gives $\delta^w = 14.6\%$. Structures, other equipment, and intellectual property keep their peacetime rates; non-durables are consumed within the year, and so is ammunition in war. Table 5 collects the rates.

Table 5: Depreciation of the military baskets in peace and in war.

Basket	Peacetime, δ_j	Combat attrition	Wartime, δ_j^w
Structures	0.020	0	0.020
Other equipment	0.116	0	0.116
Intellectual property	0.223	0	0.223
Non-durables	1.000	—	1.000
Land systems	0.155	0.228	0.383
Air systems	0.101	0.242	0.343
Naval systems	0.081	0.065	0.146
Ammunition	0.127	—	1.000

Combat attrition is the increment $\delta_j^w - \delta_j$: for land systems the mean of the Russian and Ukrainian rates in the Russia–Ukraine war, for air and naval systems the loss rates of the US armed forces in the Second World War; structures, other equipment, and intellectual property are not exposed to it. Non-durables and, in war, ammunition are consumed within the year, so their rate is one rather than a peacetime rate plus attrition.

Weights of the Baskets in Military Power The weights ϕ_j are chosen so that a government that deters with procurement only, the no-industrial-policy benchmark of Appendix C.5, holds the arsenal implied by the observed spending shares and depreciation rates: in the steady state, procurement of basket j is $p_{m,j}m_j = \theta_j\theta\,\text{GDP}$ and its stock is $M_j = m_j/\delta_j$. By the procurement conditions shown in Appendix C.5, the government chooses this arsenal if the marginal deterrence value of a stock relative to its user cost, $(\partial\mathcal{D}/\partial M_j)/\left(p_{m,j}(\rho+\delta_j)\right)$, is the same for every basket. The derivatives $\partial\mathcal{D}/\partial M_j$ depend on the weights, and we solve these $J-1$ conditions for the $J-1$ free weights ϕ_j, which sum to one. The derivatives also depend on the combat efficiency parameters $\chi^{(T)}$, which are calibrated given ϕ, so the two sets of parameters are solved for together (Appendix C.6). At the solution the peacetime steady state without subsidies coincides with the postwar steady state, so the observed composition of military spending is optimal for a government that does not use industrial policy. Table 6 collects the basket shares and the military-power weights.

Table 6: Spending shares and military-power weights of the military baskets. Entries are rounded to three decimal places.

Basket	Spending share, θ_j	Military-power weight, ϕ_j
Structures	0.030	0.023
Other equipment	0.232	0.175
Intellectual property	0.309	0.362
Non-durables	0.168	0.195
Land systems	0.007	0.007
Air systems	0.147	0.118
Naval systems	0.053	0.039
Ammunition	0.054	0.081

War Length The probabilities π_T are the frequencies of war lengths in years among the inter-state wars with at least 1,000 battle deaths in the Correlates of War data (Sarkees and Wayman, 2010), 1816–2007, with wars of six years or more collected in $T = 6$. The sensitivity κ is estimated on the same wars by a linear probability model of an indicator for a war lasting more than one year on the initiator's share of the two sides' military personnel in the year before the war, from the Correlates of War National Material Capabilities data (Singer et al., 1972; Correlates of War Project, 2026); the initiator is the side coded as such, and the other side comprises its opponents in the war's first year. On the 87 wars with personnel data the slope is -0.33 (robust standard error 0.15). Since Adver-

sary opens a war with the military power Sovereign holds at the peacetime steady state without subsidies, its share there is one half, so $a = 1$ and the war-length probabilities are the observed frequencies: the probability of a war longer than one year is $1 - \pi_1 = 0.316$.

Combat Efficiency Parameters The target path is the military-spending response to war estimated by Marzian and Trebesch (2026), the deviation of military spending from its prewar level as a share of prewar GDP over the first five years of a war, $\sigma^* = (5.3, 15, 19, 21.5, 20.5)\%$. For a war of length T we require nominal military spending in year s of the war to exceed its steady-state level by σ_s^* of steady-state GDP for $s = 0, \ldots, T-2$. The last year of the war is not targeted: its procurement follows the postwar rule and does not enter the settlement. The $T-1$ targets pin down $\chi_1^{(T)}, \ldots, \chi_{T-1}^{(T)}$. We compute them by adding these parameters to the unknowns, and the $T-1$ spending targets to the equations, of the joint war–postwar system of Appendix C.6; each war starts from the peacetime steady state without subsidies. Table 7 reports the result. The combat efficiency parameters enter the wartime problem only through the procurement conditions (115)–(116). Fixing the path of military spending therefore fixes the wartime allocation independently of $\chi^{(T)}$, and the procurement conditions can be solved for the values of $\chi^{(T)}$ that make the path optimal; the ratio conditions across baskets pin down the composition of procurement without $\chi^{(T)}$, so every basket returns the same value. For any basket j purchased in every period before the last, substituting (115) into (116) gives

$$\chi_{s+1}^{(T)} = \frac{1-\beta}{\beta^{T-s}} \frac{M_{j,s+1}}{\phi_j M_{s+1}} \left[\Lambda_s \, p_{m,j,s} - \beta(1-\delta_j^w) \, \Lambda_{s+1} \, p_{m,j,s+1} \right], \qquad s = 0, \ldots, T-3,$$

$$\chi_{T-1}^{(T)} = \frac{1-\beta}{\beta^2} \frac{M_{j,T-1}}{\phi_j M_{T-1}} \Lambda_{T-2} \, p_{m,j,T-2}.$$

Table 7: Combat efficiency parameters inverted from the wartime spending path.

T	$\chi_1^{(T)}$	$\chi_2^{(T)}$	$\chi_3^{(T)}$	$\chi_4^{(T)}$	$\chi_5^{(T)}$
2	0.493				
3	0.203	0.639			
4	0.188	0.278	0.680		
5	0.201	0.257	0.306	0.697	
6	0.213	0.269	0.271	0.342	0.683

Adversary's Economy and the Level of Deterrence We assume that Adversary's economy mirrors Sovereign's: in a war of length T its military power, consumption, and labor

supply follow Sovereign's own paths in that war started from the peacetime steady state without subsidies, $M_s^E = M_s^{(T)}(\mathbf{K}^{ss}, \mathbf{M}^{ss})$ and likewise for $C_s^{E,(T)}$ and $\tilde{L}_s^{E,(T)}$, computed once per calibration round. The economic costs $\mathcal{E}^{(T)}$ are evaluated over the T war periods and the $H = 8$ postwar periods of these paths (Appendix C.6). Every settlement term of the deterrence function then vanishes at that steady state for any value of χ_0, which is set to the probability-weighted mean over $T \geq 2$ of the parameters in the later periods of a war, $\chi_0 = 0.438$. The deterrence function therefore equals Adversary's expected economic cost of war there, and we set $\eta_t = \sum_T \pi_T \mathcal{E}^{(T)}$ so that the deterrence constraint binds at that steady state. The steady state of the no-industrial-policy benchmark is thus the postwar steady state by construction: with the calibrated ϕ_j, the arsenal implied by the data satisfies the procurement conditions, and with the calibrated enemy paths and η_t it satisfies the deterrence constraint with equality.

Table 8: The military baskets and their component sectors. Sectors are ordered within each basket by defense procurement. 'mfg.' abbreviates manufacturing, 'svcs.' services, 'equip.' equipment, 'instr.' instruments, 'prod.' product and 'exc.' except.

code	sector	code	sector
Structures (10 sectors)			
2334A0	Other residential structures	2332C0	Transportation structures & highways & streets
2332D0	Other nonresidential structures	233412	Multifamily residential structures
2332A0	Office & commercial structures	233230	Manufacturing structures
233262	Educational & vocational structures	233240	Power & communication structures
233210	Health care structures	233411	Single-family residential structures
Other equipment (105 sectors)			
334511	Search, detection, & navigation instr. mfg.	326220	Rubber & plastics hoses & belting mfg.
334220	Broadcast & wireless communications equip.	339990	All other miscellaneous mfg.
336320	Motor vehicle electrical & electronic equip. mfg.	332430	Metal can, box, & other metal container (light gauge) mfg.
334515	Electricity & signal testing instr. mfg.	336111	Automobile mfg.
334111	Electronic computer mfg.	315000	Apparel mfg.
335313	Switchgear & switchboard apparatus mfg.	333994	Industrial process furnace & oven mfg.
336350	Motor vehicle transmission & power train parts mfg.	332720	Turned prod. & screw, nut, & bolt mfg.
332200	Cutlery & handtool mfg.	33399A	Other general purpose machinery mfg.
314900	Other textile prod. mills	333130	Mining & oil & gas field machinery mfg.
33441A	Other electronic component mfg.	336112	Light truck & utility vehicle mfg.
33291A	Valve & fittings other than plumbing	33329A	Other industrial machinery mfg.
326290	Other rubber prod. mfg.	335220	Major household appliance mfg.
335314	Relay & industrial control mfg.	33131B	Aluminum prod. mfg. from purchased aluminum
334210	Telephone apparatus mfg.	332320	Ornamental & architectural metal prod. mfg.
326210	Tire mfg.	3219A0	All other wood prod. mfg.
339113	Surgical appliance & supplies mfg.	334513	Industrial process variable instr. mfg.
336120	Heavy duty truck mfg.	336500	Railroad rolling stock mfg.

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Table 8 (continued)

code	sector	code	sector
334510	Electromedical & electrotherapeutic apparatus mfg.	334610	Manufacturing & reproducing magnetic & optical media
332991	Ball & roller bearing mfg.	326110	Plastics packaging materials & unlaminated film & sheet mfg.
336390	Other motor vehicle parts mfg.	327999	Miscellaneous nonmetallic mineral prod.
333314	Optical instrument & lens mfg.	335991	Carbon & graphite prod. mfg.
33399B	Fluid power process machinery	331410	Nonferrous metal (exc. aluminum) smelting & refining
334118	Computer terminals & other computer peripheral equip. mfg.	332999	Other fabricated metal mfg.
335999	All other miscellaneous electrical equip. & component mfg.	334512	Automatic environmental control mfg.
336999	All other transportation equip. mfg.	3252A0	Synthetic rubber & artificial & synthetic fibers & filaments mfg.
333120	Construction machinery mfg.	333612	Speed changer, industrial high-speed drive, & gear mfg.
333914	Measuring, dispensing, & other pumping equip. mfg.	334514	Totalizing fluid meter & counting device mfg.
332500	Hardware mfg.	327200	Glass & glass prod. mfg.
333318	Other commercial & service industry machinery mfg.	335930	Wiring device mfg.
33451A	Watch, clock, & other measuring & controlling device mfg.	333613	Mechanical power transmission equip. mfg.
336310	Motor vehicle gasoline engine & engine parts mfg.	333316	Photographic & photocopying equip. mfg.
333611	Turbine & turbine generator set units mfg.	331420	Copper rolling, drawing, extruding & alloying
334112	Computer storage device mfg.	333517	Machine tool mfg.
334517	Irradiation apparatus mfg.	335210	Small electrical appliance mfg.
333415	Air conditioning, refrigeration, & warm air heating equip. mfg.	336991	Motorcycle, bicycle, & parts mfg.
316000	Leather & allied prod. mfg.	327910	Abrasive prod. mfg.
332119	Metal crown, closure, & other metal stamping (exc. automotive)	313100	Fiber, yarn, & thread mills
3363A0	Motor vehicle steering, suspension component (exc. spring), & brake systems mfg.	339940	Office supplies (exc. paper) mfg.
339112	Surgical & medical instrument mfg.	321910	Millwork
335312	Motor & generator mfg.	333514	Special tool, die, jig, & fixture mfg.
314120	Curtain & linen mills	33721A	Office furniture & custom architectural woodwork & millwork mfg.
339114	Dental equip. & supplies mfg.	332996	Fabricated pipe & pipe fitting mfg.
333920	Material handling equip. mfg.	326190	Other plastics prod. mfg.
333618	Other engine equip. mfg.	321100	Sawmills & wood preservation
332310	Plate work & fabricated structural prod. mfg.	336360	Motor vehicle seating & interior trim mfg.
331313	Alumina refining & primary aluminum production	333413	Industrial & commercial fan & blower & air purification equip. mfg.
313200	Fabric mills	333991	Power-driven handtool mfg.
327991	Cut stone & stone prod. mfg.	333414	Heating equip. (exc. warm air furnaces) mfg.
327100	Clay prod. & refractory mfg.	335912	Primary battery mfg.
334290	Other communications equip. mfg.	325211	Plastics material & resin mfg.
335920	Communication & energy wire & cable mfg.	326150	Urethane & other foam prod. (exc. polystyrene) mfg.

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Table 8 (continued)

code	sector	code	sector
336212	Truck trailer mfg.	326140	Polystyrene foam prod. mfg.
333912	Air & gas compressor mfg.		
Intellectual property (6 sectors)			
541700	Scientific research & development svcs.	541512	Computer systems design svcs.
541300	Architectural, engineering, & related svcs.	511200	Software publishers
541511	Custom computer programming svcs.	54151A	Other computer related svcs., including facilities management
Non-durables (54 sectors)			
324110	Petroleum refineries	311119	Other animal food mfg.
811200	Electronic & precision equip. repair & maintenance	482000	Rail transportation
325412	Pharmaceutical preparation mfg.	311910	Snack food mfg.
481000	Air transportation	322220	Paper bag & coated & treated paper mfg.
517210	Wireless telecommunications carriers (exc. satellite)	325413	In-vitro diagnostic substance mfg.
484000	Truck transportation	312110	Soft drink & ice mfg.
518200	Data processing, hosting, & related svcs.	322210	Paperboard container mfg.
31161A	Animal (exc. poultry) slaughtering, rendering, & processing	5111A0	Directory, mailing list, & other publishers
221100	Electric power generation, transmission, & distribution	311230	Breakfast cereal mfg.
221300	Water, sewage & other systems	311420	Fruit & vegetable canning, pickling, & drying
483000	Water transportation	324190	Other petroleum & coal prod. mfg.
311615	Poultry processing	312140	Distilleries
3259A0	All other chemical prod. & preparation mfg.	311940	Seasoning & dressing mfg.
517110	Wired telecommunications carriers	325610	Soap & cleaning compound mfg.
515100	Radio & television broadcasting	112A00	Animal production, exc. cattle & poultry & eggs
811300	Commercial & industrial machinery & equip. repair & maintenance	325310	Fertilizer mfg.
221200	Natural gas distribution	322230	Stationery prod. mfg.
325180	Other basic inorganic chemical mfg.	325620	Toilet preparation mfg.
323110	Printing	3118A0	Cookie, cracker, pasta, & tortilla mfg.
311513	Cheese mfg.	325320	Pesticide & other agricultural chemical mfg.
493000	Warehousing & storage	322120	Paper mills
311700	Seafood prod. preparation & packaging	325190	Other basic organic chemical mfg.
311810	Bread & bakery prod. mfg.	532400	Commercial & industrial machinery & equip. rental & leasing
532100	Automotive equip. rental & leasing	322130	Paperboard mills
517A00	Satellite, telecommunications resellers, & all other telecommunications	311225	Fats & oils refining & blending
311300	Sugar & confectionery prod. mfg.	339920	Sporting & athletic goods mfg.
311990	All other food mfg.	322299	All other converted paper prod. mfg.
Land systems (1 sector)			
336992	Military armored vehicle, tank, & tank component mfg.		
Air systems (3 sectors)			
336411	Aircraft mfg.	336412	Aircraft engine & engine parts mfg.
336413	Other aircraft parts & auxiliary equip. mfg.		
Naval systems (2 sectors)			

continued on the next page

Table 8 (continued)

code	sector	code	sector
336611	Ship building & repairing	336612	Boat building
Ammunition (3 sectors)			
336414	Guided missile & space vehicle mfg.	33641A	Propulsion units & parts for space vehicles & guided missiles
33299A	Ammunition, arms, ordnance, & accessories mfg.		

Table 9: Overhead services excluded from the military baskets, ordered by defense procurement. These purchases keep installations and personnel running rather than entering the arsenal, so no stock of them raises military power; they are dropped from the procurement bundle rather than reassigned.

code	sector	code	sector
230301	Nonresidential maintenance & repair	561700	Services to buildings & dwellings
561200	Facilities support svcs.	562000	Waste management & remediation svcs.
561300	Employment svcs.	5416A0	Environmental & other technical consulting svcs.
230302	Residential maintenance & repair	722A00	All other food & drinking places
721000	Accommodation	485000	Transit & ground passenger transportation
541800	Advertising, public relations, & related svcs.	512200	Sound recording industries
5191A0	News syndicates, libraries, archives & all other information svcs.	561400	Business support svcs.
722211	Limited-service restaurants	48A000	Scenic & sightseeing transportation & support activities for transportation
722110	Full-service restaurants	492000	Couriers & messengers
531ORE	Other real estate	541610	Management consulting svcs.
812300	Dry-cleaning & laundry svcs.	713900	Other amusement & recreation industries
811400	Personal & household goods repair & maintenance		

C.4 Sovereign's Peacetime Problem

This appendix derives the optimality conditions of the peacetime planning problem (34) and compares them with the competitive equilibrium of Appendix C.1 to prove Proposition 7.

The Lagrangian Attach to the date-t market-clearing condition of sector g the multiplier $\beta^t \Lambda_t p_{g,t}$, to the law of motion of capital $\beta^t \Lambda_t q_{g,t}$, to the law of motion of military stock j the multiplier $\beta^t \Lambda_t \varpi_{j,t}$, and to the date-$(t+1)$ deterrence constraint $\beta^{t+1} \Lambda_{t+1} \gamma_{t+1}$. Normalizing the multipliers by Λ_t in this way states them in units of the date-t consumption good. Deterrence and the public good depend on the predetermined stocks only, so the constraints that tie the periods together are the two laws of motion and the deterrence

constraint; every remaining choice is a within-period one. Substituting the aggregators for C_t, $\tilde{L}_t$, $V_{g,t}$, $X_{g,t}$, $Y_{g,t}$, $I_{g,t}$ and $m_{j,t}$, and with $I_{g,t} \geq 0$, $m_{j,t} \geq 0$ and $\gamma_{t+1} \geq 0$, the Lagrangian is

$$\begin{aligned}\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ & \log C_t - \frac{\tilde{L}_t^{\,1+1/\xi}}{1+1/\xi} + G_0 \\ & + \sum_{g\in\mathcal{G}} \Lambda_t\, p_{g,t} \Big[Y_{g,t} - c_{g,t} - \sum_b x_{b,g,t} - \sum_b i_{b,g,t} - \sum_j m_{j,g,t} \Big] \\ & + \sum_{g\in\mathcal{G}} \Lambda_t\, q_{g,t} \Big[(1-\delta_g) K_{g,t} + I_{g,t} - \frac{\varphi}{2}\Big(\frac{I_{g,t}}{K_{g,t}} - \delta_g\Big)^2 K_{g,t} - K_{g,t+1} \Big] \\ & + \sum_{j\in\mathcal{J}} \Lambda_t\, \varpi_{j,t} \Big[(1-\delta_j) M_{j,t} + m_{j,t} - M_{j,t+1} \Big] \\ & + \beta\, \Lambda_{t+1}\, \gamma_{t+1} \Big[\mathcal{D}_{t+1}(\mathbf{K}_{t+1}, \mathbf{M}_{t+1}) - \eta_{t+1} \Big] \Big\}.\end{aligned}$$

Static and Investment Conditions With the multipliers $p_{g,t}$ as the goods prices, $w_{g,t} \equiv p_{g,t}\partial Y_{g,t}/\partial L_{g,t}$ as the wages, and $q_{g,t}$ as the shadow value of capital, the planner's first-order conditions for consumption, labor, intermediate inputs, investment goods, and investment are the competitive conditions for these choices in (105)–(110), so the planner's within-period allocation coincides with the competitive one. Two of them need a step. The condition for consumption, the first below, inverts to the CES demand of (105); multiplying it by $c_{g,t}$ and summing over g, linear homogeneity of the aggregator gives $\sum_g p_{g,t} c_{g,t} = C_t$, so the consumption price index equals one and the multipliers are prices in the numeraire of Appendix C.1. The condition for investment goods, the second below, is the demand of (109), and summing it over r gives the investment condition (110). The two conditions are

$$\Big(\epsilon_g \frac{C_t}{c_{g,t}}\Big)^{\frac{1}{\sigma_C}} = p_{g,t}, \qquad p_{r,t}\, i_{g,r,t} = q_{g,t}\Big[1 - \varphi\Big(\frac{I_{g,t}}{K_{g,t}} - \delta_g\Big)\Big] \vartheta_{g,r}\, I_{g,t}.$$

Capital The stock $K_{g,t+1}$ raises date-$(t+1)$ output, affects the date-$(t+1)$ deterrence constraint, and carries over into $K_{g,t+2}$. Only the date-$(t+1)$ deterrence constraint contains $K_{g,t+1}$, so a single multiplier, γ_{t+1}, appears. With the carry-over $\partial K_{g,t+2}/\partial K_{g,t+1} = (1-\delta_g) + (\varphi/2)\big((I_{g,t+1}/K_{g,t+1})^2 - \delta_g^2\big)$ implied by the law of motion, the first-order con-

dition for $K_{g,t+1}$ is

$$q_{g,t} = \beta \frac{\Lambda_{t+1}}{\Lambda_t} \Big[\gamma_{t+1} \frac{\partial \mathcal{D}_{t+1}}{\partial K_{g,t+1}} + p_{g,t+1} \, \mathrm{MPK}_{g,t+1} + (1 - \delta_g) \, q_{g,t+1} \\ + q_{g,t+1} \frac{\varphi}{2} \Big(\Big(\frac{I_{g,t+1}}{K_{g,t+1}} \Big)^2 - \delta_g^2 \Big) \Big]. \tag{117}$$

Procurement and the Military Stocks Within each basket the planner buys goods at minimum cost, which gives the Cobb–Douglas demands of (112), so a unit of basket j costs $p_{m,j,t}$. The planner buys the basket up to the point where this cost equals the shadow value $\varpi_{j,t}$ of the stock, and buys none if the cost exceeds it. Military stocks enter neither production nor utility, so, unlike capital, they are valued for deterrence and carry-over alone. With $\gamma_{t+1} \geq 0$, $\mathcal{D}_{t+1} \geq \eta_{t+1}$, and $\gamma_{t+1}(\mathcal{D}_{t+1} - \eta_{t+1}) = 0$, the conditions for procurement and for $M_{j,t+1}$ are

$$p_{m,j,t} \geq \varpi_{j,t}, \qquad m_{j,t} \geq 0, \qquad \text{with complementary slackness,} \\ \Lambda_t \, \varpi_{j,t} = \beta \, \Lambda_{t+1} \Big[\gamma_{t+1} \frac{\partial \mathcal{D}_{t+1}}{\partial M_{j,t+1}} + (1 - \delta_j) \, \varpi_{j,t+1} \Big]. \tag{118}$$

Proof of Proposition 7. Let the planner's allocation solve problem (34), with multipliers $p_{g,t}$, $q_{g,t}$, $\varpi_{j,t}$ and γ_t. Take $p_{g,t}$ as the goods prices, $w_{g,t} = p_{g,t} \partial Y_{g,t} / \partial L_{g,t}$ as the wages, and $1/(1 + r_t) = \beta \Lambda_{t+1} / \Lambda_t$ as the bond price. Let the government purchase the planner's $m_{j,g,t}$, balance its budget with the lump-sum tax of (112), and pay the capital subsidies

$$s_{g,t} \, P^I_{g,t} = \gamma_t \frac{\partial \mathcal{D}_t}{\partial K_{g,t}}, \qquad g \in \mathcal{G}, \; t \geq 1. \tag{119}$$

By the static and investment conditions, the planner's allocation satisfies (105)–(110) at these prices, and its procurement, goods market clearing, and laws of motion satisfy (112)–(114). The two systems therefore differ in one condition only, the Euler equation for capital: where the competitive equation (111) has the subsidy $s_{g,t+1} P^I_{g,t+1}$, the planner's equation (117) has the deterrence value $\gamma_{t+1} \, \partial \mathcal{D}_{t+1} / \partial K_{g,t+1}$, which the firm does not internalize. Under (119) the two equations coincide, so the planner's allocation, together with the price system above, satisfies every condition of the competitive equilibrium of Appendix C.1; goods market clearing and the government budget then make the household's budget constraint hold by Walras' law. Solving (119) for the rate $s_{g,t}$ gives the statement of the proposition. Since $\gamma_t \geq 0$, the sign of the subsidy is the sign of $\partial \mathcal{D}_t / \partial K_{g,t}$, which the production network does not restrict. □

C.5 Sovereign's Peacetime Problem without Industrial Policy

This appendix derives the optimality conditions of the no-industrial-policy benchmark. In the benchmark the capital subsidies are zero and the government deters with procurement alone. Private investment responds to procurement in equilibrium, but the government does not internalize the resulting changes in deterrence when it chooses its purchases.

The Government's Problem The private allocation is the competitive equilibrium of Appendix C.1 with $s_{g,t} = 0$: given the government's purchases, prices and quantities satisfy (105)–(114), and sectoral capital obeys the firms' Euler equation (111) without the subsidy term. The government takes this allocation as given and chooses the purchases $m_{j,g,t}$, the baskets $m_{j,t}$, and the military stocks $M_{j,t+1}$, financing them with the lump-sum tax $T_t = \sum_j \sum_{g \in \mathcal{G}_j} p_{g,t} m_{j,g,t}$. Let $\mathcal{W}$ denote the household's lifetime utility along the competitive allocation. Since taxes are lump sum and, given the purchases, the competitive allocation is undistorted, the envelope theorem gives $\partial \mathcal{W} / \partial m_{j,g,t} = -\beta^t \Lambda_t p_{g,t}$: the purchase costs the household its price in the numeraire, valued at marginal utility, and the induced changes in prices and quantities have no first-order effect on welfare. Unlike the planner of Appendix C.4, the government has no first-order condition for sectoral capital, and the capital stocks $\mathbf{K}_{t+1}$ in the deterrence function are those chosen by the firms. Attaching, as in Appendix C.4, the multiplier $\beta^t \Lambda_t \varpi_{j,t}$ to the law of motion of stock j and $\beta^{t+1} \Lambda_{t+1} \gamma_{t+1}$ to the date-$(t+1)$ deterrence constraint, and with $m_{j,t} \geq 0$ and $\gamma_{t+1} \geq 0$, the government's Lagrangian is

$$\mathcal{L} = \mathcal{W} + \sum_{t=0}^{\infty} \beta^t \Big\{ \sum_{j \in \mathcal{J}} \Lambda_t \, \varpi_{j,t} \Big[(1 - \delta_j) M_{j,t} + m_{j,t} - M_{j,t+1} \Big] + \beta \, \Lambda_{t+1} \, \gamma_{t+1} \Big[\mathcal{D}_{t+1}(\mathbf{K}_{t+1}, \mathbf{M}_{t+1}) - \eta_{t+1} \Big] \Big\}.$$

Procurement and the Military Stocks The conditions for procurement and for $M_{j,t+1}$ are those of the planner in Appendix C.4, with the derivative $\partial \mathcal{D}_{t+1} / \partial M_{j,t+1}$ taken at given $\mathbf{K}_{t+1}$: the two problems set procurement by the same rule, and differ only in the capital stocks at which the rule is evaluated. The first-order condition for $M_{j,t+1}$ is

$$\Lambda_t \, \varpi_{j,t} = \beta \, \Lambda_{t+1} \Big[\gamma_{t+1} \frac{\partial \mathcal{D}_{t+1}}{\partial M_{j,t+1}} + (1 - \delta_j) \, \varpi_{j,t+1} \Big]. \tag{120}$$

C.6 Solution Method

This subsection describes the numerical method we use to solve the model.

Wars and the Deterrence Function We solve each war as a deterministic transition path under perfect foresight, stacking the equilibrium conditions of all its periods and solving them simultaneously, the method of Juillard (1996). The wartime capital Euler equation (111) at $s = T - 1$ involves postwar prices and returns, so a war of length $T \geq 2$ is solved together with $H = 8$ postwar periods. Procurement follows (115)–(116) until the last period of the war and the spending rule from that period on. The path is closed by setting prices, consumption, and the shadow value of capital in the period after the last postwar period to their steady-state values, with the return on capital evaluated at the terminal capital stock. In a one-period war no purchase can affect the settlement, which depends on the initial stocks alone. The paths of military power across war lengths, together with Adversary's paths, its economic costs, and the duration probabilities, give the deterrence function $\mathcal{D}(\mathbf{K}, \mathbf{M})$. Its derivatives with respect to the initial stocks follow from the implicit function theorem applied to the equilibrium conditions of each war; the derivatives with respect to the military stocks also include the change in the duration probabilities.

Calibration The combat efficiency parameters $\chi_1^{(T)}, \ldots, \chi_{T-1}^{(T)}$ of a war of length T are pinned down by the wartime spending targets σ^* of Appendix C.3: in each war year $s = 0, \ldots, T-2$, military spending exceeds its prewar level by σ_s^* of prewar GDP, in wars started from the postwar steady state. We solve for them jointly with the wartime path, adding the $T - 1$ targets to its equilibrium conditions. The military-power weights ϕ_j are pinned down by the observed composition of procurement: at the calibrated weights, the government that deters with procurement only holds, in steady state, the arsenal implied by the observed spending shares θ_j, so that its procurement conditions (120), evaluated at constant stocks, hold at that arsenal. The two sets of parameters depend on each other, since the wars that determine $\chi^{(T)}$ depend on ϕ and the procurement conditions depend on $\chi^{(T)}$. We therefore alternate between solving the procurement conditions for ϕ at given $\chi^{(T)}$ and inverting $\chi^{(T)}$ at given ϕ, until both hold. Adversary's paths, χ_0, and η follow at the calibrated weights, as described in Appendix C.3.

The Optimal Deterrence Equilibrium Given the calibrated parameters and Adversary's paths, the optimal steady state solves the peacetime first-order conditions of Appendix C.4 at constant stocks, with $I_g = \delta_g K_g$, $M_j = m_j/\delta_j$, and the multiplier on the deterrence constraint constant at ζ. When investment and procurement are positive, these

are $N + J + 1$ equations in the sectoral capital stocks, the procurement flows, and ζ, displayed below. At each candidate allocation, the static equilibrium gives the prices and the returns on capital, and the wars give $\mathcal{D}$ and its derivatives. The subsidies are then $s_g = \zeta \left(\partial \mathcal{D} / \partial K_g\right) / P_g^I$. The conditions are

$$p_g \, \mathrm{MPK}_g + \zeta \, \frac{\partial \mathcal{D}}{\partial K_g} = (\rho + \delta_g) \, P_g^I, \quad g \in \mathcal{G},$$

$$\zeta \, \frac{\partial \mathcal{D}}{\partial M_j} = (\rho + \delta_j) \, p_{m,j}, \quad j \in \mathcal{J}, \qquad \mathcal{D}(\mathbf{K}, \mathbf{M}) = \eta.$$

C.7 The Full Ranking of Optimal Capital Subsidies

Table 10 reports the optimal capital subsidy of each of the 370 sectors, a rate on the replacement value of the sector's capital stock. Sectors are ordered by decreasing subsidy, reading down the left column and then down the right column of each page; the 20 highest are listed in Table 3 in the main text.

Table 10: Optimal capital subsidies, all 370 sectors, in percent of the replacement value of the sector's capital stock, ordered by decreasing subsidy. 'mfg.' abbreviates manufacturing and 'svcs.' services.

rank	sector	s_g	rank	sector	s_g
1	Guided missile & space vehicle mfg.	11.51	24	Custom computer programming svcs.	2.16
2	Propulsion units & parts for space vehicles & guided missiles	11.46	25	Environmental & other technical consulting svcs.	2.10
3	Ship building & repairing	8.89	26	Scenic & sightseeing transportation & support activities for transportation	2.02
4	Search, detection, & navigation instr. mfg.	8.56	27	All other transportation equip. mfg.	2.01
5	Military armored vehicle, tank, & tank component mfg.	7.18	28	Printed circuit assembly (electronic assembly) mfg.	2.00
6	Ammunition, arms, ordnance, & accessories mfg.	6.18	29	Wireless telecommunications carriers (except satellite)	2.00
7	Other aircraft parts & auxiliary equip. mfg.	6.04	30	All other chemical product & preparation mfg.	1.91
8	Aircraft mfg.	5.59	31	All other miscellaneous professional, scientific, & technical svcs.	1.90
9	Truck transportation	5.51	32	Semiconductor & related device mfg.	1.86
10	Architectural, engineering, & related svcs.	5.34	33	Manufacturing & reproducing magnetic & optical media	1.83
11	Electronic & precision equip. repair & maintenance	4.63	34	Management consulting svcs.	1.83
12	Electricity & signal testing instr. mfg.	4.55	35	Specialized design svcs.	1.81
13	Other computer related svcs., including facilities management	3.95	36	Carbon & graphite product mfg.	1.81
14	Broadcast & wireless communications equip.	3.89	37	Data processing, hosting, & related svcs.	1.77
15	Water transportation	3.41	38	Computer terminals & other computer peripheral equip. mfg.	1.71
16	Switchgear & switchboard apparatus mfg.	3.40	39	Office administrative svcs.	1.70

continued on the next page

Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
17	Computer systems design svcs.	3.32	40	Communication & energy wire & cable mfg.	1.69
18	Air transportation	3.21	41	Ball & roller bearing mfg.	1.69
19	Aircraft engine & engine parts mfg.	2.84	42	Other support svcs.	1.68
20	Other electronic component mfg.	2.82	43	Biological product (except diagnostic) mfg.	1.68
21	Motor vehicle electrical & electronic equip. mfg.	2.72	44	Satellite, telecommunications resellers, & all other telecommunications	1.66
22	Scientific research & development svcs.	2.65	45	Relay & industrial control mfg.	1.65
23	Telephone apparatus mfg.	2.38	46	Software publishers	1.64

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Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
47	Other textile product mills	1.63	81	Plastics material & resin mfg.	1.26
48	Other basic inorganic chemical mfg.	1.59	82	Motor vehicle transmission & power train parts mfg.	1.26
49	Radio & television broadcasting	1.57	83	News syndicates, libraries, archives & all other information svcs.	1.25
50	Cutlery & handtool mfg.	1.57	84	Synthetic rubber & artificial & synthetic fibers & filaments mfg.	1.24
51	Computer storage device mfg.	1.56	85	Commercial & industrial machinery & equip. rental & leasing	1.24
52	Advertising, public relations, & related svcs.	1.54	86	Synthetic dye & pigment mfg.	1.22
53	Pharmaceutical preparation mfg.	1.52	87	Soap & cleaning compound mfg.	1.22
54	Employment svcs.	1.52	88	Facilities support svcs.	1.22
55	Other communications equip. mfg.	1.52	89	Valve & fittings other than plumbing	1.21
56	Fluid power process machinery	1.51	90	Management of companies & enterprises	1.21
57	Watch, clock, & other measuring & controlling device mfg.	1.49	91	Motor & generator mfg.	1.20
58	Automatic environmental control mfg.	1.47	92	Pesticide & other agricultural chemical mfg.	1.20
59	Couriers & messengers	1.45	93	Electronic computer mfg.	1.17
60	Accounting, tax preparation, bookkeeping, & payroll svcs.	1.43	94	Fertilizer mfg.	1.17
61	Medicinal & botanical mfg.	1.42	95	Wiring device mfg.	1.16
62	Industrial gas mfg.	1.42	96	All other forging, stamping, & sintering	1.16
63	Petrochemical mfg.	1.42	97	Commercial & industrial machinery & equip. repair & maintenance	1.14
64	Printing ink mfg.	1.41	98	Office supplies (except paper) mfg.	1.13
65	Motorcycle, bicycle, & parts mfg.	1.38	99	Paint & coating mfg.	1.12
66	Heavy duty truck mfg.	1.38	100	Newspaper publishers	1.12
67	Wired telecommunications carriers	1.38	101	Mechanical power transmission equip. mfg.	1.12
68	Rail transportation	1.37	102	Other motor vehicle parts mfg.	1.11
69	Business support svcs.	1.37	103	Lessors of nonfinancial intangible assets	1.11
70	Other basic organic chemical mfg.	1.37	104	Totalizing fluid meter & counting device mfg.	1.11
71	Internet publishing & broadcasting & Web search portals	1.37	105	Leather & allied product mfg.	1.11
72	Boat building	1.37	106	Petroleum & petroleum prod.	1.10
73	In-vitro diagnostic substance mfg.	1.36	107	Audio & video equip. mfg.	1.10
74	Other rubber product mfg.	1.35	108	Toilet preparation mfg.	1.10
75	Petroleum refineries	1.34	109	Spring & wire product mfg.	1.10
76	Investigation & security svcs.	1.34	110	Travel arrangement & reservation svcs.	1.09
77	Printing	1.31	111	Adhesive mfg.	1.08
78	Directory, mailing list, & other publishers	1.29	112	Apparel mfg.	1.08
79	Photographic svcs.	1.29	113	Optical instrument & lens mfg.	1.07
80	Periodical Publishers	1.28	114	All other miscellaneous electrical equip. & component mfg.	1.06

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Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
115	Hardware mfg.	1.04	149	Plastics pipe, pipe fitting, & unlaminated profile shape mfg.	0.86
116	Electric lamp bulb & part mfg.	1.03	150	Drugs & druggists' sundries	0.86
117	Speed changer, industrial high-speed drive, & gear mfg.	1.03	151	Services to buildings & dwellings	0.86
118	All other miscellaneous mfg.	1.03	152	Tire mfg.	0.85
119	Other support activities for mining	1.03	153	Lime & gypsum product mfg.	0.84
120	Turbine & turbine generator set units mfg.	1.02	154	Book publishers	0.84
121	Legal svcs.	1.01	155	Support activities for printing	0.84
122	Primary battery mfg.	1.01	156	Motor home mfg.	0.84
123	Copper rolling, drawing, extruding & alloying	1.00	157	Laminated plastics plate, sheet (except packaging), & shape mfg.	0.84
124	Storage battery mfg.	1.00	158	Travel trailer & camper mfg.	0.82
125	Alumina refining & primary aluminum production	1.00	159	Ophthalmic goods mfg.	0.82
126	Concrete pipe, brick, & block mfg.	1.00	160	Dental laboratories	0.82
127	Motor vehicle steering, suspension component (except spring), & brake systems mfg.	1.00	161	Nonferrous metal (except aluminum) smelting & refining	0.81
128	Major household appliance mfg.	1.00	162	Other animal food mfg.	0.80
129	Coating, engraving, heat treating & allied activities	1.00	163	Glass & glass product mfg.	0.80
130	Veterinary svcs.	0.99	164	Other plastics product mfg.	0.79
131	Doll, toy, & game mfg.	0.98	165	Paper bag & coated & treated paper mfg.	0.79
132	Surgical appliance & supplies mfg.	0.98	166	Textile & fabric finishing & fabric coating mills	0.77
133	Water, sewage & other systems	0.98	167	Turned product & screw, nut, & bolt mfg.	0.77
134	General & consumer goods rental	0.98	168	Oil & gas extraction	0.77
135	Jewelry & silverware mfg.	0.97	169	Polystyrene foam product mfg.	0.77
136	Seafood product preparation & packaging	0.95	170	Custom roll forming	0.76
137	Plate work & fabricated structural product mfg.	0.95	171	Motor vehicle gasoline engine & engine parts mfg.	0.74
138	Small electrical appliance mfg.	0.95	172	Curtain & linen mills	0.74
139	Cheese mfg.	0.93	173	Warehousing & storage	0.73
140	Other petroleum & coal prod. mfg.	0.93	174	Industrial process variable instr. mfg.	0.73
141	Other engine equip. mfg.	0.92	175	Aluminum product mfg. from purchased aluminum	0.73
142	Nonferrous metal (except copper & aluminum) rolling, drawing, extruding & alloying	0.90	176	Wholesale electronic markets & agents & brokers	0.73
143	Poultry processing	0.90	177	Paperboard mills	0.72
144	Household appliances & electrical & electronic goods	0.90	178	Cut stone & stone product mfg.	0.72
145	Metal crown, closure, & other metal stamping (except automotive)	0.88	179	Other nondurable goods merchant wholesalers	0.72
146	Sporting & athletic goods mfg.	0.87	180	Personal & household goods repair & maintenance	0.72
147	Machine shops	0.87	181	Fiber, yarn, & thread mills	0.71
148	Animal (except poultry) slaughtering, rendering, & processing	0.86	182	Abrasive product mfg.	0.71

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Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
183	Dry-cleaning & laundry svcs.	0.71	217	Nonstore retailers	0.59
184	Dental equip. & supplies mfg.	0.71	218	All other converted paper product mfg.	0.59
185	Paperboard container mfg.	0.70	219	Mineral wool mfg.	0.59
186	Ground or treated mineral & earth mfg.	0.69	220	Flavoring syrup & concentrate mfg.	0.59
187	Stationery product mfg.	0.69	221	Tobacco mfg.	0.59
188	Beef cattle ranching & farming, including feedlots & dual-purpose ranching & farming	0.68	222	Urethane & other foam product (except polystyrene) mfg.	0.58
189	Grocery & related product wholesalers	0.68	223	Spectator sports	0.58
190	Metal can, box, & other metal container (light gauge) mfg.	0.68	224	Soft drink & ice mfg.	0.58
191	Ferrous metal foundries	0.68	225	Electric power generation, transmission, & distribution	0.58
192	Ornamental & architectural metal prod. mfg.	0.67	226	Poultry & egg production	0.58
193	Fabric mills	0.67	227	Religious organizations	0.58
194	Plastics packaging materials & unlaminated film & sheet mfg.	0.66	228	All other food mfg.	0.57
195	Steel product mfg. from purchased steel	0.66	229	Personal care svcs.	0.57
196	Civic, social, professional, & similar organizations	0.65	230	Pulp mills	0.57
197	Other furniture related product mfg.	0.65	231	Heating equip. (except warm air furnaces) mfg.	0.57
198	Motor vehicle & motor vehicle parts & supplies	0.65	232	Grantmaking, giving, & social advocacy organizations	0.56
199	Other personal svcs.	0.65	233	Bread & bakery product mfg.	0.56
200	Plastics bottle mfg.	0.64	234	Breweries	0.56
201	Automotive equip. rental & leasing	0.64	235	Sugar & confectionery product mfg.	0.56
202	Distilleries	0.64	236	Photographic & photocopying equip. mfg.	0.55
203	Clay product & refractory mfg.	0.63	237	Snack food mfg.	0.55
204	Other durable goods merchant wholesalers	0.63	238	Truck trailer mfg.	0.55
205	Nonferrous metal foundries	0.62	239	Waste management & remediation svcs.	0.55
206	Iron & steel mills & ferroalloy mfg.	0.62	240	Natural gas distribution	0.55
207	Breakfast cereal mfg.	0.62	241	All other food & drinking places	0.55
208	Copper, nickel, lead, & zinc mining	0.61	242	Animal production, except cattle & poultry & eggs	0.55
209	Paper mills	0.61	243	Other household nonupholstered furniture	0.54
210	Asphalt paving mixture & block mfg.	0.61	244	Professional & commercial equip. & supplies	0.53
211	Other fabricated metal mfg.	0.61	245	Independent artists, writers, & performers	0.53
212	Automotive repair & maintenance (including car washes)	0.60	246	Soybean & other oilseed processing	0.52
213	Death care svcs.	0.60	247	Wet corn milling	0.52
214	Carpet & rug mills	0.60	248	Flour milling & malt mfg.	0.52
215	Wineries	0.59	249	Fats & oils refining & blending	0.52
216	Fabricated pipe & pipe fitting mfg.	0.59	250	Rubber & plastics hoses & belting mfg.	0.52

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Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
251	Offices of dentists	0.52	285	Grain farming	0.41
252	Offices of other health practitioners	0.52	286	Limited-service restaurants	0.40
253	Seasoning & dressing mfg.	0.51	287	Electromedical & electrotherapeutic apparatus mfg.	0.40
254	Cement mfg.	0.51	288	Residential maintenance & repair	0.39
255	Museums, historical sites, zoos, & parks	0.51	289	Coal mining	0.38
256	Promoters of performing arts & sports & agents for public figures	0.50	290	Other real estate	0.36
257	Nonupholstered wood household furniture mfg.	0.50	291	All other wood product mfg.	0.35
258	Fruit & vegetable canning, pickling, & drying	0.50	292	Plumbing fixture fitting & trim mfg.	0.35
259	Upholstered household furniture mfg.	0.50	293	Support activities for agriculture & forestry	0.35
260	Sanitary paper product mfg.	0.50	294	Analytical laboratory instrument mfg.	0.34
261	Outpatient care centers	0.50	295	Other crop farming	0.34
262	Dry, condensed, & evaporated dairy product mfg.	0.50	296	Iron, gold, silver, & other metal ore mining	0.33
263	Home health care svcs.	0.50	297	Sound recording industries	0.32
264	Elementary & secondary schools	0.49	298	Irradiation apparatus mfg.	0.31
265	Offices of physicians	0.49	299	Child day care svcs.	0.31
266	Nonresidential maintenance & repair	0.48	300	Individual & family svcs.	0.30
267	Coffee & tea mfg.	0.48	301	Power-driven handtool mfg.	0.30
268	Medical & diagnostic laboratories	0.48	302	Miscellaneous nonmetallic mineral prod.	0.30
269	Cookie, cracker, pasta, & tortilla mfg.	0.48	303	Multifamily residential structures	0.30
270	Pipeline transportation	0.48	304	Community food, housing, & other relief svcs., including rehabilitation svcs.	0.29
271	Fluid milk & butter mfg.	0.48	305	Greenhouse, nursery, & floriculture production	0.29
272	Dog & cat food mfg.	0.48	306	Millwork	0.29
273	Fishing, hunting & trapping	0.48	307	Single-family residential structures	0.28
274	Frozen food mfg.	0.47	308	Vegetable & melon farming	0.28
275	Ice cream & frozen dessert mfg.	0.47	309	Oilseed farming	0.28
276	Junior colleges, colleges, universities, & professional schools	0.47	310	Fruit & tree nut farming	0.28
277	Other ambulatory health care svcs.	0.47	311	Accommodation	0.26
278	Power, distribution, & specialty transformer mfg.	0.46	312	Hospitals	0.24
279	Full-service restaurants	0.46	313	Automobile mfg.	0.24
280	Performing arts companies	0.45	314	Amusement parks & arcades	0.24
281	Other educational svcs.	0.44	315	Other amusement & recreation industries	0.24
282	Dairy cattle & milk production	0.44	316	Gambling industries (except casino hotels)	0.23
283	Lighting fixture mfg.	0.43	317	Forestry & logging	0.23
284	Transit & ground passenger transportation	0.43	318	Other residential structures	0.23

continued on the next page

Table 10 (continued)

rank	sector	s_g	rank	sector	s_g
319	Machinery, equip., & supplies	0.22	345	Veneer, plywood, & engineered wood product mfg.	-0.23
320	Sawmills & wood preservation	0.20	346	Stone mining & quarrying	-0.24
321	Surgical & medical instrument mfg.	0.19	347	Industrial mold mfg.	-0.28
322	Industrial & commercial fan & blower & air purification equip. mfg.	0.18	348	Railroad rolling stock mfg.	-0.29
323	Other concrete product mfg.	0.17	349	Other general purpose machinery mfg.	-0.35
324	Cutting & machine tool accessory, rolling mill, & other metalworking machinery mfg.	0.17	350	Showcase, partition, shelving, & locker mfg.	-0.37
325	Residential mental health, substance abuse, & other residential care facilities	0.17	351	Lawn & garden equip. mfg.	-0.41
326	Ready-mix concrete mfg.	0.16	352	Special tool, die, jig, & fixture mfg.	-0.41
327	Nursing & community care facilities	0.16	353	Drilling oil & gas wells	-0.41
328	Asphalt shingle & coating materials mfg.	0.14	354	Cable & other subscription programming	-0.44
329	Motion picture & video industries	0.14	355	Machine tool mfg.	-0.46
330	Construction machinery mfg.	0.11	356	Material handling equip. mfg.	-0.48
331	Other nonmetallic mineral mining & quarrying	0.11	357	Packaging machinery mfg.	-0.53
332	Motor vehicle metal stamping	0.09	358	Other industrial machinery mfg.	-0.54
333	Air conditioning, refrigeration, & warm air heating equip. mfg.	0.09	359	Power boiler & heat exchanger mfg.	-0.55
334	Motor vehicle seating & interior trim mfg.	0.05	360	Metal tank (heavy gauge) mfg.	-0.56
335	Other commercial & service industry machinery mfg.	0.02	361	Power & communication structures	-0.58
336	Transportation structures & highways & streets	0.02	362	Air & gas compressor mfg.	-0.59
337	Measuring, dispensing, & other pumping equip. mfg.	0.02	363	Farm machinery & equip. mfg.	-0.62
338	Semiconductor machinery mfg.	-0.04	364	Other nonresidential structures	-0.62
339	Mining & oil & gas field machinery mfg.	-0.07	365	Manufacturing structures	-0.66
340	Motor vehicle body mfg.	-0.07	366	Institutional furniture mfg.	-0.88
341	Educational & vocational structures	-0.09	367	Office furniture & custom architectural woodwork & millwork mfg.	-0.96
342	Industrial process furnace & oven mfg.	-0.11	368	Office & commercial structures	-0.99
343	Wood kitchen cabinet & countertop mfg.	-0.12	369	Health care structures	-1.01
344	Light truck & utility vehicle mfg.	-0.19	370	Sign mfg.	-1.04

C.8 Capacity Experiments

We start the six-year war from the optimal peacetime steady state and raise the capital stock of one of the twenty sectors of Table 3 by 10% at the outbreak, one sector at a time, holding the combat efficiency parameters $\chi^{(T)}$ fixed; the wartime government re-optimizes its procurement. All changes are relative to the same war without the capital increase. Table 11 reports the change in the price of each military basket, averaged over the five war years in which the government procures. Table 12 reports the change in the sector's total output over the six war years and the average change in its price, and splits the rise in the quantity it sells across consumption, investment, intermediate, and military buyers.

Table 11: The change in military basket prices under a 10% increase in one sector's capital stock at the outbreak of a war of length $T = 6$.

sector (+10% K)	str	eq	int	nd	land	air	naval	ammo
Guided missile & space vehicle mfg.	-0.000	+0.001	+0.000	+0.000	+0.001	+0.001	+0.001	-0.018
Propulsion units & parts for space vehicles & guided missiles	-0.001	+0.003	+0.001	+0.001	+0.002	+0.002	+0.003	-0.059
Ship building & repairing	-0.006	+0.014	+0.004	+0.005	+0.013	+0.013	-0.435	+0.010
Search, detection, & navigation instr. mfg.	-0.041	-0.539	+0.024	+0.047	+0.089	+0.100	-0.069	+0.049
Military armored vehicle, tank, & tank component mfg.	-0.003	+0.005	-0.002	+0.005	-1.468	+0.005	+0.008	+0.007
Ammunition, arms, ordnance, & accessories mfg.	-0.004	+0.010	+0.007	+0.009	+0.011	+0.018	+0.012	-0.303
Other aircraft parts & auxiliary equip. mfg.	-0.010	+0.037	+0.010	+0.021	+0.033	-0.328	+0.035	+0.028
Aircraft mfg.	-0.032	+0.140	+0.035	+0.031	+0.063	-1.163	+0.117	+0.098
Truck transportation	-0.003	+0.008	+0.009	-0.114	+0.014	+0.021	+0.012	+0.010
Architectural, engineering, & related svcs.	-0.048	+0.133	-0.441	+0.095	+0.205	+0.239	+0.133	+0.142
Electronic & precision equip. repair & maintenance	-0.018	+0.020	+0.009	-0.186	+0.027	+0.036	+0.023	+0.021
Electricity & signal testing instr. mfg.	-0.003	-0.067	+0.002	+0.009	+0.009	+0.016	+0.010	+0.010
Other computer related svcs., including facilities management	+0.001	+0.007	-0.017	+0.004	+0.011	+0.011	+0.008	+0.007
Broadcast & wireless communications equip.	-0.006	-0.292	+0.022	+0.002	+0.069	+0.045	+0.088	-0.008
Water transportation	+0.001	+0.004	+0.001	-0.014	+0.007	+0.007	-0.030	+0.004
Switchgear & switchboard apparatus mfg.	-0.004	-0.024	+0.001	+0.004	+0.004	+0.007	+0.005	+0.004
Computer systems design svcs.	+0.012	+0.027	-0.054	+0.013	+0.032	+0.034	+0.026	+0.018
Air transportation	+0.015	+0.087	+0.057	-0.232	+0.102	-0.022	+0.093	+0.083
Aircraft engine & engine parts mfg.	+0.003	+0.015	+0.005	+0.010	+0.010	-0.071	+0.016	+0.014
Other electronic component mfg.	+0.002	-0.025	-0.002	+0.003	+0.009	+0.011	+0.009	+0.009

Percent change in each basket's price, mean over the five war years in which the government procures; in red, the largest fall in each row. str: structures; eq: other equipment; int: intellectual property; nd: non-durables.

C.9 Industrial Priorities across Conflict Scenarios

Table 13 reports the optimal capital subsidies of the twenty industries in Table 3 in the land-, air-, and naval-war scenarios of Section 5.4.

Table 12: Output, price, and the composition of the rise in sales of the shocked sector.

	% change		composition of the rise in sales (%)			
sector (+10% K)	output	price	cons.	inv.	interm.	military
Guided missile & space vehicle mfg.	+0.02	-0.02	1.1	0.3	0.8	97.7
Propulsion units & parts for space vehicles & guided missiles	+0.10	-0.40	5.8	0.0	45.4	48.8
Ship building & repairing	+0.72	-0.37	0.3	2.0	0.1	97.5
Search, detection, & navigation instr. mfg.	+1.92	-1.54	2.4	4.7	3.9	89.0
Military armored vehicle, tank, & tank component mfg.	+0.96	-1.27	4.6	0.0	2.3	93.1
Ammunition, arms, ordnance, & accessories mfg.	+0.90	-1.19	24.8	0.0	6.2	69.0
Other aircraft parts & auxiliary equip. mfg.	+0.95	-1.71	19.2	0.0	26.2	54.6
Aircraft mfg.	+1.89	-1.29	6.2	5.2	1.6	87.0
Truck transportation	+0.87	-1.47	23.5	0.0	1.6	74.9
Architectural, engineering, & related svcs.	+1.14	-1.15	5.4	7.2	4.2	83.2
Electronic & precision equip. repair & maintenance	+0.75	-1.32	12.7	0.0	0.5	86.8
Electricity & signal testing instr. mfg.	+1.27	-1.93	15.5	13.6	7.4	63.6
Other computer related svcs., including facilities management	+0.38	-0.93	26.4	0.0	1.2	72.3
Broadcast & wireless communications equip.	+1.50	-1.62	15.0	20.3	10.1	54.6
Water transportation	+0.50	-0.72	66.8	0.0	0.4	32.8
Switchgear & switchboard apparatus mfg.	+0.97	-1.25	9.2	20.2	1.4	69.2
Computer systems design svcs.	+0.51	-0.81	21.4	18.9	1.8	57.9
Air transportation	+2.16	-2.89	61.6	0.0	0.7	37.7
Aircraft engine & engine parts mfg.	+0.90	-1.75	65.4	0.2	19.9	14.5
Other electronic component mfg.	+0.20	-0.92	40.5	0.0	11.9	47.6

Output: percent change in the sector's total output over the six war years. Price: percent change in the sector's price, mean over the six war years. The last four columns split the rise in the quantity sold across buyers and sum to one hundred.

Table 13: Optimal capital subsidies of the twenty baseline priorities across conflict scenarios.

Sector	Land	Air	Naval
Guided missile and space vehicle manufacturing	11.46	11.43	11.55
Propulsion units and parts for space vehicles and guided missiles	11.35	11.36	11.48
Ship building and repairing	0.77	1.04	8.96
Search, detection, and navigation instruments manufacturing	8.55	8.03	8.99
Military armored vehicle, tank, and tank component manufacturing	6.97	3.08	2.74
Ammunition, arms, ordnance, and accessories manufacturing	6.14	6.07	6.34
Other aircraft parts and auxiliary equipment manufacturing	0.96	5.97	0.99
Aircraft manufacturing	1.05	5.51	1.10
Truck transportation	5.62	5.45	5.74
Architectural, engineering, and related services	5.55	5.29	5.69
Electronic and precision equipment repair and maintenance	4.76	4.60	4.84
Electricity and signal testing instruments manufacturing	4.87	4.55	4.99
Other computer related services, including facilities management	4.25	4.03	4.23
Broadcast and wireless communications equipment	4.09	4.04	4.21
Water transportation	3.06	2.89	3.78
Switchgear and switchboard apparatus manufacturing	3.75	3.44	3.78
Computer systems design services	3.54	3.34	3.57
Air transportation	2.44	3.84	2.32
Aircraft engine and engine parts manufacturing	1.04	2.93	1.05
Other electronic component manufacturing	3.02	2.83	3.04

Subsidy rates s_g in percent, a rate on the replacement value of the sector's capital stock. Rows are the twenty industries with the highest subsidies in the baseline (Table 3), in the baseline order.

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